

Def $(R, +, \times)$ (1) $(R, +)$ is an abelian group.(2) $\times$ is associative: $a \times (b \times c) = (a \times b) \times c \quad \forall a, b, c \in R$ (3) $\forall a, b, c \in R$

$$a \times (b + c) = (a \times b) + (a \times c)$$

$$(a + b) \times c = (a \times c) + (b \times c)$$

(4) R is commutative if multiplication is commutative: $a \times b = b \times a$ (5) R has an identity if $\exists 1 \in R$ s.t. $1 \times a = a \times 1 = a \quad \forall a \in R$.if R has 1 .

$$(1+1)(a+b) = (1+1) \times a + (1+1) \times b \\ = a+a+b+b$$

$$(1+1)(a+b) = 1(a+b) + (a+b) = a+b+a+b \\ \Rightarrow a+b = b+a$$

[if R has 1 then addition must be commutative!]we cannot have R has 1 , $+$ is non-commutative.Def Ring with 1 , $1 \neq 0$, is called division ring. if $\forall a \in R, a \neq 0 \exists b$ s.t. $ab = ba = 1$. [A commutative division ring is called a field.]we may have that a ~~ring~~ division ring does not commute too.Examples

(1) trivial ring.

let $(R, +)$ be a commutative group and def $a \times b = 0 \quad \forall a, b \in R$.(2) Ring of integer $(-n, \dots, -1, 0, 1, \dots, n)$ abelian group under addition.
 obviously not a division ring
 multiplication commutative, with identity 1 .Prop 1 let $(R, +, \times)$ be a ring

$$(a) \quad 0 \cdot a = (0+0) \cdot a = 0 \cdot a + 0 \cdot a \Rightarrow 0 \cdot a = 0 = a \cdot (0+0) = a \cdot 0 + a \cdot 0$$

(b)

$$0 = (a+(-a)) \cdot b = (a \cdot b) + (-a) \cdot b \Rightarrow (-a) \cdot b = -(a \cdot b)$$

$$0 = a \cdot (b+(-b)) = a \cdot b + a \cdot (-b) \Rightarrow a \cdot (-b) = -(a \cdot b)$$

$$(c) \quad (-a) \cdot (-b) = a \cdot (-(-b)) = a \cdot b$$

(d) $1, 1'$ identities

$$1 = 1 \cdot 1' = 1'$$

$$-a = -(1 \cdot a) = (-1) \cdot a$$

Def $(R, +, \times)$ (1) $a \in R, a \neq 0$ is called a zero divisors $\exists b \in R, b \neq 0$ s.t. $a \cdot b = 0$ or $b \cdot a = 0$

$$(\mathbb{Z}/6\mathbb{Z}) \quad [2] \neq 0 \quad [3] \neq 0 \in \mathbb{Z}/6\mathbb{Z} \quad [2] \cdot [3] = [6] = [0]$$

(2) R has identity $1 \neq 0$, $u \in R$ is called a unit in R if $\exists v \in R$ s.t. $u \cdot v = v \cdot u = 1$

$$(a \cdot b = 0 \quad v \cdot a = 1 \quad (v \cdot a) \cdot b = b \quad \text{However, } v \cdot (a \cdot b) = 0)$$

Exmples(1) Ring $\mathbb{Z}$ no zero divisors
 $1 \cdot 1 = 1 \quad -1 \cdot -1 = 1 \quad \mathbb{Z}^\times = \{1, -1\}$ (2) $\mathbb{Z}/n\mathbb{Z}$ $\mathbb{Z}/n\mathbb{Z}$ is a unit if u, n are coprimes.
 $\mathbb{Z}/6\mathbb{Z} \quad [2] \neq 0 \quad [3] \neq 0 \in \mathbb{Z}/6\mathbb{Z} \quad [2] \cdot [3] = [6] = [0]$

$$\mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

$$\gcd(a, n) = d$$

$$d \geq 1, \quad 0 \leq b < n, \quad (b) \neq 0$$

$$\text{Let } b = \frac{a}{d} \quad \frac{a}{d} \in \mathbb{Z} \text{ in } \mathbb{Z}/n\mathbb{Z}$$

$\Rightarrow \mathbb{Z}/n\mathbb{Z}$ is a field iff n is prime.

Every nonzero element is a unit. Let a nonzero element is coprime with n .

$$(f) \quad (\mathbb{R}, +, \cdot) \quad R = \{f \in C(\mathbb{R}, \mathbb{R})\} \quad f+g(x) = f(x) + g(x) \quad f \cdot g(x) = f(x)g(x)$$

$$\text{What are the units of } R, \quad f \cdot g = 1 \Leftrightarrow \begin{cases} f \neq 0 \\ g \neq 0 \end{cases}$$

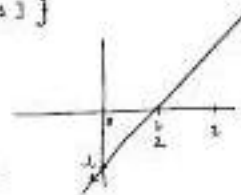
$$f \cdot g = 1 \Leftrightarrow g = \frac{1}{f}$$

$$f \text{ is not a unit and } f \neq 0, \quad f(x) = \begin{cases} 0 & \text{if } f(x) \neq 0 \\ \pm \frac{1}{f(x)} & \text{if } f(x) = 0 \end{cases} \text{ if } f \text{ is a zero divisor}$$

$$(4) \quad (\mathbb{R}, +, \cdot) \quad R = C([0, 1], \mathbb{R})$$

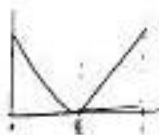
$$R = \{f \in C([0, 1], \mathbb{R}) \mid f(x) \neq 0 \forall x \in [0, 1]\}$$

$$f(x) = \frac{1}{2} \quad f(x) = 0 \text{ at } x = \frac{1}{2}$$

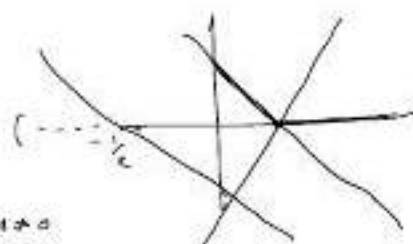


we can find any other continuous function $f \neq 0$ except 0

$$f(x) = \begin{cases} 1, & 0 \leq x \leq \frac{1}{2} \\ 0, & \frac{1}{2} \leq x \leq 1 \end{cases}$$



$$\text{Let } g(x) = 10 - x$$



Def commutative ring with $1 \neq 0$ is an integral domain if it has no zero divisors.

$\mathbb{Z}$ is a field? Nope! even not all its elements are units

Prop 2 $a, b, c \in R$ a is not a zero divisor

$$a \neq 0$$

$$ab = ac \Rightarrow a(b - c) = 0$$

$$\Rightarrow a = 0 \text{ or } b = c$$

a, b, c are elements in an integral domain and $ab = ac \Rightarrow a = 0$ or $b = c$.

$\mathbb{Z}$ is a field? Nope! even not all its elements are units

$\mathbb{Z}$ is a finite integral domain. Let $a \in \mathbb{Z}$.

by the cancellation law $x \mapsto ax$ is injective.

Since $\mathbb{Z}$ is finite the map is also surjective or injective.

$\Rightarrow \exists b$ s.t. $(ab) = 1 \Rightarrow a$ is a unit in $\mathbb{Z}$.

$\Rightarrow \mathbb{Z}$ is a field

(finite division ring is necessarily a field)

Def subring of $(R, +, \cdot)$ is subgroup of R that is closed under multiplication.

Ex (1) $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

$\mathbb{Z}$ is clearly a subgroup

$\mathbb{Z}$ is closed under $\cdot$

likewise for $\mathbb{Q} \subset \mathbb{R}$. $\mathbb{Q} \subset \mathbb{R}$ is also a subring.

$$(2) \quad \mathbb{Z} \subset \mathbb{Z}$$

$\dots -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, \dots$ $\mathbb{Z}$ is clearly a subgroup and multiplication is closed.

$\mathbb{Z}/n\mathbb{Z}$ is not a ring (or subgroup) of $\mathbb{Z}$ unless $n=1$ or $n=2$. Why would it be? lol!!!

(3) ring of continuous functions $\mathbb{R} \rightarrow \mathbb{R}$ is a subring of functions $\mathbb{R} \rightarrow \mathbb{R}$.

$$C(\mathbb{R}) \subset C(\mathbb{R}) \subset \text{Hom}(\mathbb{R}, \mathbb{R})$$

(4) if R is a subring of field F that contains identity of F then R is an integral domain.

$R \subset F$ commutative ring with no non-zero zero divisors $1 \neq 0 \in R$.
Then R is an integral domain (closed under multiplication)

Quadratic Integer Rings

Let D be a square free integer

$$\mathbb{Z}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\} \subset \mathbb{C} = \mathbb{Q}(\sqrt{D})$$

$$D \equiv 1 \pmod{4}$$

$$\mathbb{Z}\left[\frac{1 + \sqrt{D}}{2}\right] = \left\{a + b\frac{(1 + \sqrt{D})}{2} \mid a, b \in \mathbb{Z}\right\}$$

closed under multiplication.

$$\text{Def } \mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\}$$

$$\omega = \begin{cases} \sqrt{D} & \text{if } D \equiv 2, 3 \pmod{4} \\ \frac{1 + \sqrt{D}}{2} & \text{if } D \equiv 1 \pmod{4} \end{cases}$$

R be a ring

$$1. (-1)^2 = (-1) \cdot (-1) = 1 - (-1) = 1 \cdot 1 = 1$$

$$2. u \in R^* \quad b = u^{-1}$$

$$-1 = -(u \cdot b) = (-u) \cdot b$$

$$1 = (-1)^2 = (-u) \cdot b \cdot (-u) \cdot b$$

$$= (-u) \cdot b \cdot u \cdot (-b) = (-u)(-b) = u \in R^*$$

3. $u \in R$

$1 \in S \subset R$

Suppose $u \in S^* \Rightarrow u^{-1} \in S$ s.t. $uu^{-1} = 1 = u^{-1}u$

$$u \in R \quad u^{-1} \in R$$

$\mathbb{Z} \subset \mathbb{Q}$ $1 \in \mathbb{Z}$ s.t. $x > 1$

$$\frac{1}{x} \in \mathbb{Q}^* \quad \frac{1}{x} \notin \mathbb{Z}$$

4. Let $S = \{S \mid S \subset R \text{ a subring}\}$

$a \in \bigcap_{S \in S} S$ so nonempty

S are subgroups under addition. The $\bigcap_{S \in S} S$ is a subgroup.

Let $a, b \in \bigcap_{S \in S} S$ $a \cdot b \in \bigcap_{S \in S} S$ since $a \cdot b \in S \forall S \in S$

$$6. (m) S = \left\{ \frac{n}{p} \mid p \in \mathbb{Z} - \{2\} \cup \{0\} \right\}, n \in \mathbb{Z}, \gcd(n, p) = 1 \}$$

clearly nonempty

Let $a \in S$ $\exists m, n \in \mathbb{Z}$ and $p, q \in \mathbb{Z} - \{2\} \cup \{0\}$

$$a = \frac{m}{p} \quad b = \frac{n}{q} \quad \gcd(m, p) = 1, \gcd(n, q) = 1$$

$$\frac{m}{p} - \frac{n}{q} = \frac{qm - pn}{pq} \quad \frac{pq}{pq} \in \mathbb{Z} - \{2\} \cup \{0\} \quad \frac{qm - pn}{pq} \in S$$

$$a \cdot b = \frac{mn}{pq} \in S$$

b) nope

$$\frac{3}{6} + \frac{3}{6} = \frac{12}{6} = \frac{2}{1} \text{ not in } S$$

$$c) \mathbb{Q}^+ = \left\{ \frac{n}{m} \mid n, m \in \mathbb{Z}^+ \right\}$$

7. $CCR = \{z \in R \mid zr = rz \forall r \in R\}$

$CCR \subset R$ a subring

$a \in CCR$

CCR is a commutative ring

$1 \in R, 1 \neq 0$ then $1 \in CCR$ obviously

CCR is a subring

$\Rightarrow R$ has no nonzero divisors. $(a \cdot b = 0 \text{ for } a, b \neq 0 \in R) \Rightarrow (a^{-1}a) \cdot b = a^{-1}(ab) = 0 \cdot a^{-1} = 0$

$$\Rightarrow CCR^* = CCR / \{0\}$$

$$8. a \in R \quad C(a) = \{r \in R \mid ra = ar\}$$

$$1. a = a \cdot 1 = 1 \cdot a \Rightarrow 1 \in C(a)$$

$$\bigcap_{a \in R} C(a) = CCR$$

10. D is a division ring

$$z \in C(a) \Rightarrow za = az = z^{-1}az$$

$$z^{-1}a = z^{-1}(z^{-1}az) = z^{-1}(za) = az^{-1}$$

$$\Rightarrow z^{-1} \in C(a)$$

11. R is an integral domain and $x^2 = 1$. s.t. some $x \in R \Rightarrow x = \pm 1$

$$(x+1)(x-1) = 0 \quad \text{Since } R \text{ is an integral domain, } x+1=0 \text{ or } x-1=0$$

12. $S \subset R$ $1 \in S$

$a, b \in S$ $ab = ba$ since R is a field (commutative)

$a \in S$ then a is not a zero divisor since $a \in R^*$

$\Rightarrow S$ is an integral domain

$\log(100) + \log(100) = 0$

1. R commutative unitary $\mathbb{Z}$.

$$f(x) = x^2 + 3x + 2, \quad g(x) = x^2 + 2x + 1$$

$$(1) R = \mathbb{Z}$$

$$f(0) + g(0) = 3 \neq 0$$

$$f(0)g(0) = 1 \cdot 1 = 1 \neq 0$$

$$(2) R = \mathbb{Z}/6\mathbb{Z}$$

$$f(0) = 2 \neq 0$$

$$g(0) = 1 \neq 0$$

$$(3) R = \mathbb{Z}/4\mathbb{Z}$$

(4) $f(x)$ is a zero divisor $g(x) \neq 0 \in R[x]$ s.t. $f(x)g(x) = 0$

$$f(x) = \sum_{i=0}^n a_i x^i, \quad g(x) = \sum_{j=0}^m b_j x^j$$

$$f(x)g(x) = \left(\sum_{i=0}^n a_i x^i \right) \left(\sum_{j=0}^m b_j x^j \right) = \sum_{k=0}^{n+m} \left(\sum_{i+j=k} a_i b_j \right) x^k$$

$$a_n b_m = 0$$

$$a_n g(x) = a_n b_m x^{m+1} + a_n b_{m-1} x^{m+2} + \dots + a_n b_0 x^{m+1}$$

$$a_n g(x) = 0$$

$$\Rightarrow a_n b_{m-1} = 0$$

$$a_n g(x) = a_n b_{m-2} x^{m+2} + \dots + a_n b_0 x^{m+2}$$

$$\Rightarrow a_n b_{m-2} = 0 \text{ and so on, so } a_n g(x) = 0$$

(5) is true.

Suppose true for i

$$a_{n-i} g(x) = 0$$

$$\text{then } a_{n-i+1} g(x) = a_{n-i+1} g(x) (a_n x^n + \dots + a_0) = a_{n-i+1} g(x) \left(\sum_{k=0}^n a_k x^k \right) = 0$$

$$\Rightarrow a_{n-i+1} g(x) = 0$$

$$\Rightarrow b_m f(x) x^n = 0 \Rightarrow b_m f(x) = 0$$

(6) (induction)

$$(1-x)(1+x+x^2+\dots)$$

$$= (1+x+x^2+\dots) - (x+x^2+\dots)$$

$$= 1$$

$$(c) \sum_{n=0}^{\infty} a_n x^n \text{ is a unit}$$

$$\exists \sum_{n=0}^{\infty} b_n x^n \text{ s.t. } \left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right) = 1$$

$$\Rightarrow a_0 b_0 = 1$$

$$\Rightarrow a_0 b_0 = 1$$

$$\Rightarrow a_0 \text{ is a unit in } R$$

a_0 is a unit in R .

$$\sum_{n=0}^{\infty} a_n x^n = a_0 \left(1 + \left(\sum_{n=1}^{\infty} a_n x^n \right) a_0^{-1} \right)$$

$$= a_0 \left(1 - \left(-a_0^{-1} \left(\sum_{n=1}^{\infty} a_n x^n \right) \right) \right)$$

$$= a_0 (1 - u(x)) \left(a_0^{-1} (1 + u(x) + u(x)^2 + \dots) \right) = 1$$

(4) clearly if R is an integral domain,

suppose $R[x]$ has a zero divisor.

$$f(x)g(x) = 0$$

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n = 0$$

$$\Rightarrow a_0 b_0 = 0$$

Q

$$(5) F \text{ a field, } F((x)) = \left\{ \sum_{n \in \mathbb{Z}} a_n x^n \mid a_n \in F, n \in \mathbb{Z} \right\}$$

$$(1) \text{ let } f(x) = \sum_{n \in \mathbb{Z}} a_n x^n, \quad a_n \neq 0$$

$$x^{-N} \in F((x))$$

$$\text{then } x^{-N} f(x) = \sum_{n \in \mathbb{Z}} a_n x^{n+N} \text{ is a unit since } a_N \in F \text{ is a unit.}$$

$$\Rightarrow \exists g(x) \in F((x)) \text{ s.t. } x^{-N} f(x) g(x) = 1$$

$$\Rightarrow (x^{-N} f(x)) g(x) = 1$$

$$(2) v: F((x))^* \rightarrow \mathbb{Z}$$

$$v\left(\sum_{n \in \mathbb{Z}} a_n x^n\right) = N$$

$$(i) v\left(\sum_{n \in \mathbb{Z}} a_n x^n \cdot \sum_{m \in \mathbb{Z}} b_m x^m\right) = v\left(\sum_{k \in \mathbb{Z}} \left(\sum_{n+m=k} a_n b_m\right) x^k\right) = v\left(\sum_{n \in \mathbb{Z}} a_n b_{n+N} x^{n+N}\right) = v\left(\sum_{n \in \mathbb{Z}} a_n b_{n+N} x^n\right) = v\left(\sum_{n \in \mathbb{Z}} a_n x^n\right) + v\left(\sum_{n \in \mathbb{Z}} b_{n+N} x^n\right)$$

$$(ii) \text{ let } f \in \mathbb{Z}$$

$$\text{then } \sum_{n \in \mathbb{Z}} a_n x^n \in F((x)) \text{ s.t. } v\left(\sum_{n \in \mathbb{Z}} a_n x^n\right) = f$$

$$(iii) v\left(\sum_{n \in \mathbb{Z}} a_n x^n + \sum_{m \in \mathbb{Z}} b_m x^m\right) = \min\{f, g\}$$

discrete valuation

$$R = \{x \in F((x)) \mid v(x) \geq 0\} \cup \{0\} = F[[x]]$$

$$(12) G = \{g_1, \dots, g_n\}$$

$$N = g_1 + g_2 + \dots + g_n$$

$$\begin{aligned}
 \text{Let } A &= a_1q_1 + a_2q_2 + \dots + a_nq_n \in R_n \\
 AB &= (a_1q_1 + a_2q_2 + \dots + a_nq_n)(b_1 + b_2 + \dots + b_n) \\
 &= a_1(q_1b_1 + q_1b_2 + \dots + q_1b_n) + a_2(q_2b_1 + q_2b_2 + \dots + q_2b_n) \\
 &\quad + \dots + a_n(q_nb_1 + q_nb_2 + \dots + q_nb_n) \\
 &= (a_1q_1 + a_2q_2 + \dots + a_nq_n)(b_1 + b_2 + \dots + b_n) \\
 &= (a_1 + a_2 + \dots + a_n)(q_1b_1 + q_1b_2 + \dots + q_1b_n) \\
 &= (a_1 + a_2 + \dots + a_n)(q_1b_1 + \dots + q_nb_n) = NA
 \end{aligned}$$

Ring Hom. and Quotient Rings

R, S rings

$\varphi: R \rightarrow S$ hom. if

$$\varphi(a+b) = \varphi(a) + \varphi(b)$$

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \forall a, b \in R$$

$$\text{(2) } \ker \varphi = \{a \in R \mid \varphi(a) = 0\}$$

$$\text{(3) } R \cong S \iff \varphi \text{ is a bijection}$$

Examples

(1) $\varphi: \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ even to 0, odd to 1

a is odd, b is even

$$\varphi(a+b) = 1 = \varphi(a) + \varphi(b) = 1 + 0$$

both odd or both even results to 0, since even

$$\varphi(ab) = 0 \quad \varphi(a-b) = 1 = \varphi(a) - \varphi(b) = 1 - 0$$

$\ker(\varphi) = \text{all even numbers}$

$\varphi^{-1}(1) = \text{all odd numbers}$

(2) $\varphi_n: \mathbb{Z} \rightarrow \mathbb{Z} \quad \varphi_n(x) = nx$

φ_n is a homomorphism (additive)

$$\varphi_n(x+y) = n(x+y) = nx + ny = \varphi_n(x) + \varphi_n(y)$$

$\varphi_n(xy) = nxy = \varphi_n(x) \cdot y$ not hom.

(3) $\varphi: \mathbb{Q}[x] \rightarrow \mathbb{Q}$

$$\varphi(x) \mapsto f(x)$$

$$\varphi(\varphi(x) \cdot \varphi(x)) = \varphi(x) \cdot \varphi(x) = \varphi(x) \cdot \varphi(x)$$

$$\varphi(\varphi(x) + \varphi(x)) = \varphi(\varphi(x)) + \varphi(\varphi(x))$$

$$\varphi^{-1}(a) = \{f(x) \in \mathbb{Q}[x] \mid \text{constant term is } a\}$$

$$\ker \varphi = \{f(x) \in \mathbb{Q}[x] \mid f(0) = 0\}$$

Prop R, S ring $\varphi: R \rightarrow S$ hom.

(1) $\text{im } \varphi$ subring in S

$$\text{Let } \varphi(x), \varphi(y) \in \text{im } \varphi$$

$$\varphi(x-y) = \varphi(x) - \varphi(y) \in \text{im } \varphi$$

$$\varphi(xy) = \varphi(x)\varphi(y) \in \text{im } \varphi$$

(2) $\text{Let } x, y \in \ker \varphi$

$$\varphi(x-y) = \varphi(x) - \varphi(y) = 0 \Rightarrow x-y \in \ker \varphi$$

$$\varphi(xy) = \varphi(x)\varphi(y) = 0 \Rightarrow xy \in \ker \varphi$$

$$\text{if } x \in \ker \varphi \quad \forall r \in R$$

$$\ker \varphi(r) = \varphi(r)\varphi(x) = 0$$

$$\varphi(rx) = \varphi(x)\varphi(r) = 0$$

$$\begin{array}{ccc}
 R & \xrightarrow{\varphi} & \text{im } \varphi \\
 \downarrow & \nearrow & \\
 \ker \varphi & &
 \end{array}$$

Def. left, right ideal

$$rI \subseteq I \text{ (left ideal)} \quad Ir \subseteq I \text{ (right ideal)}$$

$$rI = Ir \text{ is 2 ideal. (nonempty, closed under subtraction, closed under multiplication by all elements of } R)$$

Prop R a ring, I is ideal $\Rightarrow R/I$ is a ring.

$$\text{if } (r, s) \in (R/I) \text{ i.e.}$$

$$(r+I) + (s+I) = (r+s)+I$$

$$(r+I) \cdot (s+I) = rs+I$$

then I is an ideal of R .

R/I quotient ring.

(1) A, B subring of R , B ideal.

$$A+B \text{ is a subring}$$

$$\text{Let } (a+b) + (c+d)$$

$$= (a+c) + (b+d) \in A+B$$

$$(a+b) \cdot (c+d) = a \cdot c + a \cdot d + b \cdot c + b \cdot d \in A+B$$

since B ideal

$\in B$

$$0 \in A \cap B \text{ (nonempty)}$$

$$a_1 \in A \cap B \quad a_2 \in A \cap B \Rightarrow a_1 + a_2 \in A$$

$$a_1 - a_2 \in B \text{ (B ideal)} \Rightarrow a_1 - a_2 \in A \cap B$$

$$a_1, a_2 \in A \text{ (A is a ring)}$$

$$a_1, a_2 \in B \text{ (B is an ideal)} \Rightarrow a_1 \cdot a_2 \in A \text{ (subring)}$$

$$\Rightarrow a_1 \cdot a_2 \in A \cap B \Rightarrow A \cap B \text{ is an ideal of } R$$

$$A \xrightarrow{\varphi} A+B/B$$

$$\downarrow \varphi$$

$$A/B$$

$$\ker \varphi = A \cap B$$

(2) I, J ideals of R , $I \subseteq J$.

$$\exists +1 \in J_1 \text{ nonempty}$$

$$\text{let } \bar{a}_1, \bar{a}_2 \in J/I$$

$$\bar{a}_1 - \bar{a}_2 = \overline{a_1 - a_2} \quad a_1 - a_2 \in J \quad J \text{ is closed in } (R, +) \text{ as absorbing.}$$

$$\Rightarrow \bar{a}_1 - \bar{a}_2 \in J/I$$

$$\bar{a} - \bar{a}' = \overline{a - a'} \quad +, \cdot \in J \quad \forall \cdot \text{ ideal in } R$$

$$\Rightarrow J/I \text{ is an ideal in } R/I$$

$$(R/I)/(J/I)$$

$$R/I \xrightarrow{\varphi} R/J$$

$$\varphi: R/I \rightarrow R/J$$

$$r+I \mapsto r+J$$

$$\ker \varphi = \{r+I\} = J/I$$



$$\downarrow \quad \checkmark$$

$$(R/I)/(J/I)$$

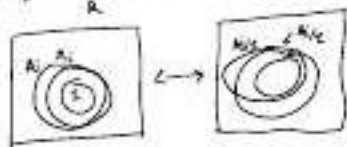
(3) let I be an ideal of R

$$A \hookrightarrow R/I$$

$$\bar{A}_I \subset R/I \text{ subring containing } I$$

$$\{\bar{A}_I\} \hookrightarrow \{\bar{A}_I/I\}$$

$$R/I$$



$$A \subset R$$

$$i(A) = A$$

$$i(A/I) = A/I \subset R/I$$

Exercises

(1) $f: \mathbb{Z} \rightarrow \mathbb{Z}$ hom

$$f(2) = a \in \mathbb{Z}$$

$$f(n)f(2) = a^2$$

$$f(1)f(2+2) = 2a$$

$$a^2 - 2a = 0$$

$$a(a-2) = 0$$

but $\mathbb{Z}$ has no zero divisors

$$\Rightarrow a = 0 \text{ or } a = 2$$

$\Rightarrow f$ is always a zero hom

Thus, $\mathbb{Z}$ and $\mathbb{Z}$ are not isomorphic.

(2) $f: \mathbb{Z}[x] \rightarrow \mathbb{Q}[x]$ is an isomorphism.

$$f(a) = b \quad a \in \mathbb{Z} \quad b \in \mathbb{Q} \quad \forall \neq 0$$

$$a \neq 1, -1$$

$$f(a') = b^{-1}$$

$$1 = f(a) \cdot f(a') = f(a \cdot a') \Rightarrow a \cdot a' = 1$$

however $\mathbb{Z}^\times = \{+1, -1\}$ Thus, contradiction.

(3) $\mathbb{Z} \xrightarrow{\varphi} \text{img}(\varphi)$

Every ring hom's kernel is an ideal.

Then, the ideals of $\mathbb{Z}$ are of the form $n\mathbb{Z}$.

$$\Rightarrow \ker \varphi = \mathbb{Z}/\ker \varphi \cong \mathbb{Z}/n\mathbb{Z}$$

ideals containing $20\mathbb{Z}$ in $\mathbb{Z}$: $2\mathbb{Z}, 4\mathbb{Z}, 6\mathbb{Z}, 10\mathbb{Z}, 15\mathbb{Z}, 20\mathbb{Z}$

(4)

$$\mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/20\mathbb{Z}$$

$$\ker \varphi = 20\mathbb{Z}$$

$$\varphi(0) = 0$$

$$\varphi(1) = a, \varphi(1+1) = \varphi(1) + \varphi(1) = 2a = 0$$

$$\Rightarrow a = 10$$

$$\text{img } \varphi = \{0, 10\}$$

$$\mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/30\mathbb{Z}$$

$$\ker \varphi = 30\mathbb{Z}$$

$$\text{img } \varphi = \{0, 10\}$$

$$\mathbb{Z}/6\mathbb{Z}$$

$$\ker \varphi = 6\mathbb{Z}$$

$$\text{img } \varphi = \{0, 5\}$$

$$\mathbb{Z}/10\mathbb{Z}$$

$$\ker \varphi = 10\mathbb{Z}$$

$$\text{img } \varphi = \{0, 3\}$$

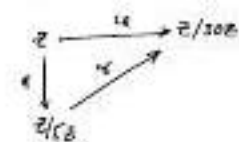
$$\mathbb{Z}/15\mathbb{Z}$$

$$\ker \varphi = 15\mathbb{Z}$$

$$\text{img } \varphi = \{0, 2\}$$

$$\mathbb{Z}$$

$$\mathbb{Z}/20\mathbb{Z}$$



$$\ker q = \{z \in \mathbb{Z} \mid \text{im } q = 0\}$$

$$\ker q = \{z \in \mathbb{Z} \mid \text{im } q = 0\}$$

$$q(0) = 0 \quad \text{with } q(z) = a$$

$$q(5) = 5a = 0$$

$$a = 0$$

$$\ker q = \{0, 5, 10, \dots\}$$

(5) $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$

$$q(a, b) = a, \quad 0 \mapsto 0$$

$$q(a, b) = b$$

$\ker q = \mathbb{Z} \times \{0\} \supset \text{im } q = \mathbb{Z}$
 $\ker q = \{0\} \times \mathbb{Z}$
 why the idea of sending even integers only on one component doesn't work!

$$q((a, 0)) = a$$

$$q((1, 0) + (1, 0)) = q((2, 0)) = 2$$

$$q((1, 0)) + q((1, 0)) = 1 + 1 = 2$$

(6) (a) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a \quad \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} + \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \mapsto a_1 + a_2 = q((a_1, b_1)) + q((a_2, b_2))$

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \cdot \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 + b_1 c_2 & a_1 b_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix} \mapsto a_1 a_2 + b_1 c_2 \neq a_1 a_2$$

not hom.

(b) $\text{tr}(M_1 M_2) \neq \text{tr}(M_1) \text{tr}(M_2)$

(c) $\det(A+B) \neq \det(A) + \det(B)$

(7) $R = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right\} \subset M_2(\mathbb{Z})$

$$q: R \rightarrow \mathbb{Z} \times \mathbb{Z} \quad \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto (a, d)$$

with $(a, b) \in \mathbb{Z} \times \mathbb{Z}$

$$q^{-1}((a, b)) = \left\{ \begin{pmatrix} a & d \\ 0 & b \end{pmatrix} \mid d \in \mathbb{Z} \right\} \Rightarrow \text{surjective}$$

$$\ker q = \left\{ \begin{pmatrix} 0 & d \\ 0 & 0 \end{pmatrix} \mid d \in \mathbb{Z} \right\}$$

8 (a) $\{(a, a) \mid a \in \mathbb{Z}\} = S$

$$(a, a) - (b, b) = (a-b, a-b) \in S$$

$$(0, 0) \in S \Rightarrow \text{nonempty}$$

$$(a', b') \cdot (a, a) = (a'a, b'a) \notin S \quad \text{not an ideal}$$

(b) $\{(2a, 2b) \mid a, b \in \mathbb{Z}\} = S = \mathbb{Z} \times \mathbb{Z}$

$$(0, 0) \in S$$

$$(2a_1, 2b_1) - (2a_2, 2b_2)$$

$$= (2(a_1 - a_2), 2(b_1 - b_2)) \in S$$

$$(a', b') \in \mathbb{Z} \times \mathbb{Z}$$

$$(a', b') \cdot (2a, 2b) = (2a'a, 2b'b) \in S \quad \checkmark \text{ ideal}$$

(c) $\{(2a, 0)\}$ clearly ideal = $\mathbb{Z} \times \{0\}$

(d) $\{(a, -a) \mid a \in \mathbb{Z}\} = S$

$$(0, 0) \in S \quad \text{nonempty}$$

$$(a, -a) - (b, -b) = (a-b, -(a-b)) \in S$$

$$(a, b) \in \mathbb{Z} \times \mathbb{Z}$$

$$(a, b) \cdot (a', -a') = (aa', -ba') \notin S \quad \text{not ideal}$$

9 (a) $\{f(x) \mid f(q) = 0 \quad \forall q \in \mathbb{Q} \cap [0, 1]\} = S$

$$0 \in S \quad \text{nonempty}$$

$$f(x) \in S, g(x) \in S \quad \text{with } q \in \mathbb{Q} \cap [0, 1]$$

$$(f-g)(q) = f(q) - g(q) = 0$$

$$\text{let } f(x) \in [0, 1]^{\mathbb{R}}$$

$$f(q) = f(q)g(q) = 0 \quad \checkmark \text{ ideal}$$

$$(10) \cdot (a) \{f(x) \in \mathbb{Z}[x] \mid f(0) = 3n \quad n \in \mathbb{Z}\} = S$$

$$f(x) \in \mathbb{Z}[x] \quad f(0) = 3 \quad \text{nonempty}$$

$$(f-g)(0) = f(0) - g(0) = 3m - 3n = 3(m-n) \in S$$

$$f \in \mathbb{Z}[x]$$

$$f \cdot g(0) = f(0) \cdot g(0)$$

$$= 3m \cdot 3n \in S \quad \checkmark \text{ ideal}$$

$$(b) \quad (2x)(3x^2 + x) = 6x^3 + 2x^2 \notin S \quad \text{thus not ideal!}$$

$$11. \quad R = C([0, 1], \mathbb{R})$$

$$q: R \rightarrow \mathbb{R}$$

$$f \mapsto \int_0^1 f(t) dt$$

$$q(f \cdot g) = \int_0^1 f(t)g(t) dt \neq \int_0^1 f(t) dt \int_0^1 g(t) dt$$

$$\text{consider } x^2 x^3$$

$$\int_0^1 t^5 dt = \left[\frac{t^6}{6} \right]_0^1 = \frac{1}{6} \quad \times$$

$$\text{However,}$$

$$\int_0^1 t^2 dt \cdot \int_0^1 t^3 dt = \left[\frac{t^3}{3} \right]_0^1 \cdot \left[\frac{t^4}{4} \right]_0^1 = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

$$(15) \quad R = \{f: X \rightarrow \mathbb{Z}/2\mathbb{Z}\}$$

$$\forall A \subseteq X \quad \chi_A: X \rightarrow \mathbb{Z}/2\mathbb{Z}$$

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$$A \mapsto \chi_A$$

$$\text{non}$$

$$\text{let } A, B \in \mathcal{P}(X)$$

$$q(A+B) = \chi_{A+B}$$

$$\chi_{A+B}(x) = \begin{cases} 1 & \text{if } x \in A+B \\ 0 & \text{otherwise} \end{cases}$$

$$q(AB) = \chi_{AB} = \begin{cases} 1 & \text{if } x \in AB \\ 0 & \text{otherwise} \end{cases} = \chi_A \cdot \chi_B$$

$$\text{suppose } A \in \mathcal{P}(X)$$

$$q(A) = \chi_A = 0 \Rightarrow A = \emptyset \quad \text{injective.}$$

$$\text{let } \chi \in R \quad \text{i.e. } \chi: X \rightarrow \mathbb{Z}/2\mathbb{Z}$$

$$\chi \neq 0 \Rightarrow \exists A \subseteq X \text{ s.t. } \chi(x) = 1.$$

$$\text{then } q^{-1}(\chi) \text{ is nonempty} \quad \text{is surjective.}$$

$$(16) \quad q: R \rightarrow S$$

$$\text{let } c \text{ be the centre of } R$$

$$q(r) = q(c) \cdot q(c) = q(c \cdot r) = q(c) \cdot q(r)$$

$$\Rightarrow q(c) \in \text{centre of } S$$

$$(17) \quad R, S \text{ nonzero rings with } 1_R, 1_S$$

$$\phi: R \rightarrow S \text{ nonzero hom}$$

$$\phi(1_R) = \phi(1_R)^2 \Rightarrow \phi(1_R)(\phi(1_R) - 1_S) = 0 \Rightarrow \phi(1_R) \text{ is a zero divisor.}$$

$$(a) \quad \nexists \phi(1_R) \neq 1_S$$

$$\text{if } S \text{ is an integral domain, then } \phi(1_R) = 1_S$$

$$(b) \quad \nexists \phi(1_R) = 1_S : u \in R^* \quad \phi(uu^{-1}) = \phi(u)\phi(u^{-1}) = 1_S = \phi(u)\phi(u^{-1})$$

$$\Rightarrow \phi(u)^{-1} = \phi(u^{-1})$$

$$(18) \cdot (a) \quad I, J \text{ ideals of } R$$

$$I \subseteq I \cap J \text{ nonempty}$$

$$a, b \in I \cap J$$

$$a-b \in I \quad a-b \in J$$

$$\Rightarrow a-b \in I \cap J$$

$$\text{let } x \in R, a \in I \cap J$$

$$xa \in I \quad xa \in J$$

$$\Rightarrow xa \in I \cap J$$

$$(b) \quad I_0 = \{I \mid I \text{ is ideal of } R\} \quad (\text{trivial})$$

$$(19) \quad I_1 \subseteq I_2 \subseteq I_3 \dots \text{ are ideals of } R.$$

$$\text{let } I_1 \text{ is an ideal. Suppose true for } k$$

$$\bigcup_{n=1}^k I_n = \left(\bigcup_{n=1}^k I_n \right) \cup I_{k+1} = I_{k+1} \text{ is an ideal of } R.$$

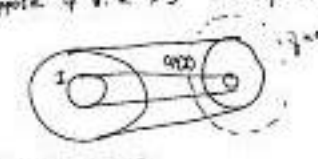
(20) I an ideal of R , $S \subseteq R$
 $0 \in I \cap S$
 $\text{let } a, b \in I \cap S$
 $a - b \in I \cap S$
 $\text{let } s \in S, a \in I \cap S$
 $sa \in I$ (since I is ideal)
 $sa \in S$ (since S is subring) $\Rightarrow sa \in I \cap S \Rightarrow I \cap S$ ideal
 $R = \mathbb{Q}, S = \mathbb{Z}$
 ideals of $\mathbb{Q}$ are $\{0\}, \mathbb{Q}$ so we cannot write $n\mathbb{Z}$ ideal in $\mathbb{Q}$
 $\text{as } I \cap \mathbb{Z} = n\mathbb{Z}$ where I is an ideal in $\mathbb{Q}$

(21) $a \in R$
 $S = \{x \in R \mid ax = 0\}$
 $\text{let } x, y \in S$
 $0 \in S$ nonempty
 $a(x-y) = 0$
 $\text{let } r \in R$
 $ax = 0 \Rightarrow rx \in S \forall r \in R$
 $\Rightarrow S$ is a right ideal.
 $S = \{x \in R \mid xa = 0\}$
 $\text{let } x, y \in S$
 $(x-y)a = 0$
 $\text{let } r \in R$
 $rx = 0 \Rightarrow x \in S \forall r \in R$
 $\Rightarrow S$ is a left ideal.
 $\Rightarrow S$ is a two-sided ideal.

(22) Suppose I is a left ideal of R .
 $S = \{x \in R \mid xa = 0 \forall a \in I\}$
 $0 \in S$
 S is a left ideal:
 $x, y \in S$
 $(x-y)a = xa - ya = 0 \forall a \in I$
 $\text{let } r \in R$
 $rx = 0 \forall x \in S \Rightarrow rx \in S$
 S is a right ideal.
 $\text{let } x \in S$
 $(xr)a = x(ra) = 0$ since $ra \in I$ (left ideal)
 $\Rightarrow S$ is a two-sided ideal.
 $\left[\begin{array}{l} \text{left annihilator of } I \text{ is left ideal, it's a two-sided ideal} \\ \text{right annihilator of } I \text{ is right ideal, it's a two-sided ideal} \end{array} \right]$

(23) $S \subseteq R, I \trianglelefteq R$
 Suppose $SI \subseteq I$
 $\bar{S} = S+I/I \cong S/I \cap I \cong S$

(24) $\varphi: R \rightarrow S$
 (a) Suppose $J \trianglelefteq S$
 $r \in R, a \in \varphi^{-1}(J)$
 $\varphi(ra) = \varphi(r)\varphi(a) \in J$ since J is an ideal.
 $\Rightarrow ra \in \varphi^{-1}(J)$
 $R \subseteq S, \varphi|_R: R \rightarrow S, J \trianglelefteq S$
 $\text{for } \forall A \subseteq S, \varphi^{-1}(A) = A \cap R$
 $\varphi^{-1}(J) = J \cap R$
 Since J is an ideal of S
 $\varphi^{-1}(J) \trianglelefteq R \Rightarrow J \cap R \trianglelefteq R$

(b) Suppose $\varphi: R \rightarrow S$ is surjective and $I \trianglelefteq R$.

 $\text{let } s \in S, a \in \varphi(I)$
 $\exists i \in I \text{ s.t. } \varphi(i) = a \quad \exists r \in R, \varphi(r) = s$
 $\varphi(sa) = \varphi(r)\varphi(i) = \varphi(r)\varphi(i) \in \varphi(S)$
 $\varphi(I) \trianglelefteq S$
 $\varphi(I)$ is not an ideal of S

(25)

$$(a+b)(a+b) = a^2 + (1+1)ab + b^2$$

$$\sum_{k=0}^n (a+b) = \left(\sum_{k=0}^{n-1} (a+b) \right) (a+b)$$

$$= \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-1-k} (a+b)$$

$$= \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-1-k} a + \sum_{k=0}^{n-1} \binom{n-1}{k} a^k b^{n-1-k} b$$

$$= \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

(26) R be a ring with identity 1 .
 $\varphi: \mathbb{Z} \rightarrow R, k \mapsto \begin{cases} 1+1+\dots+1 & k \text{ times } k > 0 \\ 0 & k=0 \\ -1-1-\dots-1 & k \text{ times } k < 0 \end{cases}$
 $\varphi(x+k) = \begin{cases} 1+1+\dots+1 & x+k > 0 \\ 0 & x+k=0 \\ -1-1-\dots-1 & x+k < 0 \end{cases}$
 $x+k > 0$
 $x, k > 0$ (WLOG) $x > 0$
 $\varphi(x+k) = 1+\dots+1 + 1+\dots+1 = \varphi(x) + \varphi(k)$
 $x, k > 0, x < 0$
 $\varphi(x+k) = 1+\dots+1 + (-1-\dots-1)$
 $= \varphi(x) + \varphi(k)$
 $x+k=0$
 $x=0, k=0$ clear
 $x < 0$
 $\varphi(x+k) = \varphi(x) + \varphi(k)$
 $\text{let } k=0$
 $\varphi(kx) = 1+1+\dots+1, k \text{ times}$

$(1+\dots)^n = 1 + \dots$
 have any more.
 Suppose ϕ is the characteristic of ring R .
 $\phi(1) = 1$ times $1 = 1$ $\phi(1) = 1$
 clearly $\forall x \in R \exists n \in \mathbb{Z} \text{ s.t. } nx = 0$
 $\phi(x) = \phi(nx) = n \phi(x) = 0$
 $\exists x \in R \text{ s.t. } \phi(x) = 0$
 $\phi(x) = 0 = 1 + \dots + 1$ times x then $n \mid 1$.
 $\forall x \in R \phi(x) = 0$
 • Boolean rings have characteristic 2.

(1) ϕ is prime.
 $(a+b)^2 = a^2 + 2ab + b^2$
 $\phi(a+b)^2 = \phi(a)^2 + 2\phi(a)\phi(b) + \phi(b)^2$
 $\phi(a+b)^2 = 0$ then $\phi(a)\phi(b) = 0$
 $\Rightarrow (a+b)^2 = a^2 + b^2$

(2) $1 \times (-1)^2 = 1$
 $\Rightarrow \text{char } 2$

Thm R, S are rings. and S is an integral domain
 $R \cong S$ Then R is also an integral domain.

Proof Suppose a is zero divisor $\exists b \in R \text{ s.t. } ab = ba = 0$. let ϕ be the isomorphism.
 $\phi(ab) = \phi(a)\phi(b) = 0$
 Now, $a, b \neq 0$ $\phi(a), \phi(b) \neq 0$
 $\Rightarrow \phi(a)$ is a zero divisor $\checkmark$

$\phi(p) = 0$ if p not prime or zero.

$p = \prod_i p_i^{q_i}$
 $\phi(p) = \prod_i \phi(p_i)^{q_i} = 0$
 then $\phi(p_i)$ is a zero divisor.

(29) R be a commutative ring.

let $a, b \in \mathcal{N}(R)$

$0 \in \mathcal{N}(R)$ nonempty $a^m = b^n = 0$ $n, m \in \mathbb{Z}^+$ let $m = \max\{m, n\}$

$(a+b)^{m^2} = \sum_{k=0}^{m^2} \binom{m^2}{k} a^k b^{m^2-k} = 0 \Rightarrow a+b \in \mathcal{N}(R)$

$\forall k < m^2 \exists l, m^2-k = ml + n$

[sharper bounds $m+n-1$
 $\binom{m+n-1}{k} a^k b^{m+n-1-k}$
 $\exists k \geq m, a^k = 0 \quad k < m, b^{m+n-1-k} = 0$

$r \in R$
 $(ra)^m = r^m a^m = 0 \Rightarrow ra \in \mathcal{N}(R)$

(30) $R/\mathcal{N}(R)$

Suppose $(\bar{a})^m = 0$
 then $a^m = 0 \Rightarrow \bar{a} = \bar{0}$

(31) -

(32) $x^m = 0$ in R $\phi: R \rightarrow S$

$0 = \phi(x^m) = \phi(x)^m$

(33) R commutative. let $p(x) \in R[x]$

(a) (1) $1(x)$ is a unit. $\exists q(x) \text{ s.t. } q(x)1(x) = 1 = 1(x)q(x)$

$p(x)q(x) = \sum_{k=0}^{m+n} \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k = 1$
 $\Rightarrow a_0 b_0 = 1$ a_0 is a unit.

$a_n b_m = 0$
 $(a_n x^{n-1} + a_{n-1} x^{n-2} + \dots + a_0) b_m = 0$
 $(a_n b_{m-2} + a_{n-1} b_{m-1} + a_{n-2} b_m) = 0$

$$a_n(a_n b_{m-1} + a_{n-1} b_m) = 0 \Rightarrow a_n^2 b_{m-1} = 0 \text{ inductively}$$

$$a_n^2 b_{m-1} = 0 \dots a_n^m b_0 = 0 \Rightarrow a_n^m = 0 \Rightarrow a_n \text{ nilpotent}$$

$$a_{n-1} a_n b_m = 0$$

$$a_{n-1} b_m (a_n b_{m-1} + a_{n-1} b_m) + a_{n-1} b_m a_{n-1} b_m = 0$$

$$a_{n-1}^2 b_m^2 = (a_{n-1} b_m)^2 = 0 \Rightarrow a_{n-1} b_m = 0$$

$$\vdots$$

$$a_{n-1} b_0 = 0 \Rightarrow a_{n-1} \text{ nilpotent}$$

inductively

$a_{n-1}, \dots, a_1$ are nilpotent.

($\Leftarrow$) suppose a_0 is a unit, $a_1, \dots, a_n$ are nilpotent

$$p(x) = a_0 + a_1 x + \dots + a_n x^n$$

$$a_1 x + \dots + a_n x^n \text{ is nilpotent}$$

$$\text{if } a, b \in R \text{ be nilpotent } a^n = 0, b^m = 0$$

$$(a+b)^{m+n-1} = \sum_{k=0}^{m+n-1} \binom{m+n-1}{k} a^k b^{m+n-1-k} = 0 \Rightarrow a+b \text{ is nilpotent}$$

$$\Rightarrow a_1 x, \dots, a_n x^n \text{ are nilpotent in } R[x]$$

$$p(x) \text{ is a unit.}$$

$$(b) (\Rightarrow) p(x) \text{ is nilpotent } \exists m \in \mathbb{Z}$$

$$(p(x))^m = 0$$

$$\left(\sum_{k=0}^n a_k x^k \right)^m = \sum_{k=0}^n (a_k x^k)^m + \dots = 0 \Rightarrow (a_k)^m = 0 \forall k$$

$$\Rightarrow a_k \text{ are nilpotent}$$

$$(\Leftarrow) a_0, \dots, a_n \text{ are nilpotent}$$

$$a_0 + a_1 x + \dots + a_n x^n \text{ is nilpotent}$$

$$(84) (a) \forall k \in \mathbb{R} \text{ s.t. } I, J \subseteq K, I+J \subseteq K$$

$$(b) I \cap J$$

$$a \in I \cap J$$

$$a \in I \text{ and } a \in J$$

$$\forall a \in I \cap J$$

$$\bullet \text{ if } i \in I, j \in J \Rightarrow ij \in I \cap J \Rightarrow I \cap J \subseteq I \cap J$$

$$(c) I = 2\mathbb{Z}, J = 4\mathbb{Z}$$

$$I \cap J = 4\mathbb{Z}$$

$$IJ = 8\mathbb{Z}$$

$$(d) R \text{ commutative and } I+J = R$$

$$a \in I \cap J$$

$$i = i+j, i \in I, j \in J$$

$$a_i + a_j = i a_i + a_j$$

$$\text{since } a \in I \text{ and } a \in J$$

$$i a_i \in I, a_j \in J$$

$$a_i a_j + a_j \in IJ \Rightarrow IJ = I \cap J$$

$$(85) (a) I, J, K \subseteq R$$

$$y \in I(J+K)$$

$$(b) y = \sum_{k=1}^n a_k b_k, a_k \in I, b_k \in J+K$$

$$b_k = c_k + d_k$$

$$= \sum_{k=1}^n (a_k c_k + a_k d_k)$$

$$(c) \in IJ + IK$$

$$y \in IJ + IK \Rightarrow y = \sum_{k=1}^n a_k b_k = \sum_{k=1}^n c_k d_k$$

$$c_k \in I, d_k \in J+K$$

$$d_k \in J+K$$

$$= \sum_{k=1}^n a_k c_k \in IJ + IK$$

$$(I+J)K = IK + JK \text{ same as above}$$

$$(b) J \subseteq I,$$

$$(\Leftarrow) a \in I \cap (J+K)$$

$$a \in I \text{ and } a \in J+K, a = j+k, j \in J, k \in K$$

$$j \in I, k \in K (J \subseteq I)$$

$$\therefore a = j+k \in I+K$$

$$a-j+k=j+(a-j) \in J+(I \cap K)$$

$$(2) a \in J+(I \cap K)$$

$$a=j+b \quad b \in I, b \in K$$

$$\text{into } (J \subseteq I)$$

$$a \in I, a \in J+K \Rightarrow a \in I \cap (J+K)$$

$$(36) I \subseteq \mathbb{Z}[x]$$

$$I^n = \{a_n x^n + \dots + a_{n+m} x^{n+m} \mid a_i \in \mathbb{Z}, m \geq 0\}$$

$$I = \{a_1 x + a_2 x^2 + \dots + a_m x^m \mid a_i \in \mathbb{Z}, m \geq 0\}$$

$$I \cdot I = \{a_1 x^2 + \dots + a_{m_2} x^{m_2+2} \mid a_i \in \mathbb{Z}, m \geq 0\}$$

suppose true for n

$$I^{n+1} = I \cdot I^n \quad \text{then true!}$$

$$(37) \mathbb{P}^2 / \mathbb{P}^m \mathbb{Z}$$

$$p\mathbb{N} + p^m \mathbb{Z}$$

$$\sum_{k=1}^n \overline{p n_k} \cdot \overline{p m_k} = \overline{p^2 (nm)_k}$$

$$\text{let } (\mathbb{P}^2 / \mathbb{P}^m \mathbb{Z})^m \text{ then } a = \sum_k \overline{p^n \left(\prod_{i=1}^m (n_i) \right)_k} \in (\mathbb{P}^2 / \mathbb{P}^m \mathbb{Z})^m = \overline{0} \in \mathbb{Z} / \mathbb{P}^m \mathbb{Z}$$

Properties of Ideals

set $A \subseteq R$, R Ring

(1) (A) smallest ideal containing A .

$$(2) RA = \{r_1 a_1 + \dots + r_n a_n \mid r_i \in R, a_i \in A, n \in \mathbb{Z}^+\} \quad \{RA=0 \mid A=\emptyset\}$$

$$AR = \{a_1 r_1 + \dots + a_n r_n \mid \dots\}$$

(3) ideal generated by one element is called principal ideal.

(4) finite elements : finitely generated ideal.

RA = left ideal generated by A ($A \subseteq R, RA$ is an ideal and $\forall x \in A, x \in RA, x \in R$)

AR = right
if R is commutative.
 $AR = RA = RAR = (A)$

R commutative : $a \in R$ (a) is the principal ideal.

if R is noncommutative

$$(a) = RAR = \{ \sum r_i a s_i \mid r_i, s_i \in R \}$$

Examples

- (a) $\mathbb{Z} = (1), \mathbb{Z} = (1)$ are principal
- (b) $a\mathbb{Z} = \{an \mid n \in \mathbb{Z}\} = (a) \quad \forall a \in \mathbb{Z}$, every ideal of $\mathbb{Z}$ is principal.
 $a, n \in \mathbb{Z}^+, na \in \mathbb{Z} \Leftrightarrow a \mid n$ in $\mathbb{Z}$
 $(nm) = (a)$ $\Leftrightarrow \text{divides } (nm) \quad (nm) = (a) \Leftrightarrow \text{equal } (nm) = (a)$
- (c) (a, b) is not a principal ideal in $\mathbb{Z}[x]$
 $(x, x^2) = \{x p(x) + x^2 q(x) \mid p(x), q(x) \in \mathbb{Z}[x]\}$ it is a proper ideal without generator.
if $(x, x^2) = (0) = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ then $\mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ are constants
or $\text{any } (x) + \text{any } (x^2) = \text{any } (x) = (x)$ or $\mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ are constants
 $\mathbb{Z}\mathbb{Z}(0) = \{x, x^2\} \mid \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ not possible
 $\{x, x^2\} = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ not possible by degree
if $\mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ then $\mathbb{Z}\mathbb{Z}(0) = \mathbb{Z}\mathbb{Z}(0)$ is principal.

Prop 8 $I \subseteq R$

- (1) $1 \in I \Leftrightarrow I = R \Rightarrow I$ contains a unit
- (2) Suppose I contains a $a \in R^*, a^{-1} \in R$
 $a^{-1} a = 1 \in I \Rightarrow I = R$
- (3) R commutative.
(a) R is a field, $R \setminus \{0\} = R^*$ if $I \subseteq R, I \neq 0, I$ contains a unit $1 \in R$.
- (4) Suppose only ideals of R are 0 and R .
if $a \in R, R \setminus \{0\} = R^*$ if $a \in R, a \neq 0 \Rightarrow a$ is a unit $\Rightarrow R$ is a field.

Lemma if R is a field, let S be a ring, $\phi: R \rightarrow S$ non trivial ring hom.
then $\phi(R) \Rightarrow \text{help } 0$ by previous prop.

Prop 9 $M \subseteq S$ is maximal if only ideals containing M are M and S .

Prop 10 R with $1 \neq 0$, every proper ideal is contained in a maximal ideal.

Proof let I be a proper ideal.
Now, let S be the collection of all proper ideal containing I . $I \in S$.
Specially, $S \subseteq R, S$ is a partially ordered set. let C be a chain in S .
let $J = \bigcup_{A \in C} A$ $I \subseteq J, J \subseteq R$ nonempty. let $a, b \in J$ then $a, b \in A$ for some $A \in C$.
Then C is a chain $A \subseteq B \Leftrightarrow B \subseteq A$. Then $a, b \in A \Rightarrow a, b \in B \Rightarrow a, b \in J$.
Thus all $A \in C$ are ideal under $\cup$. J is an ideal.
Let $J = R$ for some $A \in C$. then A is a maximal ideal.
So, J is a proper ideal. then by Zorn's lemma J is a maximal ideal containing I . $\square$

$\pi(g_i) \in \pi(\langle g_i \rangle) \forall i, g_i$
 π descends to a unique homomorphism
 $\pi: R/\langle g_i \rangle \rightarrow \pi(R/\langle g_i \rangle)$

(4) $R = \{g_1, \dots, g_n\}$ a commutative ring

reducing

$$r_1 + r_2 + \dots + r_n$$

$$r = \sum_{i=1}^n r_i g_i \quad r_i \in R \text{ or } g_i \in R$$

$$r(g_1 + \dots + g_n) = \sum_{i=1}^n r(g_i \cdot 1) = r(g_1 + \dots + g_n)$$

$$= \sum_{i=1}^n \sum_{j=1}^n r_i g_j = \sum_{i=1}^n r_i g_i$$

$$= \sum_{i=1}^n \sum_{j=1}^n r_i g_j = \sum_{i=1}^n r_i \sum_{j=1}^n g_j = 0$$

(5) Assume R commutative.

$\{g\}$ is maximal $\Leftrightarrow R/\langle g \rangle = R$ is a field.

(6) $\{g\}$ is an ideal $\Leftrightarrow R/\langle g \rangle$ is a field. Then its only ideals are $\{0\}$ and $R/\langle g \rangle$.
 $\{g\}$ is maximal $\Leftrightarrow R/\langle g \rangle$ is a field.

(7) R is a division ring

Let I be an ideal. $I \neq \{0\}$

$I \cap R = I = R$ if

(8) Suppose only ideals of R are $\{0\}$ and R .
 Then for any $a \in R$, $\langle a \rangle = R$. $\Rightarrow \exists r \in R$ s.t. $ra = 1$.
 $(r) = R$ $\exists b \in R$ s.t. $rb = 1$.
 $\Rightarrow (ab)(r) = (1)r = r = a$.

(9) R commutative, with 1.

(10) $\langle a \rangle$ is a prime ideal $\Leftrightarrow R/\langle a \rangle$ is an integral domain.

(11) R is an integral domain since $R/\langle a \rangle \cong R$.

$$R/\langle a \rangle \cong R$$

(12) $\langle a \rangle$ is a maximal ideal $\Leftrightarrow R/\langle a \rangle$ is a field $\Leftrightarrow R$ is a field.

(13) R is an integral domain.

(14) $\langle a \rangle = \langle b \rangle$ if $a, b \in R$

$ra = b \Rightarrow r' = ra = b \Rightarrow r' = a \Rightarrow r = 1$ is a unit.

(15) The unit $1 \in R$ is a unit

$a = ua' \Rightarrow a = ua' \Rightarrow a = a' \Rightarrow \langle a \rangle = \langle a' \rangle$

(16) $R = \mathbb{Z}[\frac{1}{2}]$

$I = \{f(x) \in R \mid f(\frac{1}{2}) = 0\}$

$I \neq \emptyset$ nonempty

$$f(x) = g(x) = \frac{1}{2} \Rightarrow \frac{1}{2} = 0 \in \mathbb{Z}$$

(17) $f(x) \in I$ ideal

I is prime $\Leftrightarrow R/I$ is an integral domain

$$f(x) = (\sqrt{x} - \frac{1}{\sqrt{2}}) (\sqrt{x} - \frac{1}{\sqrt{2}}) \quad g(x) = (\sqrt{x} + \frac{1}{\sqrt{2}}) (\sqrt{x} + \frac{1}{\sqrt{2}})$$

$$f(x) \cdot g(x) = (x - \frac{1}{2}) (x - \frac{1}{2}) = 0 \quad R/I \text{ is not an integral domain!}$$

(18) R commutative.

P is a prime ideal $\Leftrightarrow R/P$ is an integral domain. Further suppose P contains no zero divisors

$a, b \in R$ s.t. $ab = 0$

$\bar{a}\bar{b} = 0$ R/P is an integral domain $\Rightarrow \bar{a} = 0$ or $\bar{b} = 0$

(*) $\bar{a} = 0$ and $\bar{b} = 0 \Rightarrow a, b \in P$

$\Rightarrow a = 0$ or $b = 0$ since P domain

(**) $\bar{a} \neq 0, \bar{b} = 0$

$\Rightarrow a \notin P, b \in P$

Since P contains no zero divisors

$\Rightarrow a = 0$

$\Rightarrow R$ is an integral domain

(19) R commutative.

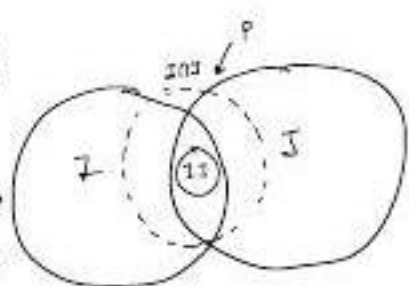
$I \subset P$ Prime ideal

Let $a \in I, b \in J$

$ab \in I \cap J$ $I \subset P$

$\Rightarrow a \in P$ or $b \in P$

$\Rightarrow I \subset P$ or $J \subset P$



(20) R commutative. $I = (a_1, \dots, a_n)$ $J = (b_1, \dots, b_m)$ ideals in R

What does an element of I, J look like?

$$x \in I \quad x = \sum_{i=1}^n r_i a_i \quad a_i \in \{a_1, \dots, a_n\} \quad r_i \in R$$

$$y \in J \quad y = \sum_{j=1}^m r'_j b_j \quad b_j \in \{b_1, \dots, b_m\} \quad r'_j \in R$$

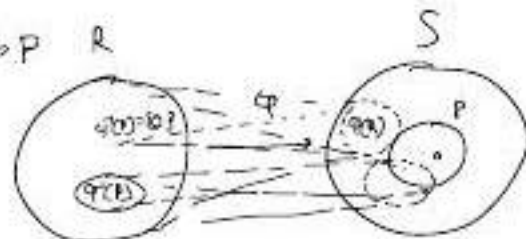
$\forall x, y \in I, J$ respectively

$$xy = \sum_{i,j} r_i r'_j a_i b_j$$

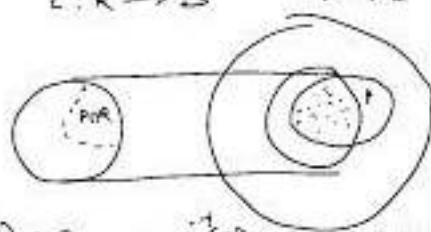
Then, $IJ = \langle a_i b_j \rangle_{i,j=1}^{n,m}$

(21) $\phi: R \rightarrow S$ be a homomorphism of commutative rings.

(a) Suppose P is a prime ideal of R .
 $a, b \in R$ $a, b \in \varphi^{-1}(P)$ $0 \in P$ $\varphi(0) = 0 \in \varphi^{-1}(P)$
 $\varphi(a-b) = \varphi(a) - \varphi(b) \in P \Rightarrow a-b \in \varphi^{-1}(P)$ $r \in R, \varphi(r) = \varphi(r)\varphi(a) \in P$
 $\Rightarrow r \in \varphi^{-1}(P)$ or $a \in \varphi^{-1}(P) \Rightarrow \varphi^{-1}(P)$ is a prime ideal of R .
 $\varphi(ab) = \varphi(a)\varphi(b) \in P$
 $\Rightarrow \varphi(a) \in P$ or $\varphi(b) \in P$
 $\Rightarrow a \in \varphi^{-1}(P)$ or $b \in \varphi^{-1}(P) \Rightarrow \varphi^{-1}(P)$ is a prime ideal of R .
 $\varphi(R) \subseteq P \Rightarrow \varphi^{-1}(P) = R$.



If $R \subseteq S$ $\varphi: R \rightarrow S$ $P \subseteq S$ prime



$$P \cap R = \varphi^{-1}(P) \cap R \text{ or } \varphi^{-1}(P) = R = P \cap R$$

(b) M is maximal ideal of S . φ is surjective.
 M is prime $\Rightarrow \varphi(M)$ is prime or $\varphi^{-1}(M) = R \oplus (Q \text{ is surjective})$

$$\bar{\varphi}: R/\varphi^{-1}(M) \rightarrow S/M$$

$$a + \varphi^{-1}(M) \mapsto \varphi(a) + M$$

Let $\bar{x} \in \ker \bar{\varphi}$

$$\bar{\varphi}(\bar{x}) = \varphi(x) + M = 0 + M$$

$$\Rightarrow \varphi(x) \in M \quad \varphi^{-1}(\varphi(x)) \subseteq \varphi^{-1}(M) \Rightarrow \bar{x} = \bar{0} \Rightarrow \text{injective}$$

$\bar{\varphi}$ is surjective

thus, $R/\varphi^{-1}(M)$ is a field $\Rightarrow \varphi^{-1}(M)$ is maximal.

(14) R commutative x indeterminate.
 $f(x)$ monic in $R[x]$ of degree $n \geq 1$. $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$

(a) Let $p(x) \in R[x]$

$$\overline{f(x)} = \bar{0} = \bar{x}^n + \bar{a}_{n-1}\bar{x}^{n-1} + \dots + \bar{a}_0$$

$$\Rightarrow \bar{x}^n = -(\bar{a}_{n-1}\bar{x}^{n-1} + \dots + \bar{a}_0)$$

$$\bar{x}^{n+1} = \bar{x} \bar{x}^n = -(\bar{a}_{n-1}\bar{x}^n + \dots + \bar{a}_0\bar{x})$$

$$= -(\bar{a}_{n-1}(-\bar{a}_{n-1}\bar{x}^{n-1} - \dots - \bar{a}_0) + \bar{a}_0\bar{x}^{n-1} + \dots + \bar{a}_0\bar{x})$$

$$= \bar{a}_{n-1}\bar{x}^{n-1} + \dots + \bar{a}_0\bar{x}$$

Suppose now for k

$$\bar{x}^{n+k+1} = \bar{x}^{n+k}\bar{x} = (\bar{a}_{n-1}\bar{x}^{n+k-1} + \dots + \bar{a}_0\bar{x}^k) \cdot \bar{x} = \bar{a}_{n-1}\bar{x}^{n+k} + \dots + \bar{a}_0\bar{x}^{k+1}$$

$$\text{then } \overline{p(x)} = \bar{a}_{n-1}\bar{x}^{n-1} + \dots + \bar{a}_0$$

(b) $p(x) \neq q(x) \in R[x]$ with $\deg(p), \deg(q) < n$

$$\text{Suppose } \overline{p(x)} = \overline{q(x)} \Rightarrow \overline{p(x) - q(x)} = \bar{0} \Rightarrow p(x) - q(x) \in (f(x))$$

$$p(x) - q(x) = p(x)x^k + b_{k+1}x^{k+1} + \dots + b_n x^n$$

$$\text{However, no degree } \geq n \text{ is possible, so } p(x) = q(x)$$

(c) $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ $\deg(f) = n$

$$\overline{f(x)} = \bar{0} \Rightarrow \bar{x}^n + \bar{a}_{n-1}\bar{x}^{n-1} + \dots + \bar{a}_0 = \bar{0} \Rightarrow \bar{f(x)} \text{ is a zero-divisor.}$$

(d) $f(x) = x^n - a$ $a^n = a$, $n \in \mathbb{Z}^+$

$$\overline{f(x)} = \bar{x}^n - \bar{a} = \bar{0}$$

$$\Rightarrow \bar{x}^n = \bar{a}$$

$$\bar{x}^{n^2} = (\bar{x}^n)^n = \bar{a}^n = \bar{a} = \bar{0}$$

(e) f prime $R = \mathbb{F}_p$ $f(x) = x^n - a$ $a \in \mathbb{F}_p$

$$\bar{0} = \overline{f(x)} = \bar{x}^n - \bar{a}$$

$$\Rightarrow \bar{x}^n = \bar{a}$$

$$\overline{(x-a)^n} = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k} (-1)^{n-k}$$

$$= \bar{x}^n - \bar{a} \quad (\bar{a}^n = \bar{a} \text{ because } a^n = a)$$

$$= \bar{0} - \bar{0} = \bar{0}$$

(15) $x^3 - 1 \in \mathbb{Z}[x]$

$$(a) \overline{x^3 - 1} = \bar{0} \Rightarrow \bar{x}^3 = \bar{1}$$

$$\bar{x}^3 - 1 = (\bar{x} - 1)(\bar{x}^2 + \bar{x} + 1)$$

$$= (\bar{x} - 1)(\bar{x}^2 + \bar{x} + 1) = \bar{0}$$

$$= (\bar{x} - 1)(\bar{x}^2 + \bar{x} + 1) = \bar{0}$$

$$(b) (\bar{x} - 1)(\bar{x}^2 + \bar{x} + 1) = \bar{0} \Rightarrow \bar{x}^3 = \bar{1}$$

(16) $\bar{0} = \overline{x^3 - 1} \in \mathbb{Z}[x]$

$$\overline{x^3 - 1} = \bar{0} \Rightarrow \bar{x}^3 = \bar{1}$$

$$\bar{x}^3 = \bar{1}$$

$$\bar{x}^3 - 1 = \bar{0}$$

$$\Rightarrow \bar{x}^3 = \bar{1}$$

$$\overline{x^3 - 1} = \bar{0} \Rightarrow \bar{x}^3 = \bar{1}$$

$$= \bar{0} - \bar{0} = \bar{0}$$

6) $\mathbb{R}[x]/(x^2+1)$ $x^2+1 = (x-i)(x+i) = 0$ not domain

6) $\bar{I} = \bar{x}(2-\bar{x}^2)$

(14) R integral domain

$R[x]$ commutative with 1.

$$\left(\sum_{n=0}^{\infty} a_n x^n\right) \left(\sum_{m=0}^{\infty} b_m x^m\right) \in (x)$$

$$a_0 b_0 = 0 \Rightarrow a_0 = 0 \text{ or } b_0 = 0$$

$$\Rightarrow \text{either } \sum_{n=0}^{\infty} a_n x^n \in (x) \text{ or } \sum_{m=0}^{\infty} b_m x^m \in (x) \Rightarrow (x) \text{ is a prime ideal.}$$

$$(x) \text{ is maximal } \Leftrightarrow R[x]/(x) \text{ is a field.}$$

$$E_0: R[x] \rightarrow R$$

$$\ker E_0 = (x)$$

$$\Leftrightarrow R[x]/(x) \cong R \Rightarrow R \text{ is a field.}$$

(15) R is commutative with 1 (finite)

let P be a prime ideal.

$\Rightarrow R/P$ is an integral domain with identity (finite)

$$\Rightarrow \varphi_a(R/P) = R/P \quad \forall a \in R/P$$

$$\exists \bar{b} \in R/P \text{ s.t. } \bar{a}\bar{b} = 1 \Rightarrow R/P \text{ is a field } \Leftrightarrow P \text{ is maximal.}$$

(20) R nonzero finite commutative ring.

$$\begin{matrix} \varphi: R \rightarrow R \\ a \mapsto a \cdot a \end{matrix}$$

let $a \in R$ $\exists n \in \mathbb{Z}^+$

$$\varphi^n(a) = 0 \Rightarrow a^n = 0 = 0a^n = 0 \cdot a^n = 0 \quad (\text{no nilpotency since no zero divisors})$$

(21) finite ring with 1 & 0 has nonzero divisors is a division ring is a field.

(nonzero commutative)

(25) P prime $\Rightarrow R/P$ is an integral domain $\Rightarrow R/P \cong \mathbb{Z}/p\mathbb{Z}$ is a field.

(boolean)

$\Rightarrow R/P$ is a field $\Rightarrow P$ is maximal.

(26) let $\{a_1, a_2, \dots, a_n\}$ be a finitely generated ideal of boolean

$$\{a_1, a_2, \dots, a_n\} = (a_1 + a_2 + \dots + a_n + a_1 a_2 + \dots + a_1 a_2 \dots a_n)$$

$$(a_1 + a_2 + \dots + a_n + a_1 a_2 + \dots + a_1 a_2 \dots a_n)(a_i) = a_i$$

$$(a_1 + a_2 + \dots + a_n)(a_i) = a_i$$

$$(a_1 + a_2 + \dots + a_n + a_1 a_2 + \dots + a_1 a_2 \dots a_n)(a_1 + a_2 + \dots + a_n) = a_1 + a_2 + \dots + a_n$$

$$(a_1 + a_2 + \dots + a_n) = 1 \Rightarrow (a_1 + a_2 + \dots + a_n) \in (a_1 + a_2 + \dots + a_n)$$

(27) obvious

(28) Suppose R commutative $a^n = a$ $n > 1$ depending on a . P ideal prime.

$\Rightarrow R/P$ is an integral domain.

Suppose $\bar{I}$ is an ideal of R/P

$$\text{let } \bar{a} \in \bar{I} \quad \bar{a}^m = \bar{a} \quad m \geq 1$$

$$\text{now } \bar{r} \in R/P \quad (\bar{r}\bar{a})^m = \bar{r}\bar{a} \Rightarrow \bar{r}^m = \bar{r} \quad \Leftrightarrow \Rightarrow \text{the only ideals of } R/P \text{ are } \{0\} \text{ and } R/P$$

$$\bar{r}^l = \bar{r} \quad l \neq m$$

$$\Rightarrow R/P \text{ is a field.}$$

(29) $\mathcal{N}(R) \subseteq P$ for P prime ideal

R/P is an integral domain.

if $\bar{a}$ is nilpotent $\Rightarrow$ either $\bar{a} = 0$ or a zero divisor

$$\Rightarrow \text{since domain, } \bar{a} = 0 \Rightarrow a \in P$$

$$\Rightarrow \mathcal{N}(R) \subseteq \bigcap_{P \text{ prime}} P$$

(30) $a \in \mathcal{N}(R)$ R commutative with $1 \neq 0$.

$$ab \in \mathcal{N}(R) \quad \forall b \in R$$

$$\text{let } 2^m > m \quad 2^l = m + n \quad \text{all positive integers.}$$

$$(1-ab)^{2^m} = 1 - (ab)^{2^m} = 1 - (ab)^m (ab)^n = 1$$

(31) R commutative $N = (a_1, a_2, \dots, a_n)$ $a_i \in \mathcal{N}(R)$

$$x \in N \quad x = \sum_{i=1}^n r_i a_i \quad m = \max\{m_i \mid a_i^{m_i} = 0\}$$

$$\prod_{i=1}^n x_i = \prod_{i=1}^n r_i a_i = 0$$

$$\mathcal{N}(R) = (a_1, \dots, a_n) \text{ then } \mathcal{N}(R)^m = 0 \text{ for some } m \in \mathbb{Z}^+$$

(32) P prime $(1) = P^n$

$$\text{ending } = (1 - 1) \in P^n$$

$$(1 - 1)^p = 1^p - 1^p = 0 \Rightarrow 1 - 1 \text{ is nilpotent } \Rightarrow (1 - 1) \in P^n \text{ is a finitely generated ideal}$$

$\Rightarrow$ ending is a nilpotent ideal.

(33) $I \triangleleft R$ ideal

$$\text{rad } I = \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{Z}^+\}$$

$$0 \in \text{rad } I$$

$a^0 \in I \Rightarrow 1^0 \in I$
 $(a-b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \in I \Rightarrow a-b \in \text{rad } I$

$\text{rad } R$
 $(\text{rad } R)^2 \subseteq I \Rightarrow \text{rad } R \subseteq \text{rad } I$

$I \subseteq \text{rad } I$ (obviously)
 $\text{Let } \bar{a} \in \overline{R/I} \Rightarrow \bar{a}^n = \bar{0} \Rightarrow a^n \in I \Rightarrow \bar{a} \in \text{rad } I/I$

31. $I \triangleleft R$ commutative, it called radical ideal if $\text{rad } I = I$.

(a) $P \triangleleft R$ prime.
 $\text{rad } P = \{r \in R \mid r^n \in P \text{ for some } n \in \mathbb{Z}^+\}$
 $\text{if } r \in R \text{ s.t. } r^n \in P$
 $(r^n)^m \in P$
 $\Rightarrow r^{nm} \in P \text{ or } r \in P \text{ or } r \in P$

$\text{rad } P = P$

(b) $n \geq 1$ integer
 $\Rightarrow \text{rad } R = \bigcap_{p \mid n} \mathbb{Z}/p\mathbb{Z} = 0$ if $n = p_1^{e_1} \dots p_k^{e_k}$ then every $p_i \mid n$

$\Rightarrow \bigcap_{p_i \mid n} \mathbb{Z}/p_i\mathbb{Z} = 0$

if $n = p_1^{e_1} \dots p_k^{e_k}$ then there does not exist $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ s.t. every prime divisor of n is a divisor of a .

$\Rightarrow \text{rad } (\mathbb{Z}/n\mathbb{Z}) = 0$

(n) is a radical ideal of $\mathbb{Z} \Leftrightarrow \text{rad } (\mathbb{Z}/n\mathbb{Z}) = 0$ as n is a product of distinct primes.

(32). $I \triangleleft R$ commutative.

$\text{Jac } I = \bigcap_{M \in \mathcal{M}} M$ $M \triangleleft R$ maximal

$\text{Jac } R = R$
 $\text{if } I = 0$, $\text{Jac } 0$ is called Jacobson radical. $\text{Jac } 0 = \bigcap_{M \in \mathcal{M}} M = \bigcap_{M \in \mathcal{M}} M$ all maximal ideals

$\text{Jac } I = \bigcap_{M \in \mathcal{M}} M$
 $\text{Jac } 0 = \bigcap_{M \in \mathcal{M}} M$ all maximal ideals of R/I .
 $\bar{M} = M/I$
 $\bigcap_{M \in \mathcal{M}} (M/I) = \bigcap_{M \in \mathcal{M}} M/I = \text{Jac } I/I$

(a) $\text{Jac } I = \bigcap_{M \in \mathcal{M}} M$ maximal $\Rightarrow I \subseteq \text{Jac } I$
 $\text{mutually annihilating and idempotents of ideals of } R \text{ are ideals}$
 $\text{Jac } I \subseteq R$

(b) $\text{rad } I \subseteq \text{Jac } I$
 $\text{Let } M$ be a maximal ideal containing I .
 M is a radical ideal containing I .
 $\text{Let } \text{rad } I \subseteq I$.
 $\text{Let } a \in \text{rad } I \Rightarrow a^n \in I \Rightarrow a^n \in M$
 $\Rightarrow a \in \text{rad } M = M$
 $\Rightarrow \text{rad } I \subseteq M$
 $\Rightarrow \text{rad } I \subseteq \text{Jac } I$

(c) $n \geq 1$ integer. $n = p_1^{e_1} \dots p_k^{e_k}$ $(n) \subseteq (n)$
 $\text{Jac } (n) = \bigcap_{p \mid n} (p) = (p_1 \dots p_k)$

(33).

$R = \mathbb{C}[x, y] \xrightarrow{F} R \xrightarrow{G} R$
 $\downarrow$
 $\mathbb{C}[x, y]/M_c$

(a) M_c is a maximal ideal of R
 $E(M_c) = \{a \in \mathbb{C}[x, y] \mid f(a) = 0 \forall f \in M_c\}$ is nonempty
 $\text{By IVT, if } f \text{ has } n \text{ roots, we can pick finitely many } f_i \in M_c \text{ s.t. } f_i(a) = 0 \text{ covering } E(M_c)$
 $\text{Then } 1 = \sum_{i=1}^n f_i^2 = 0 \text{ for } 1 \notin M_c$
 $\text{where } 0 \in M_c \Rightarrow M_c \subseteq M_c \Rightarrow M_c = M_c$

(b) If $a, b \in \mathbb{C}[x, y]$ $\nexists M_c = M_c$
 $f(a) = f(b) = 0 \Rightarrow M_c, M_c$ are not prime ideals
 hence not maximal
 $\begin{cases} f(x) = (x - \sqrt{a})(x - \sqrt{b}) \\ g(x) = (x + \sqrt{a})(x + \sqrt{b}) \end{cases} \Rightarrow \overline{f(x)} \overline{g(x)} = (x - a)(x - b) = 0$
 R/M_c not an integral domain

(c) consider Weierstrass function $W(x)$. $\text{Let } c \in \mathbb{C}[x, y]$, $W(c) = W(c) - W(c)$
 $W(c) \in M_c$ however $W(c) \notin (c)$ $\text{take } W(c) = 0$
 $\lim_{x \rightarrow c} \frac{W(x)}{x-c} = \lim_{x \rightarrow c} g(x) = g(c)$

(d) Suppose $M_c = (f_1, \dots, f_n)$ $f_i \in R \forall i$

$F(x) = \sum_{i=1}^n f_i^2 \in M_c$

$\text{if } g = \frac{1}{\sqrt{F(x)}} \Rightarrow g \in M_c \text{ if } g = \sum_{i=1}^n h_i f_i$

$\|g\|_2 \leq \left\| \sum_{i=1}^n h_i \right\|_2 \left\| \sum_{i=1}^n f_i^2 \right\|_2 = \left\| \sum_{i=1}^n h_i \right\|_2 \sqrt{F(x)}$

$\Rightarrow \frac{1}{\sqrt{F(x)}} \leq \left\| \sum_{i=1}^n h_i \right\|_2 \Rightarrow \exists h_i \in \mathbb{C}[x, y] \text{ s.t. } h_i \neq 0$

(34). $R = \mathbb{C}(R, R)$

$M_c = \{f \in R \mid f(a) = 0\}$

(a) $I = \mathbb{C}_c(R, R)$ $a \in I$ nonempty

$q_1, q_2 \in I$

$\text{supp}(q_1 + q_2) \subseteq \text{supp } q_1 \cup \text{supp } q_2$

$x \in \text{supp}(q_1 + q_2)$

$q_1(x) + q_2(x) \neq 0$

$\Rightarrow x \in \text{supp } q_1 \cup \text{supp } q_2$

$\text{Let } f \in R, q_i \in I$

$\text{if } x \in R \setminus \text{supp } q_i \Rightarrow x \in R \setminus \text{supp } q_i$

$R \setminus \text{supp } q_i \subseteq R \setminus \text{supp } (q_i)$

$\Rightarrow \text{supp } q_i \subseteq \text{supp } q_i$

$\Rightarrow I$ is an ideal.

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases} \quad g(x) = \begin{cases} x & \text{if } x \leq 0 \\ 0 & \text{if } x > 0 \end{cases} \quad f, g \notin I$$

$$f(x) - g(x) = 0 \quad \forall x \Rightarrow f, g \in I \Rightarrow I \text{ is not prime.}$$

(b) let M maximal $\subset I \subset M$.

$\forall c \in R$ we can find $f \in I$ s.t. $c \in \text{supp}(f)$ ($\text{supp}(f) \neq \emptyset$)

$\Rightarrow Z(H)$ is empty $\Rightarrow M \neq N_L \quad \forall c \in R$

(5f) R a local ring. M is a unique maximal ideal.

All proper ideals are contained in M

let $x \in R - M$ (x) = R

$$\exists r \in R^* \quad r = ax \quad a \in R \quad r^{-1}r = (a^{-1})x = 1 \Rightarrow x \text{ is a unit.}$$

R commutative ring with 1.

$R - R^* = M \subset R$ all the nonzero elements of R/M are representatives of R^* $\forall c \in R/M \quad x \in M \Rightarrow x \in R - R^* \Rightarrow x \in R - R^*$

$R/M - \{0\} = R^* \Rightarrow M$ is maximal.

$I \subset R$ then I does not contain a unit

$I \subset R - R^* = M \Rightarrow R$ is a local ring.

$$(2d) \quad R = \left\{ a + \frac{b}{2} \mid a, b \in \mathbb{Z} \right\} \quad 1 \in R$$

$$R^* = \left\{ a + \frac{b}{2} \mid a, b \in \mathbb{Z}, a \text{ odd} \right\}$$

$$R - R^* = \left\{ a + \frac{b}{2} \mid a, b \in \mathbb{Z}, a \text{ even} \right\} = (2) \Rightarrow (2) \text{ is unique maximal, and } R \text{ is a local ring.}$$

(2e) K a field v valuation $v: K^* \rightarrow \mathbb{Z}$ (i) $v(xy) = v(x) + v(y)$ (ii) $v(x^{-1}) = -v(x)$ (iii) $v(x+y) \geq \min\{v(x), v(y)\}$ if $x, y \neq 0$

R valuation ring. $A_K = \{x \in K \mid v(x) \geq 0\} \cup \{0\}$

(a) Since $\Rightarrow$ is surjective $\exists x, y \in R \quad v(x) = v(y) = k$
 $\exists z \in R \quad v(z) = k$
 $\Rightarrow z \in A_K$

$$(r) \subseteq A_K \quad (v(x) \geq 0)$$

$$a \in A_K \quad \exists r' \in R \quad s.t. \quad v(r') = v(a)$$

$$K^*/\{1, -1\} \cong \mathbb{Z} \Rightarrow \bar{a} = \overline{r' \cdot r}$$

$$\{a, -a\} = \{r' \cdot r, -r' \cdot r\}$$

$$\Rightarrow (a) = (-a) \Rightarrow a \in (r)$$

$$\text{if } A_K = (r) \quad \text{let } v(r) = 1.$$

$$A_{K+1} = (r') \subseteq (r) = A_K$$

$$(b) \text{ then } 1, -1 \in \mathbb{Z}, \quad v(2) \geq 1 \Rightarrow \exists r \quad v(r) = v(2) = 1$$

$$I = (r) = A_K \quad \text{for some } r \in \mathbb{Z}$$

(c) $\mathbb{Z}$ has only unit in R

$$R - R^* = R - \{1\} = A_K \Rightarrow R \text{ is a local ring.}$$

(4a) (i) $\Rightarrow$ (ii) R has exactly one prime ideal $\Rightarrow R$ is a local ring.
 $\Rightarrow R - R^*$ is a set of units
 $\Rightarrow \mathcal{P}(R) = \{0, R\}$

(i) $\Rightarrow$ (ii) $\mathcal{P}(R)$ is an ideal

$\Rightarrow \mathcal{P}(R)$ is a set of nonunits

$\Rightarrow \mathcal{P}(R)$ is maximal $\Rightarrow R/\mathcal{P}(R)$ is a field

(ii) $\Rightarrow$ (i) $R/\mathcal{P}(R)$ is a field for $\mathcal{P}(R)$ is a maximal ideal.

and $\mathcal{P}(R) \cap P$

$\Rightarrow \mathcal{P}(R)$ is the only prime ideal.

(2) $\mathcal{Q} \subset R$ ideal is primary

$\Rightarrow$ nonzero $a, b \in \mathcal{Q}$ and $a \in \mathcal{Q}$

$\Rightarrow a \in \mathcal{Q}$ for some $n \in \mathbb{N}$

$\Rightarrow a, b \in \mathcal{Q} \Rightarrow a^n, b^n \in \mathcal{Q}$ for some $n, m \in \mathbb{N}$

$\Rightarrow \mathcal{Q}$ is primary $\Leftrightarrow \text{rad}(\mathcal{Q})$ is prime.

(1) suppose $ab \in \mathcal{Q}$ $a \notin \mathcal{Q}$

$ab \in \text{rad}(\mathcal{Q}) \Rightarrow a \in \text{rad}(\mathcal{Q})$ or $b \in \text{rad}(\mathcal{Q})$

$$\Rightarrow a^n \in \mathcal{Q} \quad \text{or} \quad b^m \in \mathcal{Q}$$

$\Rightarrow \mathcal{Q}$ is primary.

$\Rightarrow$ suppose $\mathcal{Q}$ is primary.

$ab \in \text{rad}(\mathcal{Q})$

$$(ab)^n \in \mathcal{Q} \Rightarrow a^n b^n \in \mathcal{Q}$$

$$\Rightarrow (a^n)^m \in \mathcal{Q} \quad \text{and} \quad (b^n)^m \in \mathcal{Q}$$

$$\Rightarrow a \in \text{rad}(\mathcal{Q}) \quad \text{and} \quad b \in \text{rad}(\mathcal{Q})$$

$\Rightarrow \text{rad}(\mathcal{Q})$ is prime.

(a) $\{f(x)\}$ are all ideals of $\mathbb{Z}$

(n) is primary $\Leftrightarrow \text{rad}(n)$ is prime.

$$\Leftrightarrow \text{rad}(n) = (p) \text{ or } 0 \text{ for } p \text{ prime.}$$

$$\Leftrightarrow (n) = (p^k) \text{ or } 0 \text{ for some } k \in \mathbb{N}. \quad \{x^k \in (n) \mid n \mid p^k\}$$

(b) P is a prime ideal $\Rightarrow \text{rad}(P)$ prime $\Rightarrow P$ is primary.

(c) $\mathcal{Q}$ is primary. Suppose $\bar{a}, \bar{b}$ are zero divisors in $R/\mathcal{Q}$ $\Rightarrow a, b \in \mathcal{Q}$

$$\Rightarrow a^n \in \mathcal{Q} \quad b^m \in \mathcal{Q} \Rightarrow a^n \cdot b^m \in \mathcal{Q}$$

$\Rightarrow \bar{a}^n, \bar{b}^m$ are nilpotent

$$\bar{a}^n, \bar{b}^m \text{ are nilpotent}$$

$$\Rightarrow a^n \in \mathcal{Q} \quad b^m \in \mathcal{Q}$$

$\Rightarrow \mathcal{Q}$ is primary.

Ring of fractions (R commutative)

if a is not a zero divisor nor 0, and $ab = ac \Rightarrow (a \neq 0 \Rightarrow b = c)$

if a is a zero divisor then we cannot always cancel $ab = ac$ to get $b = c$.

$c = 0 \quad ac = 0 \quad ab = 0$ (zero divisor) $b \neq 0$ b is clearly nonzero.

Goal: a commutative ring R is always a subring of a larger ring $\mathcal{Q}$ where every nonzero element is invertible in $\mathcal{Q}$.

(*) if R is an integral domain $\exists \mathcal{Q} \supset R$ s.t. $\forall a \in R \quad a \in \mathcal{Q}^*$
 in this ring $\mathcal{Q} \supset R$ s.t. $\forall a \neq 0, a$ not zero divisor $a \in \mathcal{Q}^*$

$$\begin{array}{ccc} Q & \xrightarrow{\varphi} & \mathbb{Q}(R) \subseteq F \\ \uparrow \iota & & \uparrow \iota \\ R & \xrightarrow{\varphi} & R \end{array}$$

$$f(a) = \begin{matrix} a & \xrightarrow{q} & f(a) & \cdots & f(n) \\ \downarrow & \nearrow & & & \\ \{a_1, \dots, a_n\} & & & & \end{matrix} \quad \begin{cases} q(a_i) = i & \forall i \\ q(a_i) = q(a_j) \end{cases}$$

intuitively if $q \Rightarrow \exists x \in \mathbb{Z} \text{ s.t. } q(x) = (a_1, \dots, a_n) \text{ w/ } a_1, \dots, a_n \in \mathbb{Z}$
 uniqueness of a model $(a_1, \dots, a_n)$ is implied by the isomorphism.

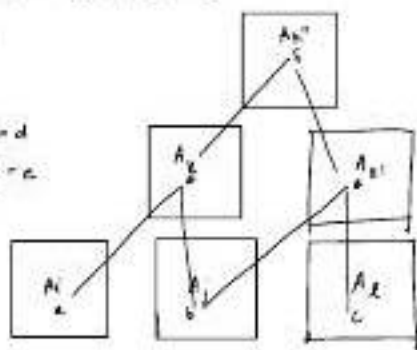
$\Rightarrow \text{let } n_i = \frac{n}{n_i} \quad (n_i, n_j) = 1 \quad \forall i, j = 1, \dots, k$
 $\Rightarrow a_i n_i + b_i n_i^2$
 $a_i n_i^2 \equiv 1 \pmod{n_i}$
 $1 \pmod{n} = (c_1 n_1^2, \dots, c_k n_k^2)$
 $x \pmod{n} = (a_1, \dots, a_k) \quad (c_1 n_1^2, \dots, c_k n_k^2) = (a_1 c_1 n_1^2, \dots, a_k c_k n_k^2)$
 $\Rightarrow x = a_1 c_1 n_1^2 + \dots + a_k c_k n_k^2 \pmod{n}$
 (c) $x \equiv 1 \pmod{8} \quad x \equiv 2 \pmod{25} \quad x \equiv 3 \pmod{81}$
 $n = 8 \cdot 25 \cdot 81 \quad (8, 25)(25, 81)(8, 81) = 1$
 $\frac{x}{8} \equiv \frac{x}{25} \cdot \frac{x}{81} \cdot \frac{x}{81} \pmod{81} \quad (C.C.T.)$
 $\frac{8 \cdot 25 \cdot 81}{8} = 25 \cdot 81 \quad 25 \cdot 81 \pmod{81} \equiv 1 \cdot 1 \pmod{81}$
 $\frac{8 \cdot 25 \cdot 81}{25} = 8 \cdot 81 \quad 8 \cdot 81 \pmod{25} \equiv 8 \cdot 6 \pmod{25}$
 $\equiv 48 \pmod{25}$
 $\equiv 23 \pmod{25}$
 $\frac{8 \cdot 25 \cdot 81}{81} = 8 \cdot 25 \quad 8 \cdot 25 \pmod{81} \equiv 200 \pmod{81}$
 $\equiv 38 \pmod{81}$
 $x = (1 + 2 \cdot 23 + 3 \cdot 38) \pmod{8 \cdot 25 \cdot 81}$
 $\{^* \quad \varphi(n) = |\mathbb{Z}/n\mathbb{Z}^*| \}$

4. $f(x) \equiv f_1(x) \pmod{n_1} \quad f(x) \equiv f_2(x) \pmod{n_2} \quad \dots \quad f(x) \equiv f_k(x) \pmod{n_k} \quad (n_i, n_j) = 1 \quad i \neq j$
 consider coefficients of $x^i \quad i \in \{0, 1, \dots, k\}$
 $a_i \equiv a_{f_1} \pmod{n_1} \quad a_i \equiv a_{f_2} \pmod{n_2} \quad \dots \quad a_i \equiv a_{f_k} \pmod{n_k}$
 solve by CRT, $\exists a$ solution, $f(x) \in \mathbb{Z}[x]$
 $\{^* \quad f(x) \text{ are monic} \}$
 hence $a_i \equiv 1 \pmod{n_1} \quad \dots \quad a_i \equiv 1 \pmod{n_k}$
 by CRT, $\exists a_i \equiv 1 \pmod{n_i}$

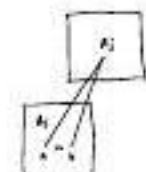
5. $M, n \in \mathbb{Z}^+ \quad (M, n) = 1$
 with $\mathbb{Z}/n\mathbb{Z}$ are isomorphic to $\mathbb{Z}/n\mathbb{Z}$ as a $\mathbb{Z}/n\mathbb{Z}$ module
 of natural isomorphism map properties: is also isomorphism as $\mathbb{Z}/n\mathbb{Z}$

6. $\mathbb{Z} \oplus \mathbb{Z}$ is a subset with partial order, free abelian $\mathbb{Z} \oplus \mathbb{Z}$ is an additive abelian group

(a) Suppose $\mathbb{Z}$ is a directed set: $i, j \in \mathbb{Z} \quad \exists k \in \mathbb{Z} \quad i \leq k \quad j \leq k$
 for $i, j \in \mathbb{Z} \quad (n_j) \quad \exists f_{ij}: A_i \rightarrow A_j$ s.t. preserving order:
 (i) $f_{ik} \circ f_{ij} = \text{id}_{A_i}$ whenever $i \leq j \leq k$
 (ii) $f_{ii} = \text{id}_{A_i}$
 let $B = \bigcup_{i \in \mathbb{Z}} A_i \quad x, y \in B \quad x \in A_i \quad y \in A_j \quad i, j \in \mathbb{Z} \quad \exists k \in \mathbb{Z} \quad i \leq k \quad j \leq k \quad \text{for } x, y \in A_k$
 (a) Reflexivity: $a \in A_i \quad f_{ii}(a) = a \Rightarrow a \sim a$
 Symmetry: $a \sim b \quad a \in A_i \quad b \in A_j \Rightarrow \exists k \in \mathbb{Z} \quad i \leq k \quad j \leq k \quad f_{ik}(a) = f_{jk}(b)$
 $\Rightarrow b \sim a$
 Transitivity: $a \sim b, b \sim c$
 $a \in A_i \quad b \in A_j \quad c \in A_k$
 $a \sim b \Rightarrow \exists k \quad i \leq k \quad j \leq k \quad f_{ik}(a) = f_{jk}(b) = d$
 $b \sim c \Rightarrow \exists l \quad j \leq l \quad k \leq l \quad f_{jl}(b) = f_{kl}(c) = e$
 Since $\mathbb{Z}$ is a directed set $\exists m \in \mathbb{Z} \quad k \leq m \quad l \leq m$
 and $(\exists x \in A_k) \Rightarrow f_{km} \circ f_{ik} = f_{lm} \circ f_{jl}$
 $f_{km}(d) = f_{lm}(e) = f_{km}(f_{jk}(b)) = f_{lm}(f_{kl}(c))$
 $\Rightarrow a \sim c$ is an equivalence class



(b) $\{A_i\}_{i \in \mathbb{Z}} = \bigcup_{i \in \mathbb{Z}} A_i = A$
 let $a \in A \quad f_i: A_i \rightarrow A \quad f_i(a) = \bar{a}$
 Suppose $f_i: A_i \rightarrow A_j \quad i, j \in \mathbb{Z} \quad i \leq j$
 Suppose $f_i(a) = \bar{a} \Rightarrow \exists k \in \mathbb{Z} \quad i \leq k \quad j \leq k \quad f_{ik}(a) = f_{jk}(b)$
 $\Rightarrow \bar{a} = \bar{b}$



(c) Suppose $\{f_{ij}\}$ are homomorphisms (group)
 let $a \in A_i \quad b \in A_j \quad i, j \in \mathbb{Z} \quad i \leq k \quad j \leq k$
 $\Rightarrow f_{ik}(a) = f_{jk}(b)$
 $\Rightarrow (a, b) \mapsto (f_{ik}(a), f_{jk}(b))$
 $\Rightarrow (a, b) \mapsto (f_{ik}(a), f_{jk}(b))$
 $\Rightarrow (a, b) \mapsto (f_{ik}(a), f_{jk}(b))$
 $\Rightarrow (a, b) \mapsto (f_{ik}(a), f_{jk}(b))$

$\bar{a} + \bar{b} = \overline{f_{ik}(a) + f_{jk}(b)}$
 $= \overline{f_{im}(a) + f_{jm}(b)}$
 $= \overline{f_{im}(a) + f_{jm}(b)}$
 $= \overline{f_{im}(a) + f_{jm}(b)}$
 $= \bar{a} + \bar{b}$

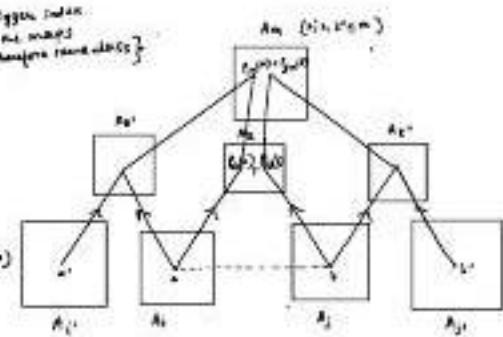
$f_i: A_i \rightarrow A \quad a, b \in A_i$
 $f_i(a + b) = \overline{(a + b)} = \bar{a} + \bar{b} = \overline{f_i(a) + f_i(b)} = \overline{f_{im}(a) + f_{im}(b)} = \overline{f_{im}(a + b)} = \bar{a} + \bar{b} = \bar{a} + \bar{b}$

7. $\{^* \quad A_i \text{ are commutative rings with } 1, \text{ and all } f_{ij} \text{ are ring homomorphisms} \}$
 then we $\bar{a} \times \bar{b} = \overline{f_{ik}(a) \cdot f_{jk}(b)}$
 $= \overline{f_{im}(a) \cdot f_{jm}(b)}$
 $= \overline{f_{im}(a) \cdot f_{jm}(b)}$
 $= \overline{f_{im}(a) \cdot f_{jm}(b)}$
 $= \bar{a} \times \bar{b}$ is well defined

$\forall \bar{a} \in A \quad \bar{a} \times \bar{1} = \overline{f_{ik}(a) \cdot f_{jk}(1)}$
 $= \overline{f_{im}(a) \cdot f_{jm}(1)}$
 $= \overline{f_{im}(a) \cdot f_{jm}(1)}$
 $= \bar{a} + \bar{0} = \bar{a}$

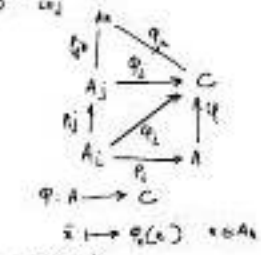
$\bar{a} \times \bar{b} = \overline{f_{ik}(a) \cdot f_{jk}(b)}$
 $= \overline{f_{jk}(b) \cdot f_{ik}(a)}$ since A_k commutative ring
 $= \bar{b} \times \bar{a}$

$\Rightarrow A$ is a commutative ring with 1.



$p_i: A_i \rightarrow A$
 $p_i(a \cdot b) = (a \cdot b)_i = p_{iA}(a) p_{iA}(b)$
 $= \bar{a} \cdot \bar{b}$

$\Rightarrow$ ring isom.



uniqueness:
 $\bar{a} = \bar{b} \quad \forall a \in A_i \quad \forall b \in A_j \quad \exists x \in A_i \cap A_j \quad p_{iA}(ax) = p_{jA}(bx)$
 $q(\bar{a}) = q(p_{iA}(ax))$
 $= q_i(p_{iA}(ax))$
 $= q_i(p_{jA}(bx))$
 $= q_j(b) = q(\bar{b})$
 $q(\bar{a} = \bar{b}) = q(p_{iA}(a) + p_{jA}(b))$
 $= q_i(p_{iA}(a) + p_{jA}(b))$
 $= q_i(p_{iA}(a)) + q_i(p_{jA}(b))$
 $= p_{iA}(a) + p_{jA}(b) = q(a) + q(b)$

Suppose $\exists q': A \rightarrow C$
 $\text{st } q' \circ p_i = q_i = q \circ p_i$
 $\forall a \in A_i \quad q'(a) = q \circ p_i(a) = q_i(a) = q \circ \text{image}$

$\mathcal{I} = \{ (u, R) \mid u \in (A_i, b) \text{ for } R \text{ on } (u, b) \}$
 $u \in u \in V \quad \forall u \in \mathcal{I}$
for each $u \in \mathcal{I}$ let $A_{u, \mathcal{I}} = \{ f \in \mathcal{I} \mid (u, f) \in \mathcal{I} \}$
for $v \in \mathcal{I}$ let $p_{u, v}: A_{u, \mathcal{I}} \rightarrow A_{v, \mathcal{I}}$
 $f \mapsto f|_v$

$A = \bigcup_{u \in \mathcal{I}} A_{u, \mathcal{I}}$
 A is an abelian group
 $E_{u, \mathcal{I}}: A_{u, \mathcal{I}} \rightarrow R$
 $f \mapsto f(u)$
 $E_{u, \mathcal{I}}$ is surjective.
From $\mathcal{I}$ not injective, $E_{u, \mathcal{I}} \neq \{0\}$



$E_{u, \mathcal{I}} = q \circ p_{u, \mathcal{I}} \Rightarrow q$ is surjective
 $\Rightarrow q$ is non-injective or $p_{u, \mathcal{I}}$ is non-injective or both.
 $\bar{g} \in A \quad q(\bar{g}) = g(u) = a \Rightarrow \exists f \in A_{u, \mathcal{I}} \quad E_{u, \mathcal{I}}(f) = f(u) = g(u) \Rightarrow f = \bar{g} = p_u(f)$
 $\Rightarrow p_u$ is surjective $\forall u \in \mathcal{I}$.
Since $E_{u, \mathcal{I}}$ is non-injective, $\exists f \in A_{u, \mathcal{I}} \quad \text{st } E_{u, \mathcal{I}}(f) = 0 \Rightarrow f(u) = 0(u)$
 $f \neq 0 \Rightarrow p_u(f) = \bar{f} = \bar{0}$.

$\Rightarrow p_u$ is non-injective.

(50). $\mathcal{I}$ partially ordered. A_i is a group $\forall i \in \mathcal{I}$.

Suppose $\forall i, j \in \mathcal{I} \quad \exists \mu_{ji}: A_j \rightarrow A_i \text{ st } \mu_{ji} \circ \mu_{ij} = \text{id}_{A_i}$
(i) $\mu_{ji} \circ \mu_{ij} = \text{id}_{A_i}$ whenever $i, j \in \mathcal{I}$
(ii) $\mu_{ii} = \text{id}_{A_i} \quad \forall i \in \mathcal{I}$



let $P = \{ (a_i)_{i \in \mathcal{I}} \in \prod_{i \in \mathcal{I}} A_i \mid \mu_{ji}(a_j) = a_i \text{ whenever } i, j \in \mathcal{I} \}$

(a) Define all μ_{ji} and group hom. $P \subseteq \prod_{i \in \mathcal{I}} A_i$

$(a_i)_{i \in \mathcal{I}} \in P$ then $\mu_{ji}(a_j) = a_i \quad P \neq \emptyset$

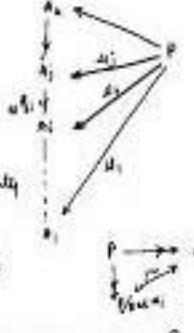
suppose $(a_i)_{i \in \mathcal{I}} = (b_i)_{i \in \mathcal{I}} \in P$

$(a_i)^{-1}(a_i)_{i \in \mathcal{I}} = (b_i^{-1}b_i)_{i \in \mathcal{I}} \quad \mu_{ji}(b_i^{-1}b_i) = \mu_{ji}(b_i^{-1}) \mu_{ji}(b_i)$
 $= (b_i^{-1} \cdot a_i) \quad \forall i, j$
 $\Rightarrow (b_i)^{-1}(a_i)_{i \in \mathcal{I}} \in P \Rightarrow P$ is a subgroup of $\prod_{i \in \mathcal{I}} A_i$

(b) $\mathcal{I} = \mathbb{N}^+$ for each $i \in \mathcal{I}$ let $\mu_i: P \rightarrow A_i$

suppose μ_{ji} are injective.

at $a_i \in A_i$ then μ_{ji} is injective
 $\exists a_j \text{ st } \mu_{ji}(a_j) = a_i$
 $\exists a_{i+1} \text{ st } \mu_{i+1,i}(a_{i+1}) = a_i$



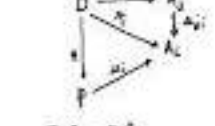
$(a_1, \dots, a_i, \mu_{ji}(a_i), \dots)$
 $\forall i \in \mathcal{I}$ let $a_i = \mu_{ji}(a_i)$ (compatibility property)
 $\forall i > j \quad \exists a_j \text{ st } \mu_{ji}(a_j) = a_i$ (surjectivity)
 $(\mu_{ji}(a_i), \dots, \mu_{ji}(a_i), a_i, \dots)$
 $\mu_i(a_i) = a_i \Rightarrow \mu_i$ is surjective.

$P/P \cong A_i \Rightarrow A_i$ is a quotient group of P .

(c) if all A_i are comm. rings with 1 and μ_{ji} are all ring homs. that are unital.

with $1 = (1_i)_{i \in \mathcal{I}}$ then all maps are unital.
 $1 \in P$
 $1, x$ are unadjoined as group.
 $\mu_i((a_j)_{j \in \mathcal{I}} + (b_j)_{j \in \mathcal{I}}) = a_i + b_i = \mu_i(a_j)_{j \in \mathcal{I}} + \mu_i(b_j)_{j \in \mathcal{I}}$
 $\mu_i((a_j)_{j \in \mathcal{I}} \cdot (b_j)_{j \in \mathcal{I}}) = a_i \cdot b_i = \mu_i(a_j)_{j \in \mathcal{I}} \cdot \mu_i(b_j)_{j \in \mathcal{I}}$
 $\mu_i((a_j)_{j \in \mathcal{I}}) = a_i \quad \text{st } \mu_{ij}$ are ring homs. and unital.

(d) if D is any group, $\forall i \in \mathcal{I} \quad \exists \phi_i: D \rightarrow A_i \quad \phi_i = \mu_{ji} \circ \phi_j$ whenever $i, j \in \mathcal{I}$



$\pi: D \rightarrow P$
 $x \mapsto (\phi_i(x))_{i \in \mathcal{I}} \quad (\phi_i(x))_{i \in \mathcal{I}} \in P$ since $(\phi_i(x))_{i \in \mathcal{I}} = (\mu_{ji} \phi_j(x))_{i \in \mathcal{I}}$
 $\text{suppose } \pi(x) = \pi(y) \quad \forall i \in \mathcal{I} \quad (\pi_i \text{ group hom. unadjoined})$

$$(\pi(x))_{i \in I} = (\pi_i(x))_{i \in I}$$

suppose $x, y \in D$

$$\pi(x \cdot y) = (\pi_i(x \cdot y))_{i \in I} = (\pi_i(x) \cdot \pi_i(y))_{i \in I} = (\pi_i(x))_{i \in I} \cdot (\pi_i(y))_{i \in I} = \pi(x) \cdot \pi(y)$$

$$\pi_i(\pi(x)) = \pi_i(\pi_i(x))_{i \in I} = \pi_i(x) = \pi_i(x)$$

$$\bigcap_{i \in I} \pi_i^{-1} \pi_i(x) = \pi_i^{-1}(\pi_i(x))_{i \in I} = \pi_i^{-1}(\pi_i(x)) = x$$

$$(1) \quad I \in \mathbb{R}^+ \quad A = \mathbb{R}/p\mathbb{Z} \quad \pi_i: \mathbb{R}/p\mathbb{Z} \rightarrow \mathbb{R}/p\mathbb{Z} \quad i \in I$$

$$\pi_i: \mathbb{R}/p\mathbb{Z} \rightarrow \mathbb{R}/p\mathbb{Z} \quad i \in I$$

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$$(2) \quad (a_i)_{i \in I} \in \mathbb{Z}_p \quad \text{ s.t. } a_i \bmod p^i = a_j \bmod p^j \quad \text{ for } i < j$$

$$(a_i)_{i \in I} = a + \sum_{k=1}^{\infty} b_k p^k \quad b_k \in \{0, 1, \dots, p-1\}$$

$$+ : \mathbb{Z}_p \times \mathbb{Z}_p \rightarrow \mathbb{Z}_p$$

$$\pi: \mathbb{Z}_p \times \mathbb{Z}_p \rightarrow \mathbb{Z}_p$$

$$+ : \sum_{k=1}^{\infty} a_k p^k + \sum_{k=1}^{\infty} b_k p^k \mapsto \sum_{k=1}^{\infty} (a_k + b_k) p^k$$

$$+ : \sum_{k=1}^{\infty} a_k p^k \mapsto \sum_{k=1}^{\infty} (\sum_{i=1}^k a_i) p^k$$

$$(3) \quad \mathbb{Z}_p \subseteq \mathbb{Z}_p[[t]] \quad \text{ since } \mathbb{Z}_p \text{ is an integral domain } \Rightarrow \mathbb{Z}_p[[t]] \text{ is an integral domain.}$$

$$\text{consider } \pi: \mathbb{Z} \rightarrow \mathbb{Z}_p$$

$$\pi: \mathbb{Z} \rightarrow \mathbb{Z}_p$$

$$\pi: \mathbb{Z} \rightarrow \mathbb{Z}_p$$

$$(4) \quad \text{suppose } b_0 + b_1 p + b_2 p^2 + \dots \quad \text{ is a unit in } \mathbb{Z}_p$$

$$b_0 + b_1 p + b_2 p^2 + \dots \quad \text{ is a unit in } \mathbb{Z}_p$$

$$(5) \quad \text{suppose } b_0 \neq 0 \Rightarrow b_0 \text{ is a unit in } \mathbb{Z}_p$$

$$(6) \quad \mathbb{Z}_p = \{b_0 + b_1 p + \dots \mid b_i \in \{0, 1, \dots, p-1\}\} \quad \text{ is a local ring with unique maximal ideal } \mathfrak{m}_p$$

$$(7) \quad \text{for each } \pi_i: \mathbb{Z}_p \rightarrow \mathbb{Z}/p^i\mathbb{Z}$$

$$\pi_i(\mathbb{Z}_p) = \mathbb{Z}/p^i\mathbb{Z}$$

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if \$a, b\$ are unique, if we require \$r > 0\$
 (2) if \$F\$ is a field, \$F[x]\$ is an euclidean domain. let norm = deg. \$f(x) \to \deg(f)\$
 (standard long division algorithm)

(3) \$\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}\$ \$D \in \mathbb{Z}\$ is not a perfect square.
 it is a subring of \$\mathbb{C}\$ (obvious as \$D \neq 0\$)
 if \$a, b \in \mathbb{Q}\$ then \$a, b\$ not both zero.
 then \$(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - b^2 D \in \mathbb{Q}\$

if \$a, b \in \mathbb{Q}\$, then \$\frac{a^2 - b^2 D}{a^2 + b^2} \in \mathbb{Q}\$
 if \$D \in \mathbb{Z}\$ is a field (quadratic field)
 any rational number \$D = f^2 D'\$ for some \$f \in \mathbb{Q}\$ and unique \$D' \in \mathbb{Z}\$ not divisible by any integer greater than 1.

\$D = \frac{a}{b} = \frac{\prod p_i^{e_i}}{\prod p_i^{f_i}} = \frac{1}{\prod p_i^{f_i}}\$ (cancel out all the even powers)
 if \$D \in \mathbb{Z}\$, then \$D = \pm \prod p_i^{e_i}\$
 we could have \$-1\$ or \$1\$ if \$\prod p_i^{e_i}\$ are distinct.
 \$D = \pm \prod p_i^{e_i}\$ is the square free part of \$D\$ which is not a perfect square in \$\mathbb{Z}\$.

if \$D = \pm 1\$ then \$\mathbb{Q}(\sqrt{D}) = \mathbb{Q}(\sqrt{D'})\$ so we can assume \$D\$ is a square free integer.

quadratic integer rings
 let \$D\$ be a square free integer. (\$D \neq \pm 1\$ times product of distinct primes or \$-1\$)

\$\mathbb{Z}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\} \subseteq \mathbb{Q}(\sqrt{D})\$

\$D \equiv 1 \pmod{4}\$
 \$\mathbb{Z}[\frac{1+\sqrt{D}}{2}] = \{a + b(\frac{1+\sqrt{D}}{2}) \mid a, b \in \mathbb{Z}\}\$ is also a subring.

if \$D \equiv 1 \pmod{4}\$, then \$(\frac{1+\sqrt{D}}{2})^2 = \frac{1+2\sqrt{D}+D}{4} = \frac{D+1}{4} + \frac{\sqrt{D}}{2} \in \mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$
 if \$D \equiv 0 \pmod{4}\$, then \$(\frac{\sqrt{D}}{2})^2 = \frac{D}{4} \in \mathbb{Z}[\frac{\sqrt{D}}{2}]\$

if \$D \equiv 3 \pmod{4}\$, then \$(\frac{1+\sqrt{D}}{2})^2 = \frac{1+2\sqrt{D}+D}{4} = \frac{D+1}{4} + \frac{\sqrt{D}}{2} \notin \mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$

if \$D \equiv 2 \pmod{4}\$, then \$(\frac{\sqrt{D}}{2})^2 = \frac{D}{4} \notin \mathbb{Z}[\frac{\sqrt{D}}{2}]\$

if \$D \equiv 1 \pmod{4}\$, then \$\mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$ is a subring of \$\mathbb{Q}(\sqrt{D})\$

\$D = -1\$ \$\mathbb{Z}[i]\$ gaussian integers = \$\{a + bi \mid a, b \in \mathbb{Z}\}\$

\$D = -4\$ \$\mathbb{Z}[2i]\$

\$N(a + bi) = a^2 + b^2 \in \mathbb{Z}\$ is known \$(a + bi)(a - bi) = a^2 + b^2\$

\$N\$ is multiplicative.

if \$D \equiv 1 \pmod{4}\$, then \$N(\frac{1+\sqrt{D}}{2}) = \frac{D+1}{4} \in \mathbb{Z}\$

if \$D \equiv 0 \pmod{4}\$, then \$N(\frac{\sqrt{D}}{2}) = \frac{D}{4} \in \mathbb{Z}\$

if \$D \equiv 3 \pmod{4}\$, then \$N(\frac{1+\sqrt{D}}{2}) = \frac{D+1}{4} \in \mathbb{Z}\$

if \$D \equiv 2 \pmod{4}\$, then \$N(\frac{\sqrt{D}}{2}) = \frac{D}{4} \notin \mathbb{Z}\$

\$D\$ is a unit in \$\mathbb{Q}\$ and \$N(D) = D^2\$.

\$D \in \mathbb{Z}\$, \$D \neq \pm 1\$, then \$D\$ is not a unit in \$\mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$ or \$\mathbb{Z}[\frac{\sqrt{D}}{2}]\$

if \$D \equiv 1 \pmod{4}\$, then \$D\$ is not a unit in \$\mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$

if \$D \equiv 0 \pmod{4}\$, then \$D\$ is not a unit in \$\mathbb{Z}[\frac{\sqrt{D}}{2}]\$

if \$D \equiv 3 \pmod{4}\$, then \$D\$ is not a unit in \$\mathbb{Z}[\frac{1+\sqrt{D}}{2}]\$

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Let $I = \{0\}$ is principal. Suppose $I \neq \{0\}$. $\{N(d) \mid d \in I\} \subseteq \mathbb{Z}^+ \cup \{0\}$. $\Rightarrow$ $\exists$ a least element $d \in I$ minimal norm. $\Rightarrow (d) \subseteq I$. Suppose $a \in I$ not by minimality of d .
Then, $a, d \in I$, $2d \in I$, $a - d \in I$. $\Rightarrow a = qd + r$, $N(r) < N(d)$.
 $\Rightarrow a - qd \in I$, $\Rightarrow N(r) = 0$ by minimality of d .
 $\Rightarrow r = 0$.

Example

(1) $R = \mathbb{Z}[i]$. $(2, i)$ is not principal in $\mathbb{Z}[i]$ & not Euclidean.
In $\mathbb{Z}[i]$ every ideal is principal.

(2) $R = \mathbb{Z}[\sqrt{-5}]$. $N(a + b\sqrt{-5}) = a^2 + 5b^2$. $\Rightarrow I = (2, 1 + \sqrt{-5})$

Suppose $I = (d + e\sqrt{-5})$. $2 = d(a + b\sqrt{-5})$, $1 + \sqrt{-5} = f(a + b\sqrt{-5})$

$$2 = N(d) \cdot N(a + b\sqrt{-5}) \quad 1 + \sqrt{-5} = N(f) \cdot N(a + b\sqrt{-5})$$

$$\Rightarrow 2^2 = 4 = N(d) \cdot N(a + b\sqrt{-5}) \quad (a^2 + 5b^2) = 1, 2, 3, \dots$$

$$\{ (a^2 + 5b^2) \mid N(d) = 4 \} = \{ 4 \} \Rightarrow d = \pm 2 \quad \{ (a^2 + 5b^2) \mid N(f) = 2 \} = \{ 2 \}$$

$$\{ (a^2 + 5b^2) = 1 \} = \{ 1 \} \Rightarrow a + b\sqrt{-5} = \pm 1$$

$$\Rightarrow (a + b\sqrt{-5}) = \pm 1 \Rightarrow 1 \in (d, 2 + \sqrt{-5})$$

$$\Rightarrow 1 = d\beta + \gamma(2 + \sqrt{-5}) \quad \text{for some } d, \beta, \gamma \in R.$$

$$(2 - \sqrt{-5}) = d\beta + \gamma(2 + \sqrt{-5})$$

$$\Rightarrow (2 - \sqrt{-5}) \in (d) \text{ which is a contradiction!}$$

$\Rightarrow \mathbb{Z}[\sqrt{-5}]$ is not an Euclidean domain!

Def: R unimodular ring. $a, b \in R$ s.t.

(1) a is a multiple of b if $\exists x \in R$ s.t. $a = bx$. i.e.

(2) $\gcd(a, b) = d$ if

(i) $d \mid a$ and $d \mid b$

(ii) $\{ d' \mid d' \mid a \text{ and } d' \mid b \} \Rightarrow d' \mid d$.

$b \mid a$ in R $\Leftrightarrow (a, b) \subseteq (b) \Leftrightarrow (a) \subseteq (b)$

$\{ I = (a, b) \}$ and $(a, b) \Rightarrow I \subseteq (d)$

$\{ (d') \supseteq I \}$ then $(d) \subseteq (d')$

Prop 1: $\{0, 1\}$ are nonzero elements in commutative ring R with 1.

$$(a, b) = (d) \Rightarrow d = \gcd(a, b)$$

Prop 2: $\{ (d) = (d') \mid d, d' \in \{0, 1\} \}$

$$d = ad', d' = bd, a \in R, b \in R$$

$$d = abd \Rightarrow d(1 - ab) = 0 \Rightarrow ab = 1$$

$$\text{if } d = \gcd(a, b) = d \text{ then } d' \mid d \text{ and } d \mid d' \Rightarrow (d) = (d') \Rightarrow d = ud' \text{ for } u \in R^*$$

Theorem 4: Let R be a Euclidean domain. $a, b \in R$, $a \neq 0$.

Let $d = r_n$ last nonzero remainder in Euclidean Algorithm for a and b .

Then (1) $d = \gcd(a, b)$

$$(2) (d) = (r_n)$$

Def: $a, b \in R$ $a = qb + r_0$ (1)

$$b = q_1 r_0 + r_1 \quad (2)$$

$$r_0 = q_2 r_1 + r_2 \quad (3)$$

$$\vdots$$

$$r_{n-2} = q_{n-1} r_{n-1} + r_n$$

$$r_{n-1} = q_n r_n + 0$$

(1) $K = \mathbb{Z}$. Suppose $r_n \mid r_{n-1}$ $\forall i \leq K$ for some $K < n$.

$$r_{n-(k+1)} = q_{n-k} r_{n-k} + r_{n-k-1}$$

$$\Rightarrow r_n \mid r_{n-(k+1)}$$

$$\Rightarrow r_n \mid a, r_n \mid b \Rightarrow (a, b) \subseteq (r_n)$$

$$(2) r_0 = a - qb \in (a, b)$$

$$r_1 = b - q_1 r_0 \in (b, r_0) \subseteq (a, b)$$

Suppose $r_{k-1}, r_k \in (a, b)$

$$r_2 = r_0 - q_2 r_1$$

$$r_{k+1} = r_{k-1} - q_{k+1} r_k \in (r_{k-1}, r_k) \subseteq (a, b)$$

$$\Rightarrow (r_n) \subseteq (a, b)$$

$$\Rightarrow (r_n) = (a, b) \Rightarrow r_n = \gcd(a, b)$$

$$\Rightarrow r_n = ax + by \quad x, y \in R$$

$$ax + by = N \text{ is solvable in } R \text{ (Euclidean)}$$

$$\Leftrightarrow N \in (d) \quad d = \gcd(a, b) \quad (d, a, b) \mid N$$

For any integral domain let $\tilde{R} = R^* \cup \{0\}$. $\tilde{R} \subseteq R - \tilde{R}$ is called a side divisor.

$\{ x \in R \mid \exists a \in \tilde{R} \text{ s.t. } x \mid a \text{ in } R \}$. $x = qu + t$ u neither zero or a unit.

Prop 5: Let R be an integral domain that is not a field. Suppose R is Euclidean, then there are minimal side divisors in R .

Def: $u \in R - \tilde{R}$ s.t. $N(u) = \min \{ N(a) \mid a \in R - \tilde{R}, N(a) \neq 0 \}$ then $u \in R$.

$x = qu + r$ $\Rightarrow 0 \leq N(r) < N(u)$ $\Rightarrow r = 0$ $\Rightarrow x \in \tilde{R}$.

$\Rightarrow u$ is a universal side divisor.

Example: $\tilde{R} = \mathbb{Z}[(1 + \sqrt{-5})/2]$

$$\tilde{R} = \{0, \pm 1\} \text{ (we know)}$$

Suppose u is a side divisor.

$$N(x + y(1 + \sqrt{-5})/2) = a^2 + ab + b^2$$

$$\{ a, b \in \mathbb{Z} \mid b \neq 0 \}$$

$$a^2 + ab + b^2 \geq 1$$

$$a^2 + ab + \frac{b^2}{4} + \frac{3b^2}{4} \geq 1$$

$$N(x) = (a + \frac{b}{2})^2 + \frac{3b^2}{4} \geq 1 \quad \forall x \in R \text{ with } x \neq 0$$

$$\Rightarrow$$
 with norms we get $a, b \in \mathbb{Z}$ then $N(x) = a^2$

$$\{ N(x) = 1 \mid x \in \mathbb{Z} \} \Rightarrow x = \pm 1$$

$$N(x) = 4 \Rightarrow x = \pm 2$$

$$\{ x = 2 \Rightarrow u \mid 2 - 0 \text{ or } u \mid 2 \pm 2 \Rightarrow u \in R$$

$$\Rightarrow u \mid 2 \text{ or } u \mid 3$$

$$\{ 3 = u^2 \text{ for } u \in R$$

$$4 = N(u)N(v) \Rightarrow N(u) = 2 \text{ or } 4$$

$$\Rightarrow u \mid 2 \text{ or } u \mid 3$$

$$\Rightarrow u \mid 2 \text{ or } u \mid 3$$

$$\Rightarrow u \mid 2 \text{ or } u \mid 3$$

$\Rightarrow u \in \{ \pm 1, \pm 2 \}$
 $\frac{1}{2} z = qd \Rightarrow d = N(d)N(q)$
 $N(d) = \pm 1$ or ± 3 otherwise for $N(d)$
 $\Rightarrow d = \pm 1$ or ± 3
 $\Rightarrow u \in \{ \pm 1, \pm 3 \}$
 suppose $z = 1 + \frac{\sqrt{13}}{2}$ then $\pm 2 \mid \frac{1 + \sqrt{13}}{2}$ or $\pm 3 \mid \frac{1 + \sqrt{13}}{2}$
 $\pm 2 \mid \frac{\sqrt{13}}{2}$ or $\pm 3 \mid \frac{\sqrt{13}}{2}$
 $\pm 2 \mid 1 + \frac{\sqrt{13}}{2}$ or $\pm 3 \mid 1 + \frac{\sqrt{13}}{2}$
 all impossible!
 $\Rightarrow \mathbb{Z}[\frac{1 + \sqrt{13}}{2}]$ is not an euclidean domain!

Exercises

(a) $a=20, b=13$
 $20 = 2 \cdot 13 - 6$
 $13 = 2 \cdot 6 + 1$
 $2 = 2 \cdot 1 + 0 \Rightarrow 1 = \gcd$
 $1 = 2 \cdot 10 + 13 \cdot 18$
 (b) $a=69, b=89$
 $89 = 1 \cdot 69 + 20$
 $69 = 3 \cdot 20 + 9$
 $20 = 2 \cdot 9 + 2$
 $9 = 4 \cdot 2 + 1$
 $2 = 2 \cdot 1 + 0 \Rightarrow 1 = \gcd$
 $1 = 89 \cdot 13 + 69 \cdot 17$
 (c) $a=11301, b=4773$
 $11301 = 2 \cdot 4773 + 1755$
 $4773 = 2 \cdot 1755 + 1263$
 $1755 = 1 \cdot 1263 + 492$
 $1263 = 2 \cdot 492 + 279$
 $492 = 1 \cdot 279 + 213$
 $279 = 1 \cdot 213 + 66$
 $213 = 3 \cdot 66 + 15$
 $66 = 4 \cdot 15 + 6$
 $15 = 2 \cdot 6 + 3$
 $6 = 2 \cdot 3 + 0 \Rightarrow 3 = \gcd$
 $3 = 11301 \cdot 13 + 4773 \cdot 77$

$1 = 13 - (13 \cdot 2 - 20) \cdot 2$
 $1 = 13 - 4 \cdot 13 + 0 \pmod{20}$
 $= -3 \cdot 13 \pmod{20} \Rightarrow 13^{-1} \pmod{20} = 3$

(2) (a) $a=13, n=20$
 $13(13) + 20 \cdot 2 = 1 \Rightarrow \gcd(13, 20) = 1$
 $13^{-1} \pmod{20} = 17$

(b) $a=69, n=89 \Rightarrow \gcd(69, 89) = 1$
 $89 = 1 \cdot 69 + 20$
 $69 = 3 \cdot 20 + 9$
 $20 = 2 \cdot 9 + 2$
 $9 = 4 \cdot 2 + 1$
 $1 = 89 \cdot 13 + 69 \cdot 17$

(3) Let R be an Euclidean domain.
 Let $\min \{ N(x) \mid 0 \neq x \in R \}$
 Let $a \in R, a \neq 0, N(a) = m$
 $\forall b \in R, \exists q, r \in R$
 $b = qa + r$ with either $r = 0$ or $N(r) < N(a)$
 $\Rightarrow r = 0$
 $\Rightarrow b = qa$
 $\Rightarrow a$ is a unit.
 Suppose $a \neq 0 \in R$ let $N(a) = 0$
 $\forall x \in R, b = qa + r$ is necessarily 0, since $N(a) > N(r) \Rightarrow r = 0$
 $\Rightarrow b = qa \Rightarrow a$ is a unit.

(4) A Euclidean domain:

(a) Suppose $(a, b) = 1$ and $a \mid bc, a \nmid b, a \mid c \Rightarrow a \mid c$
 Suppose $a \mid bc$ with $a \neq 0, b \neq 0$
 let $(a, b) = d = ax + by = d$
 $\left(\frac{a}{d} \right)x + \left(\frac{b}{d} \right)y = 1 \Rightarrow \left(\frac{a}{d}, \frac{b}{d} \right) = 1$
 Since $a \mid bc \Rightarrow \frac{a}{d} \mid \frac{b}{d}c \Rightarrow \frac{a}{d} \mid c$
 $\left\{ \frac{a}{d} \mid \frac{b}{d}c = \frac{a}{d} \cdot \frac{b}{d} \cdot \frac{c}{d} \right\}$

(b) $ax + by = N, a, b \neq 0$
 Suppose $(x_0, y_0) \in \mathbb{Z} \times \mathbb{Z}$ i.e. $ax_0 + by_0 = N$
 $a(x - x_0) + b(y - y_0) = 0$
 $a(x - x_0) = -b(y - y_0)$
 $(a, b) = (d)$ gcd
 $\left(\frac{a}{d}, \frac{b}{d} \right) = (1)$
 $\frac{a}{d} \mid \frac{b}{d}(y - y_0) \Rightarrow \frac{a}{d} \mid (y - y_0)$ and $x - x_0 = m \cdot \frac{b}{d}$
 where m ranges over integers.

(b) (a) $2x + 4y = 5$ i.e. $2 \mid 5$
 if 5 has no solution.

(b) $17x + 23y = 5$
 $5 = (17 \cdot 1 - 23 \cdot 1) \Rightarrow y = 1 \pmod{17}$
 let $y_0 = 1 \Rightarrow x_0 = -5$
 $x = -5 + m \cdot 23, y = 1 + m \cdot 17$

(c) let $a, b \in \mathbb{Z}^+, a, b \neq 0, (a, b) = 1, a, b > 0$
 $\forall N \in \mathbb{Z}, \exists x, y \in \mathbb{Z}$ s.t. $ax + by = N$
 $bx \equiv N \pmod{a}$
 $x \equiv b^{-1}N \pmod{a}$
 $x \in \{ 0, \dots, a-1 \}$
 $x = \frac{N - by}{a} \in \mathbb{Z} \Rightarrow N - by \equiv 0 \pmod{a}$
 $\Rightarrow N \equiv by \pmod{a}$
 $\Rightarrow N \equiv b \cdot y \pmod{a}$
 $\Rightarrow N \equiv b \cdot y \pmod{a}$
 $\Rightarrow N \equiv b \cdot y \pmod{a}$

Suppose $ax + by = ab - a - b, x, y \geq 0$
 $bx = -b \pmod{a}$
 $x \equiv -1 \pmod{a}$
 similarly $y \equiv -1 \pmod{b}$
 let $y = (a-1) \Rightarrow x = (b-1)$
 $a(b-1) + b(a-1) = ab - a - b + 1 \neq ab - a - b$

(7) $(85, 1+13i) \quad N(85+13i) = 85^2 + 13^2$
 $85 = (1+13i)(7i) + (6+7i) \quad N(85+13i) = 85^2 + 13^2 = N(1+13i) = 16+169$
 $1+13i = (6+7i) + (3+5i)$
 $6+7i = (3+5i)(2i) + (4-7i)$
 $3+5i = (4-7i)(i) + i$
 $4-7i = (i)(4-7i)$
 $\Rightarrow \gcd(85, 1+13i) = i$

8. let $P = \mathbb{Q}(\sqrt{5})$ with $\mathcal{O} = \mathbb{Z}[\omega] \quad \omega = \begin{cases} \frac{1+\sqrt{5}}{2} & \text{if } 5 \equiv 1 \pmod{4} \\ \frac{1+\sqrt{5}}{2} & \text{if } 5 \equiv 3 \pmod{4} \end{cases}$
 $N: \mathbb{Q}(\sqrt{5}) \rightarrow \mathbb{Q}$
 $\alpha \mapsto \begin{cases} a^2 - 5b^2 \in \mathbb{Z} & \text{if } 5 \equiv 1 \pmod{4} \\ a^2 + ab + \frac{1-5}{4}b^2 & \text{if } 5 \equiv 3 \pmod{4} \end{cases}$

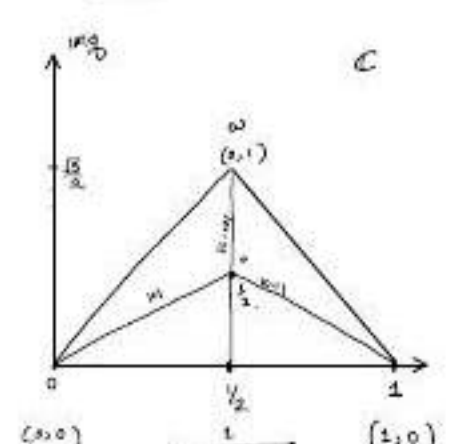
(*) Suppose $D = -2 \quad \text{in } \mathcal{O} = \mathbb{Z}[\omega] \quad N(a+b\omega) = a^2 + b^2$
 let $\alpha, \beta \in \mathcal{O} \quad \alpha = a+b\omega \quad \beta = c+d\omega$
 in $\mathbb{Q}(\omega)$, $\frac{\alpha}{\beta} = \frac{a+b\omega}{c+d\omega} \cdot \frac{c-d\omega}{c-d\omega} = \frac{(ac+bd) + (cb-ad)\omega}{c^2+d^2} = x+y\omega$
 $x, y \in \mathbb{Q} \quad \exists r, q \in \mathbb{Z} \quad x = \frac{r}{2}, y = \frac{q}{2}$

$\frac{\alpha}{\beta} = (r+q\omega) + (s-r\omega)\omega = (r+q)\omega + s - r\omega = (r+q-s)\omega + s$
 $N(\frac{\alpha}{\beta}) = (r-p)^2 + (s-p)^2 \leq \frac{1}{4} + \frac{1}{4} \leq \frac{1}{2}$
 Now $N(\frac{\alpha}{\beta}) \leq \frac{1}{2} N(\beta) < N(\beta)$
 $\Rightarrow \mathbb{Z}[\omega]$ is Euclidean domain.

$D = -2 \quad N(a+b\sqrt{2}) = a^2 + 2b^2$
 let $\alpha, \beta \in \mathbb{Z}[\sqrt{2}] \quad \alpha = a+b\sqrt{2} \quad \beta = c+d\sqrt{2}$
 in $\mathbb{Q}(\sqrt{2})$, $\frac{\alpha}{\beta} = \frac{a+b\sqrt{2}}{c+d\sqrt{2}} \cdot \frac{c-d\sqrt{2}}{c-d\sqrt{2}} = \frac{(ac+2bd) + (bc-ad)\sqrt{2}}{c^2-2d^2} = x+y\sqrt{2}$
 $x, y \in \mathbb{Q} \quad \exists r, q \in \mathbb{Z} \quad x = \frac{r}{4}, y = \frac{q}{2}$
 $\frac{\alpha}{\beta} = (r+q\sqrt{2}) + (s-r\sqrt{2})\sqrt{2} = (r+q\sqrt{2}) + (s-r)2 = (r+q+2s-2r)\sqrt{2} + (r+q)$
 $N(\frac{\alpha}{\beta}) = (r-p)^2 + 2(s-p)^2 \leq \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$
 $N(\frac{\alpha}{\beta}) = N(\frac{\alpha}{\beta}) N(\beta) \leq \frac{3}{4} N(\beta) < N(\beta)$
 $\Rightarrow \mathbb{Z}[\sqrt{2}]$ is an Euclidean domain.

injection from $\left[\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{smallmatrix} \right]_1$

$D = 3, 7, 11 \quad -3 \equiv 1 \pmod{4}, -7 \equiv 1 \pmod{4}, -11 \equiv 1 \pmod{4} \pmod{4}$
 $\mathcal{O} = \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$



$c = \frac{1}{2} + i \frac{10^{1/2}-1}{4\sqrt{10}} = \frac{1}{2} + i \frac{|D|-1}{4\sqrt{D}}$
 $N(c) = |c|^2 = \frac{(1+10)^2}{16 \cdot 10}$

$\Rightarrow \forall \alpha, \beta \in \mathbb{Q}(\omega) \quad \exists (r, q) \in \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$
 $\text{ s.t. } N(\alpha\omega - (r, q)) \leq \frac{(1+10)^2}{16 \cdot 10}$
 let $\alpha, \beta \in \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$
 $\frac{\alpha}{\beta} = \frac{a+b\omega}{c+d\omega} = x+y\omega = (r+p)\omega + (s-r)\omega = (r+p-s)\omega + s$
 $N(\frac{\alpha}{\beta}) = N((r+p-s)\omega + s) \leq \frac{(1+10)^2}{16 \cdot 10} \quad \beta \in \mathbb{Q}(\sqrt{5}) \cap \mathbb{Q}(\omega)$
 Now, $\beta \omega - (r+p-s)\omega = s \Rightarrow s \in \mathbb{Z}[\omega]$
 $N(\frac{\alpha}{\beta}) = N(s) N(\frac{\alpha}{\beta\omega}) \leq N(\frac{\alpha}{\beta\omega}) \frac{(1+10)^2}{16 \cdot 10} < N(\frac{\alpha}{\beta\omega})$
 $\Rightarrow \mathbb{Z}[\omega] \quad \text{with } \omega = \frac{1+\sqrt{D}}{2} \quad \text{whenever } D = 3, 7, 11$

(8) $D = -43, -67, -163$
 Suppose ω is a side divisor in $\mathbb{Z}[\frac{1+\sqrt{D}}{2}] \subset \mathbb{R}$
 let $N(a+b(\frac{1+\sqrt{D}}{2})) = a^2 + ab + \frac{1-D}{4}b^2$
 $\{0, 1\} = \mathbb{R}$
 $\forall a, b \in \mathbb{R} \quad b \neq 0$
 $N(\frac{a}{b}) = a^2 + ab + \frac{1-D}{4}b^2 = (a+\frac{b}{2})^2 + (\frac{1-D}{4})b^2 \geq 1$
 The smallest nonzero values of N on $\mathbb{R}$ are: $1, 4, 9$
 $\frac{1}{4}, \frac{9}{4}, \frac{25}{4}$
 $\forall x \in \mathbb{Z}, u \mid 2 = 0 \text{ or } u \mid 3$
 $\forall 2 \nmid u \quad u, \beta \in \mathbb{R}$
 $6 = N(u)N(\beta) \Rightarrow \text{either } N(u) = 1 \text{ or } N(\beta) = 1$
 $\Rightarrow u = \pm 1 \text{ or } \beta = \pm 1$
 Now only divisors of 2 in $\mathbb{R}$ are $\{\pm 1, \pm 2\}$
 Now only divisors of 3 in $\mathbb{R}$ are $\{\pm 1, \pm 3\}$
 $\Rightarrow u = \pm 1 \text{ or } \pm 3$
 $\forall u \nmid 2 \quad u \nmid 3$
 Now $u \mid 12 \text{ or } u \mid 18$
 $N(u \mid 12) = 4 N(u) \text{ or } N(u) = 4 N(u)$
 $N(u \mid 18) = 9 N(u) \text{ or } N(u) = 9 N(u)$
 $N(u \mid 12) = 12 \Rightarrow 4, 9 \nmid 12, 11$
 $N(u \mid 18) = 11 \Rightarrow 4, 9 \nmid 11$
 $N(u) = 1$
 side divisor

$$\begin{aligned} & \begin{cases} x=3, & u|3 & \text{or } u|2 & \text{or } u|4 \\ \downarrow & \downarrow & \downarrow \\ \text{not possible} & \text{not possible} & u=2 \end{cases} \end{aligned}$$

$$\begin{aligned} 16 &= N(u)N(b) \\ 2 &\text{ or } 8 & \text{not possible} \\ 4 &\text{ or } 4 & \Rightarrow u=\pm 2 \text{ not possible a side divisor} \\ 16 &\text{ or } 1 & \Rightarrow N(u)=16 \Rightarrow 11 \Rightarrow u \nmid w \text{ or } u \nmid w \end{aligned}$$

$\Rightarrow R$ has no side divisors $\Rightarrow \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ is not Euclidean.

$$D=-37, \quad \omega = \frac{1+\sqrt{-37}}{2}, \quad R = \mathbb{Z}[\omega]$$

$$N(a+b\omega) = a^2+ab+17b^2$$

$$\tilde{R} = \{0, \pm 1\}$$

Suppose $u \in R - \tilde{R}$ is a side divisor
 $b \neq 0 \Rightarrow a^2+ab+17b^2 = (a+\frac{b}{2})^2 + (\frac{3b}{4})^2 \geq 17$
 The smallest nonzero values of N on R are $\{1, 4, 9, 16\}$ ($b=0$)

$$\begin{aligned} N(u)=1 & \Rightarrow u=\pm 1 \\ N(u)=4 & \Rightarrow u=\pm 2 \\ N(u)=9 & \Rightarrow u=\pm 3 \\ N(u)=16 & \Rightarrow u=\pm 4 \end{aligned}$$

$$i) \quad x=2 \quad u|2 \text{ or } u|3 \text{ in } R$$

$$\text{suppose } 2=u\beta \quad N(2)=4=N(u)N(\beta) \Rightarrow u=\pm 2$$

$$\text{suppose } 3=u\beta \Rightarrow u=\pm 3$$

$$\begin{aligned} & \begin{cases} x=3 & u|3 & \text{or } u|2 & \text{or } u|4 \\ \Rightarrow u=\pm 3 & \Rightarrow u=\pm 2 & 4=u\beta \\ & & 16=N(u)N(\beta) \\ & & \Rightarrow u=\pm 4 \end{cases} \end{aligned}$$

$$i) \quad x=4 \quad u|4 \text{ or } u|3$$

$$\begin{aligned} & \begin{cases} u \nmid 4 & u \nmid 3 \\ \text{or } u|5 \Rightarrow 5=u\beta & 25=N(u)N(\beta) \Rightarrow u=\pm 5 \\ & 25=5 \cdot 1 \end{cases} \end{aligned}$$

$$\Rightarrow u \in \{\pm 2, \pm 3, \pm 4\}$$

$$u \nmid w, u \nmid w-1, u \nmid w+1$$

$\Rightarrow \mathbb{Z}[\omega]$ is not euclidean.

$$D=-163, \quad \omega = \frac{1+\sqrt{-163}}{2}, \quad R = \mathbb{Z}[\omega]$$

$$N(a+b\omega) = a^2+ab+\frac{1}{4}b^2 = (a+\frac{b}{2})^2 + (\frac{\sqrt{-163}b}{4})^2 \geq 1$$

$$\min N = \{1, 4, 9, 16, 25, 36\}$$

$$u \in \{1, 2, 3, 4, 5, 6\}$$

$$x=2 \quad u|2 \text{ or } u|3$$

$$x=3 \quad u|2 \text{ or } u|3 \text{ or } u|4 \text{ or } u|5$$

$$x=4 \quad u|3 \text{ or } u|4 \text{ or } u|5 \text{ or } u|6$$

$$x=5 \quad u|4 \text{ or } u|5 \text{ or } u|6$$

$$x=6 \quad u|5 \text{ or } u|6 \text{ or } u|7$$

$$u \in \{1, 2, 3, 4, 5, 6\}$$

$$u \nmid w, u \nmid w-1, u \nmid w+1$$

$$41 = N(u)N(\beta) \Rightarrow u \nmid 41$$

$$41 = N(u)N(\beta) \Rightarrow u \nmid 41$$

$$41 = N(u)N(\beta) \Rightarrow u \nmid 41$$

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$$41 = N(u)N(\beta) \Rightarrow u \nmid 41$$

$$41 = N(u)N(\beta) \Rightarrow u \nmid 41$$

Then, one of a, b is even and other is odd

Suppose both a, b are even then $\frac{a}{2}, \frac{b}{2} \in \mathbb{Z}$
Suppose both odd $\frac{a+b}{2} = \frac{a}{2} + \frac{b}{2} \in \mathbb{R}$

$$\frac{1}{2} - \frac{((a-1) + b\sqrt{5})}{2} = \frac{a+b\sqrt{5}}{2} - \frac{(a-1)}{2} + \frac{b\sqrt{5}}{2}$$

$$= \frac{1}{2}$$

$$N(\frac{1}{2}) = \frac{1}{4} < 1$$

Suppose $c=3$. a^2+15b^2 is not divisible by 3 since $a^2 \equiv 1 \pmod{3}$ or $0 \pmod{3}$.
 If $a \equiv 0 \pmod{3}$, $b \not\equiv 0 \pmod{3}$, then $a^2+15b^2 \equiv 0 \pmod{3}$ but $a^2 \equiv 0 \pmod{3}$ and $15b^2 \equiv 0 \pmod{3}$.
 If $a \not\equiv 0 \pmod{3}$, $b \equiv 0 \pmod{3}$, then $a^2+15b^2 \equiv a^2 \pmod{3}$ and $a^2 \not\equiv 0 \pmod{3}$.

$$a^2+15b^2 = 3q+r \text{ with } r=1 \text{ or } 2$$

$$\text{Then } \frac{1}{2} = \frac{a+b\sqrt{5}}{2} \in \mathbb{Q} \text{ then}$$

$$N\left(\frac{a+b\sqrt{5}}{2}\right) = \frac{a^2+5b^2}{4} = \frac{3q+r}{4} = \frac{1}{4}$$

$$= N\left(\frac{3q+r}{4}\right) = \frac{3q+r}{16} = \frac{1}{16} < \frac{1}{4}$$

Suppose $c=4$. Then both a, b cannot be odd.

Suppose one of a, b is even. Then $a^2+15b^2 \equiv 1 \pmod{4}$.
 $a^2+15b^2 \equiv 1 \pmod{4} \Rightarrow a^2+15b^2 = 4q+r$ with $r=1$ or 3 .
 Then $\frac{1}{2} = \frac{a+b\sqrt{5}}{2} \in \mathbb{Q}$ with $r=1$ or 3 .

$$N\left(\frac{1}{2}\right) = \frac{1}{4} = \frac{1}{16} < \frac{1}{4}$$

Suppose both odd.

$$a^2+15b^2 \equiv 1 \pmod{8}$$

$$\Rightarrow a^2+15b^2 = 8q+4 \text{ for some } q \in \mathbb{Z}$$

$$\text{Then } \frac{1}{2} = \frac{a+b\sqrt{5}}{2} \in \mathbb{Q}$$

$$N\left(\frac{1}{2}\right) = \frac{1}{4} = \frac{8q+4}{16} = \frac{2q+1}{4} < \frac{1}{4}$$

$\Rightarrow R$ has a D-M norm $\Rightarrow R$ is P.I.D.

1. R is P.S.O.

$$(a,b) = (a) + (b) = (c) \Rightarrow \gcd(a,b) = c$$

2. R is D. $a, b \in R$. $(a) \cap (b) = (c)$

with $a \mid c$ and $b \mid c$.

$$\frac{1}{2} a \mid c \text{ and } \frac{1}{2} b \mid c \Rightarrow \frac{1}{2} (a) \subseteq (c) \cap \frac{1}{2} (b) \subseteq (c) \Rightarrow \frac{1}{2} c \mid c$$

$$\Rightarrow c = \frac{1}{2} c \Rightarrow c = 0$$

3. R is a P.S.O. $R/\langle p \rangle$ is a field. $\langle p \rangle$ is prime since every prime is maximal.
 so only ideals in $R/\langle p \rangle$ are $\{0\}$ and $\langle p \rangle = R/\langle p \rangle \Rightarrow R/\langle p \rangle$ is P.S.O.

4. R is an integral domain. Suppose

$$(1) \exists a, b \in R, a, b \neq 0 \text{ s.t. } \gcd(a,b) = 1 \text{ but } a \nmid b \text{ and } b \nmid a$$

$$(2) \exists a_1, a_2, \dots \text{ are nonzero s.t. } a_i \mid a_{i+1} \forall i \text{ and } a_i \nmid a_{i+2} \forall i \in \mathbb{N}$$

Suppose $I \neq 0$ is an ideal of R .

Suppose $a_i \in I, a_i \neq 0 \Rightarrow \frac{1}{2} (a_i) \subseteq I$ then we are done.

Suppose $(a_i) \subseteq I$. Let $a_i \in I \setminus (a_{i+1})$. $\frac{1}{2} (a_i) \subseteq I \Rightarrow \gcd(a_i, a_{i+1}) = (a_{i+1}) \Rightarrow \frac{1}{2} (a_i) \subseteq (a_{i+1})$

$\frac{1}{2} (a_i) \subseteq (a_{i+1})$ then we are done. Suppose $(a_i) \subseteq I$. Then we have a sequence

$$(a_i) \subseteq (a_{i+1}) \subseteq \dots \Rightarrow \exists a_i \mid a_{i+1} \forall i \Rightarrow \exists N \text{ s.t. } (a_N) = (a_{N+1}) = \dots$$

by (2)

$$\text{claim } I = (a_N)$$

$$\text{we are done and } \frac{1}{2} \in (a_N) \text{ then } (a_N) = (d)$$

$$\text{with } (a_N) \subseteq (d)$$

$$\frac{1}{2} (a_N) \subseteq (d) \text{ then we have contradiction}$$

$$\text{with } \frac{1}{2} (d) \subseteq (a_N) \Rightarrow \text{chain doesn't terminate}$$

(5) let $R = \mathbb{Z}[\sqrt{5}]$

$$I_1 = (1, \sqrt{5}) \quad I_2 = (2, 1+\sqrt{5}) \quad I_3 = (3, 1+\sqrt{5})$$

(a) Suppose $I_1 = (a+b\sqrt{5})$

$$1 = a+b\sqrt{5} \text{ for some } a, b \in \mathbb{Z}$$

$$N(1) = 1 = N(a+b\sqrt{5})$$

$$a^2+5b^2 = 1$$

$$\frac{1}{2} a^2+5b^2 = 1 \Rightarrow (a+b\sqrt{5}) = (\pm 1) = R$$

$$2d + (1+\sqrt{5})^2 = 1$$

$$2d + 6 + 2\sqrt{5} = 1 \Rightarrow 2\sqrt{5} \text{ is a multiple of } 2 \text{ impossible}$$

$$\frac{1}{2} a^2+5b^2 = 1 \Rightarrow N(1) = 2 \text{ since } a^2+5b^2 \geq 5 \text{ for } b \neq 0$$

$$\text{so } a^2 = 1 \text{ and } b = 0$$

$$\frac{1}{2} a^2+5b^2 = 1 \Rightarrow N(1) = 1 \Rightarrow a = \pm 1$$

$$\Rightarrow \pm 2 = a + b\sqrt{5} \Rightarrow 2 \nmid (1+\sqrt{5})^2$$

$$\text{then } I_2 = (2, 1+\sqrt{5}) = (1+\sqrt{5})$$

$$c = d(1+\sqrt{5}) \text{ for some } d \in R$$

$$N(1) = 1 = N(d(1+\sqrt{5})) \text{ since } a^2+5b^2 > 0$$

$$\Rightarrow a^2+5b^2 \in \{1, 3, 5\}$$

$$\frac{1}{2} a^2+5b^2 = 1 \Rightarrow (a+b\sqrt{5}) = (\pm 1)$$

$$\Rightarrow 3d + (2+\sqrt{5})^2 = 1$$

$$3d + 9 + 4\sqrt{5} = 1 \Rightarrow 4\sqrt{5} \text{ is a multiple of } 3 \text{ which is impossible}$$

$$\frac{1}{2} a^2+5b^2 = 3 \Rightarrow N(1) = 3 \Rightarrow a^2+5b^2 = 3$$

$$\Rightarrow a^2 = 3 \text{ impossible}$$

$$\frac{1}{2} a^2+5b^2 = 5$$

$$\Rightarrow d(1) = 5 \Rightarrow d = \pm 5$$

$$\text{or } b\sqrt{5} = \pm 5$$

$$\Rightarrow 2 \nmid (1+\sqrt{5}) \text{ impossible}$$

$$\text{then } (1, 2+\sqrt{5}) = (2+\sqrt{5})$$

$$c = N(d) a + b\sqrt{5}$$

$$\frac{1}{2} a^2+5b^2 = 1 \Rightarrow (a+b\sqrt{5}) = (\pm 1)$$

$$\Rightarrow 3d + 1 + \sqrt{5} = \pm 1 \Rightarrow d = 0$$

$$3d + (1+\sqrt{5}) + (2+\sqrt{5}) = (1+\sqrt{5})$$

$$\Rightarrow 3d + 3 + 2\sqrt{5} = 1 + \sqrt{5} \text{ impossible (cannot divide the coefficients)}$$

$$\frac{1}{2} a^2+5b^2 = 5 \Rightarrow N(1) = 5 \Rightarrow a^2+5b^2 = 5$$

$$\Rightarrow a^2 = 5$$

$$\frac{1}{2} a^2+5b^2 = 5 \Rightarrow N(1) = 5 \Rightarrow a = \pm 5$$

$$\Rightarrow (a+b\sqrt{5}) = (\pm 5)$$

$$\Rightarrow 2 \nmid (1+\sqrt{5}) \text{ impossible}$$

$$\Rightarrow I_1, I_2, I_3 \text{ are nonprincipal ideals}$$

$$\text{3) } I_1^2 = (1, \sqrt{5})^2 = (1, 2\sqrt{5}) = \{x(1+\sqrt{5}) + y(1-\sqrt{5}) \mid x, y \in R\} = (1)$$

$$\text{4) } I_2 I_3 = (2, 1+\sqrt{5})(3, 1+\sqrt{5})$$

$$a_1 = 6 \quad a_2 = 4 + 2\sqrt{5} \quad a_3 = 3 + \sqrt{5} \quad a_4 = (1 + \sqrt{5})(2 + \sqrt{5})$$

$$= 2 + 3\sqrt{5} - 5 = -3 + 3\sqrt{5} = -3(1 - \sqrt{5})$$

$$a_1 = (1 + \sqrt{5})(1 - \sqrt{5})$$

$$a_2 = (1 + \sqrt{5})(1 - \sqrt{5})$$

$$a_3 = (-2 + \sqrt{5})(1 - \sqrt{5})$$

$$a_4 = -3(1 - \sqrt{5}) \Rightarrow I_2 I_3 = (1 - \sqrt{5})$$

$$I_2 I_3' = (2, 1 + \sqrt{5})(3, 2 - \sqrt{5})$$

$$a_1 = (1 + \sqrt{5})(1 - \sqrt{5})$$

$$a_2 = 4 - 3\sqrt{5} = (-1 - \sqrt{5})(1 + \sqrt{5})$$

$$a_3 = 3 + 3\sqrt{5} = 3(1 + \sqrt{5})$$

$$a_4 = (1 + \sqrt{5})(2 - \sqrt{5}) \Rightarrow I_2 I_3' = (1 + \sqrt{5})$$

$$I_2 I_3 I_3' = (1 - \sqrt{5})(1 + \sqrt{5})$$

$$= (6)$$

(4) R integral domain. Suppose every prime ideal in R is principal.

(a) $\mathcal{S} = \{I \subseteq R \mid I \text{ is not principal and } I \neq \emptyset\} \neq \emptyset$

(i) $A \in \mathcal{S}$ (obvious)

(ii) $\nexists A \subseteq B$ and $B \in \mathcal{S} \Rightarrow B = A$ (obvious sets under inclusion)

(iii) $\nexists A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$ (obvious)

$\mathcal{S}$ is partially ordered.

Suppose $\mathcal{C}$ is a chain in $\mathcal{S}$: $\forall A, B \in \mathcal{C} \quad A \subseteq B$ or $B \subseteq A$

$\bigcup_{A \in \mathcal{C}} A$ is nonprincipal and contains all elements of $\mathcal{C}$ under inclusion

$\bigcup \mathcal{C}$ is the upper bound of $\mathcal{C}$.

By Zorn's lemma $\exists I \in \mathcal{S}$ maximal nonprincipal ideal in $\mathcal{S}$.

(b) Let $a, b \in R$ $a \notin I$ $a \notin I$ and $b \notin I$. Let $I_a = (I, a)$ $I_b = (I, b)$

$$\text{def } J = \{r \in R \mid r \in I_a \cap I_b\}$$

$$\text{claim: } \{r \in R \mid r \in I\} = J$$

$$(1) \text{ Suppose } a, b \in \{r \in R \mid r \in I\}$$

$$a, I_a = \{i + a \in R \mid i \in I, i \in I\} \subseteq I$$

$$(2) \text{ Suppose } b \in J \quad b, I_b \subseteq I$$

$$b, I_b = (I, b) \subseteq I \Rightarrow b \in \{r \in R \mid r \in I\}$$

$a \in I$ J is nonempty

$$\forall r, s \in J \Rightarrow \begin{cases} r, s \in I_a \cap I_b \\ \Rightarrow (r, s) \subseteq I_a \cap I_b \\ \Rightarrow r, s \in I \\ \Rightarrow (r - s) \subseteq I_a \cap I_b \\ \Rightarrow r - s \in I \end{cases}$$

$\Rightarrow J$ is an ideal.

$$I \not\subseteq I_a \Rightarrow I_a \text{ is principal (by maximality } I \text{ is nonprincipal)}$$

$$I_a = (d)$$

$$I \subseteq J \text{ since } I \subseteq I_a \cap I_b$$

$$I \subseteq J \Rightarrow I \subseteq I$$

$$I \not\subseteq I_b \text{ then } I_b \text{ is principal } \Rightarrow I_b = (e) \text{ principal}$$

$$(d, e) = I_a I_b = (I + (a)) (I + (b)) = I + J(a) \subseteq I$$

$$\begin{cases} I \subseteq I \\ (d, e) \in I \text{ for all } d, e \in I \end{cases}$$

$$(c) \text{ not } \Rightarrow \text{not } I \subseteq I$$

$$I \subseteq I \Rightarrow I \subseteq I$$

$\Rightarrow I = I_a I_b$ is a contradiction to the assumption that I is nonprincipal

$\Rightarrow R$ is P.I.D.

Remark: The reason why every prime ideal is principal is because we could not find a nonprincipal prime ideal and everything would break down.

(3) R integral domain. $\nexists (a, b) = (d)$ $\forall a, b \in R$ for some $d \in R$ then R is called a Bezout domain.

(a) (i) R is a Bezout domain if $\forall a, b \in R \quad (a, b) = (d)$ $d \in R$.

$$(1) d \mid a \text{ and } d \mid b$$

$$(2) \nexists d' \mid a \text{ and } d' \mid b \Rightarrow (a, b) = (d) \subseteq (d') \Rightarrow d' \mid d$$

$$\Rightarrow d \text{ is } \gcd(a, b). \quad (d) = (a, b) = (a) + (b) \quad \text{[commutative ring with } 1 \neq 0]$$

(ii) Suppose $d \mid \gcd(a, b)$ for $a, b \in R$.

and $d \nmid a + by$ for some $a, b \in R$.

$$(1) d \mid a \text{ and } d \mid b \Rightarrow d \mid (a) \quad d \mid (b)$$

$$\Rightarrow (a, b) \subseteq (d)$$

$$(2) d \nmid a + by \in (a, b) \Rightarrow (d) \subseteq (a, b) \Rightarrow (d) = (a, b)$$

(b) Suppose I is a finitely generated ideal.

Let $I = (a_1, \dots, a_n)$ be the generating set.

$$(a_1) = (a_1, a_2) \text{ since } R \text{ Bezout domain } \Rightarrow d_1 \mid \gcd(a_1, a_2) \text{ and } d_1 \mid a_1 + a_2 \in I$$

$$\nexists (a_1) = I \text{ then we are done}$$

Suppose not then we have a chain of ideals with inclusion:

$$(a_1) \subseteq (a_1, a_2) \subseteq (a_1, a_2, a_3) \subseteq \dots$$

$$\text{Since } \{a_1, \dots, a_n\} \text{ is finite the chain terminates } \Rightarrow \exists (d) \supseteq (a_1, \dots, a_n) \subseteq (d)$$

$$\Rightarrow I = (d) \text{ (since } d \in I)$$

(c) Let $F = D^{-1}R$.

$$\text{Let } \frac{a}{b} \in F \Rightarrow (a, b) = (\gcd(a, b)) = (d) \text{ and } a + by = d \quad a, b \in R$$

$$\Rightarrow \left(\frac{a}{b}\right) = \left(\frac{d}{b}\right) = 1 \Rightarrow \gcd\left(\frac{a}{b}, \frac{1}{b}\right) = 1$$

$$\frac{a}{b} = \frac{a}{d} \cdot \frac{d}{b} = \left(\frac{a}{d}\right)$$

$$\left(\frac{a}{d}\right)$$

(4) R is a P.I.D. D is a multiplicatively closed subset of R .

$D^{-1}R$ is an integral domain.

Let $I = D^{-1}J$ where J is an ideal in $\overline{R} \subseteq R$. Since R is P.I.D. $\overline{R} \subseteq R$ is P.I.D.

$\Rightarrow \overline{R}$ is principal $\Rightarrow R$ is principal.

$$\left[\begin{array}{l} \text{if } A \text{ contains } a^{-1} \text{ and } a \in D \Rightarrow A = (a) \\ A \neq (1) \text{ or } (a) \Rightarrow A \text{ contains elements of the form } \frac{a}{d} \text{ where } a \notin D. \\ \Rightarrow \left\{ \frac{a}{d} \in R \mid \frac{a}{d} \in A \right\} \text{ is an ideal of } R. \end{array} \right]$$

Unique Factorization Domain

(1) Let R be an integral domain.

(a) Suppose $r \in R$ is nonzero and square free. Then r is irreducible in R if whenever

$r = ab$ with $a, b \in R$, at least one of a, b is a unit in R , otherwise r is reducible.

(b) The nonzero element $p \in R$ is called prime if (p) is a prime ideal.

(c) $a, b \in R$ are said to be associates in R if $a \mid b$ and $b \mid a$.

Prop 1 In an integral domain a prime element is always irreducible.

Proof: Suppose $p \in R$ is prime, (p) is a prime ideal.

Suppose $r = ab$ with $a, b \in R$ and $a \notin (p)$ and $b \notin (p)$. Suppose $a \notin (p)$ $\Rightarrow a \notin pR$ for some

$p \in R$. Then, $p \nmid a$ and $p \mid ab \Rightarrow b \in (p) \Rightarrow b \in pR$ is irreducible.

[Remark: A consequence: Let p be prime in R then $p \mid a$ and $p \nmid b$ then $a \nmid b$ and $b \nmid a$]

[The ideal (p) is maximal if R is a P.I.D. and the only ideals containing (p) are (p) and R]

Def 11 p is irreducible in $\mathbb{Z}[R]$. p is not prime since $(p+1)(p-1) = p^2 - 1$ is divisible by p .
 but neither $(p+1)$ or $(p-1)$ are divisible by p .

Def 12 In a P.I.D a nonzero element is a prime or it is irreducible.

Th 13 (i) $\Rightarrow$ (ii) If R is a P.I.D then R is an integral domain.

(ii) $\Rightarrow$ (i) If R is any ideal containing (p) then $R = (p)$ or $R = R$ since R is a P.I.D.
 if $R = (p)$ then either p is a unit or p is a prime $\Rightarrow (p) = (p)$ or $(p) = (1)$.
 if $R = R$ then p is a unit or p is a prime $\Rightarrow (p) = (p)$ or $(p) = (1)$ is maximal $\Rightarrow (p)$ is prime.

Example

$\bullet \mathbb{Z}[R]$ is not P.I.D. This is irreducible but not prime.

Def 14 U.F.D. is an integral domain in which every non zero element can be factored into primes.

(i) $p_1, p_2, \dots, p_n$ are irreducible in R (not necessarily distinct)

(ii) the decomposition is unique up to associates.

$\{q_1, \dots, q_m\}$ is another factorization into irreducibles then $m=n$ and $q_i = u_i p_i$ for some unit u_i .

Example

(i) $\mathbb{Z}$ is a U.F.D. since every element is a unit.

(ii) $\mathbb{Z}[R]$ is not U.F.D.

(iii) $R[x]$ is U.F.D. whenever R is U.F.D.

$\mathbb{Z}[x]$ is U.F.D. since $\mathbb{Z}$ is U.F.D.

(iv) $R = \mathbb{Z}[x] = \{a + bx \mid a, b \in \mathbb{Z}\}$ where (a, b) is an integral domain but not U.F.D. $2, 3, 4$ are irreducibles which are not associates. since $12 \in R$ and $12 = 2 \cdot 3 = 4 \cdot 3$ has two distinct factorizations in R . $\mathbb{Z}[x]$ is not a U.F.D.

(v) $\mathbb{Z}[x]$ is a domain but not U.F.D. since $6 = 2 \cdot 3 = (1+2x)(1-2x)$ gives two distinct factorizations of 6 into irreducibles.

$(6) = (2)(3)$

$= (1+2x)(1-2x)$

$= (1+2x)(1-2x)$

$= (1+2x)(1-2x)$

Th 14 In a U.F.D. a nonzero element p is prime or it is irreducible.

(i) $\Rightarrow$ (ii) If R is a U.F.D. then R is an integral domain.

(ii) $\Rightarrow$ (i) Let p be irreducible in R . Suppose $ab \in (p)$ where $a, b \notin (p)$ for some $a, b \in R$.

Then a and b have unique factorizations into irreducibles up to associates.

Then p must be associated to one of the irreducibles in the factorization of a or b . Suppose p is associated to one of the irreducibles in the factorization of a . Then $a = p \cdot u$ for some unit u . Then $ab = p \cdot u \cdot b$ and $p \mid ab$ implies $p \mid b$.

Th 15 Let a, b be two nonzero elements of U.F.D. R . Suppose

$a = u_1 p_1^{e_1} \dots p_n^{e_n}$ $b = v_1 q_1^{f_1} \dots q_m^{f_m}$

are prime factorizations for a and b . u, v are units, $p_1, \dots, p_n$ are primes and distinct, $q_1, \dots, q_m$ are primes and distinct. Then $\gcd(a, b) = u_1^{e_1} v_1^{f_1} \dots$ (if all exponents are 1).

Th 16 $d \mid a$ since $a \mid p_1^{e_1} \dots p_n^{e_n}$

$a \mid b$ since $b \mid q_1^{f_1} \dots q_m^{f_m}$

Suppose $a \mid b$ and $d \mid b$ is $d \mid p_1^{e_1} \dots p_n^{e_n}$ is the prime factorization of d .

Since each $p_i \mid d$ so $p_i \mid a$ and $p_i \mid b$ so p_i is associated to a prime q_i .

$\Rightarrow \{p_1, \dots, p_n\} \subseteq \{q_1, \dots, q_m\}$ up to associates.

Let $q_i \in \{p_1, \dots, p_n\}$ so $d \mid a$ so $d \mid \gcd(a, b)$.

Th 17 Every P.I.D. is U.F.D.

Th 18 Let R be a P.I.D. Let $p \in R$ and p is not a unit. If p is irreducible then we are done.

Suppose p is not irreducible. Then $p = p_1 p_2$ where neither p_1 nor p_2 are units. If both elements are irreducible then we are done. If not, then one of p_1, p_2 is reducible. If p_1 is reducible then $p_1 = p_1' p_2'$ and so $p = p_1' p_2' p_2$.

Repeat the factorization process. Since R is a P.I.D. the process terminates. Then $p = p_1' p_2' p_3$ where p_1', p_2', p_3 are units.

By induction the process terminates in finite steps. Then p is a product of primes. p is a product of primes.

$(p) \subseteq (p_1) \subseteq (p_1 p_2) \subseteq \dots \subseteq (p)$

By induction (ii) from (i) (P.I.D.) $p_1, p_2, \dots, p_n$ are a sequence of elements of P.I.D.

so $p_1, p_2, \dots, p_n$ are

Claim: the decomposition is unique.

Let p be a product of irreducible factors of R .

$\{p_1, \dots, p_n\}$ and $\{q_1, \dots, q_m\}$ are two sets of irreducibles. If p is a unit so q is a unit.

which is contradiction.

Suppose $n < m$ and we have two factorizations for

$p = p_1 p_2 \dots p_n = q_1 q_2 \dots q_m$ where (p_i, q_j) are not necessarily distinct.

(p_i, q_j) are irreducibles.

Then $p_1 \mid q_1 \dots q_m$ so $p_1 \mid q_i$ for some $i \in \{1, \dots, m\}$ since p_1 is irreducible.

Then $p_1 = u q_i$ for some unit u . Then p_1 is associated to q_i .

Canceling p_1 we obtain

$p_2 \dots p_n = u q_1 \dots q_m$ where q_i is a unit.

where q_i is a unit. Then each factor of $p_2, \dots, p_n$ divides $q_1, \dots, q_m$ similarly (no associates).

with factors of $p_2, \dots, p_n$ and with factors of $q_1, \dots, q_m$. Then $p_2, \dots, p_n$ are already associated to $q_1, \dots, q_m$ and we are done.

analogous, thus the induction is complete and we are done.

Corollary 18 (Fundamental Theorem of Arithmetic) $\mathbb{Z}$ is a U.F.D.

Th 19 $\mathbb{Z}$ is a U.F.D. if and only if $\mathbb{Z}$ is a P.I.D.

Th 20 Let R be a P.I.D. $\exists$ a multiplicative subring S of R which is not a unit.

Th 21 R is U.F.D. $\Leftrightarrow N(R) = \mathbb{Z} \cup \{0\}$ with $N(p) = 0$ and $N(q) = 1$ if q is a unit.

$\Rightarrow N(p) = \mathbb{Z} \setminus \{0\}$ if p is not a unit. $p_1, \dots, p_n$ are irreducibles in R .

$N(p) = N(p_1) N(p_2) \dots N(p_n)$ is multiplicative.

Suppose $a, b \in R$ $a, b \neq 0$. Then $(a, b) = (p)$ [for P.I.D.]

$\Rightarrow a \in (p)$ and $b \in (p)$. Since $a \in (p)$ for some $a \in R$ so a is not a unit in R . $[N(a) \geq 2]$

$\Rightarrow N(a) = N(b) N(c) \geq N(c)$ $\Rightarrow N(c) \geq 2$

$\Rightarrow N(c) = N(d) N(e) \geq N(e)$ $\Rightarrow N(e) \geq 2$

$\Rightarrow N(e) = N(f) N(g) \geq N(g)$ $\Rightarrow N(g) \geq 2$

$\Rightarrow N(g) = N(h) N(i) \geq N(i)$ $\Rightarrow N(i) \geq 2$

$\Rightarrow N(i) = N(j) N(k) \geq N(k)$ $\Rightarrow N(k) \geq 2$

$\Rightarrow N(k) = N(l) N(m) \geq N(m)$ $\Rightarrow N(m) \geq 2$

$\Rightarrow N(m) = N(n) N(o) \geq N(o)$ $\Rightarrow N(o) \geq 2$

$\Rightarrow N(o) = N(p) N(q) \geq N(p)$ $\Rightarrow N(p) \geq 2$

$\Rightarrow N(p) = N(r) N(s) \geq N(r)$ $\Rightarrow N(r) \geq 2$

$\Rightarrow N(r) = N(t) N(u) \geq N(t)$ $\Rightarrow N(t) \geq 2$

$\Rightarrow N(t) = N(v) N(w) \geq N(v)$ $\Rightarrow N(v) \geq 2$

$\Rightarrow N(v) = N(x) N(y) \geq N(x)$ $\Rightarrow N(x) \geq 2$

$\Rightarrow N(x) = N(z) N(a) \geq N(z)$ $\Rightarrow N(z) \geq 2$

$\Rightarrow N(z) = N(b) N(c) \geq N(b)$ $\Rightarrow N(b) \geq 2$

$\Rightarrow N(b) = N(d) N(e) \geq N(d)$ $\Rightarrow N(d) \geq 2$

$\Rightarrow N(d) = N(f) N(g) \geq N(f)$ $\Rightarrow N(f) \geq 2$

$\Rightarrow N(f) = N(h) N(i) \geq N(h)$ $\Rightarrow N(h) \geq 2$

$\Rightarrow N(h) = N(j) N(k) \geq N(j)$ $\Rightarrow N(j) \geq 2$

$\Rightarrow N(j) = N(l) N(m) \geq N(l)$ $\Rightarrow N(l) \geq 2$

$\Rightarrow N(l) = N(n) N(o) \geq N(n)$ $\Rightarrow N(n) \geq 2$

$\Rightarrow N(n) = N(p) N(q) \geq N(p)$ $\Rightarrow N(p) \geq 2$

$\Rightarrow N(p) = N(r) N(s) \geq N(r)$ $\Rightarrow N(r) \geq 2$

$\Rightarrow N(r) = N(t) N(u) \geq N(t)$ $\Rightarrow N(t) \geq 2$

$\Rightarrow N(t) = N(v) N(w) \geq N(v)$ $\Rightarrow N(v) \geq 2$

$\Rightarrow N(v) = N(x) N(y) \geq N(x)$ $\Rightarrow N(x) \geq 2$

$\Rightarrow N(x) = N(z) N(a) \geq N(z)$ $\Rightarrow N(z) \geq 2$

$\Rightarrow N(z) = N(b) N(c) \geq N(b)$ $\Rightarrow N(b) \geq 2$

$\Rightarrow N(b) = N(d) N(e) \geq N(d)$ $\Rightarrow N(d) \geq 2$

$\Rightarrow N(d) = N(f) N(g) \geq N(f)$ $\Rightarrow N(f) \geq 2$

Consider $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is KED and UFD, $\mathbb{Z}[i]$ is Euclidean w.r.t. $\mathbb{Z}[i]$ is KED and UFD. $\mathbb{Z}[i]$ is Euclidean w.r.t. $\mathbb{Z}[i]$ is KED and UFD. $\mathbb{Z}[i]$ is Euclidean w.r.t. $\mathbb{Z}[i]$ is KED and UFD.

Let $z = a+bi$. $N(z) = z\bar{z} = a^2+b^2$. $\bar{z}$ factors in $\mathbb{Z}[i]$ into two irreducibles (or a^2+b^2 is reducible). $\bar{z}$ factors in $\mathbb{Z}[i]$ into two irreducibles (or a^2+b^2 is reducible).

only case where conjugate irreducibles are associated.

if $p \equiv 1 \pmod{4}$, $p = (a+bi)(a-bi)$. Suppose $p \equiv 1 \pmod{4}$. $-i(a-bi) = b-ai \neq a-bi$.

$a^2 \equiv 0 \pmod{4}$ or $1 \pmod{4}$. If a is odd prime in $\mathbb{Z}$ that is sum of two squares must be $0 \pmod{4}$.

Thus, p is prime in $\mathbb{Z}$ with $p \equiv 3 \pmod{4}$ then p is not sum of two squares and p is irreducible.

Lemma 12. prime $p \in \mathbb{Z}$, $p \nmid n+1$ and $p \nmid n^2-1$ then $p \nmid n^2-1$ and $p \nmid n^2-1$.

Proof. $p \nmid n+1$ is clear. If $p \nmid n^2-1$ then $p \nmid (n+1)$ or $p \nmid (n-1)$ so $p \nmid n-1$ is clear.

By Lemma 12, if $p \equiv 3 \pmod{4}$ is a prime then $p \nmid n^2-1$ in $\mathbb{Z}$ for some n .

if $p \nmid (n+1)(n-1)$ then p is irreducible in $\mathbb{Z}[i]$ since $p \nmid (n+1)$ or $p \nmid (n-1)$.

Prop 8. prime $p \equiv 1 \pmod{4}$ and $p \equiv 1 \pmod{4}$. representation of p as sum of two squares is unique up to swapping or changing signs of a and b .

(a) irreducible elements of $\mathbb{Z}[i]$:

(1) $(1+i)$ is prime.

(2) p prime in $\mathbb{Z}$ with $p \equiv 3 \pmod{4}$ is prime in $\mathbb{Z}[i]$.

(3) $a+bi, a-bi$ distinct irreducible factors of $p = a^2+b^2 = (a+bi)(a-bi)$ for $p \equiv 1 \pmod{4}$.

no a^2+b^2 can $n = N(a+bi)$ $a+bi \in \mathbb{Z}[i]$.

$a+bi = u_1 u_2 \dots u_k$ u_i irreducible (unique up to associates).

Then by Prop 8, n is a norm 203 if every prime divisor of n that is congruent to $1 \pmod{4}$ occurs to even exponent.

Corollary 10. If n is a positive integer.

$n = 2^k p_1^{a_1} \dots p_r^{a_r} q_1^{b_1} \dots q_s^{b_s}$ $p_i, q_i \equiv 1 \pmod{4}$ $q_i \equiv 3 \pmod{4}$.

Then n is a sum of two squares iff k is even and a_i, b_i are even.

Proof. $p \equiv 1 \pmod{4} \mid a^2+b^2$ for $p \nmid a$ and $p \nmid b$ since p is irreducible in $\mathbb{Z}[i]$ (Prop 8).

This is equivalent to $a+bi = p^k (a'+bi')$ for some a', b' and k is even.

Suppose n is a sum of two squares. Then factorization of $n = (a+bi)(a-bi)$ in $\mathbb{Z}[i]$.

with $a+bi = p_1^{a_1} \dots p_r^{a_r} q_1^{b_1} \dots q_s^{b_s}$ and $a-bi = p_1^{a'_1} \dots p_r^{a'_r} q_1^{b'_1} \dots q_s^{b'_s}$.

Then a_i, a'_i is determined by $(a_i+1)(a_i-1) \dots (a_i-1)$ distinct elements.

$a+bi$ in $\mathbb{Z}[i]$ of norm n , up to units.

Since a and b are just units in $\mathbb{Z}[i]$, there are $4(a+1)(a-1) \dots (a-1)$ representations.

Summary.

Fields $\mathbb{C} \supset \mathbb{R} \supset \mathbb{Q} \supset \mathbb{F}_p \supset \mathbb{F}_q \supset \mathbb{F}_r \supset \mathbb{F}_s \supset \mathbb{F}_t \supset \mathbb{F}_u \supset \mathbb{F}_v \supset \mathbb{F}_w \supset \mathbb{F}_x \supset \mathbb{F}_y \supset \mathbb{F}_z \supset \mathbb{F}_1 \supset \mathbb{F}_0$.

Examples.

(1) $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. any nonzero integer has a unique prime factorization.

Let $m \in \mathbb{Z} \setminus \{0\}$, $m = 2^k 3^l 5^m \dots$ $(2, 3) = (3, 2) = 1$, $2 \nmid 3$, $3 \nmid 2$.

Def. $\phi: \mathbb{Z} \rightarrow \mathbb{Z}$ $\phi(x) = x^2$ $\phi(2) = 4$, $\phi(3) = 9$.

Claim: ϕ is a multiplicative bijection.

Proof. ϕ is a multiplicative bijection is unique.

Suppose $m, n \in \mathbb{Z} \setminus \{0\}$, $m = 2^k 3^l 5^m \dots$, $n = 2^p 3^q 5^r \dots$, $m, n \in \mathbb{Z} \setminus \{0\}$.

$\phi(mn) = \phi(2^{k+p} 3^{l+q} 5^{m+r} \dots)$

$= \phi(2^k 3^l 5^m \dots) \phi(2^p 3^q 5^r \dots)$

$= \phi(m) \phi(n)$ ϕ is multiplicative.

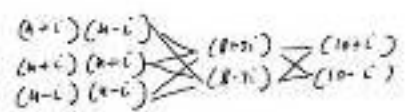
Def. $\phi: \mathbb{Z} \rightarrow \mathbb{Z}$ $\phi(x) = x^2$ $\phi(2) = 4$, $\phi(3) = 9$.

Claim: ϕ is a multiplicative bijection.

$\Rightarrow p_i = a_i$
 In particular, $\{a_1, \dots, a_m\} \subseteq \{a_1, \dots, a_m\}$
 $\Rightarrow d_i = p_1^{a_{i1}} \dots p_m^{a_{im}}$ where a is composed of $\{a_1, \dots, a_m\} = \{e_1, \dots, e_m\}$
 $\Rightarrow d_i = p_1^{a_{i1}} \dots p_m^{a_{im}} \geq p_1^{e_1} \dots p_m^{e_m} = d$
 $\Rightarrow d \mid d_i$
 $\Rightarrow d$ is the least common multiple.
 $d = p_1^{\max(a_{11}, \dots, a_{1m})} \dots p_m^{\max(a_{m1}, \dots, a_{mm})}$
 $= p_1^{\max(a_1)} \dots p_m^{\max(a_m)}$ using parameters

(5) $2130793 = 13^2 \cdot 25 \cdot 101$

$13 \mid 2130793 \Rightarrow 13 \mid 2130793$
 $2130793 \div 13 = 163915$
 $163915 \div 13 = 12608$
 $12608 \div 13 = 969$
 $969 \div 13 = 74$
 $74 \div 13 = 5$
 $5 \div 13 = 0$
 $13 \mid 2130793$
 $2130793 = 13^2 \cdot 25 \cdot 101$
 $12 = (3)(2)(2)$ representations



(6) Suppose $n = \left(\frac{a}{b}\right)^2 + \left(\frac{c}{d}\right)^2 \Rightarrow (a,b) = 1, (c,d) = 1$

$n = \frac{(a,b)^2 + (c,d)^2}{(b,d)^2}$
 $(a,b)^2 + (c,d)^2$ is a sum of integer squares therefore has a factorization of the form: $2^k p_1^{a_1} \dots p_m^{a_m} q_1^{b_1} \dots q_n^{b_n}$ where $p_i \equiv 1 \pmod{4}$, $q_i \equiv 3 \pmod{4}$ distinct primes.
 $(a,b)^2 + (c,d)^2$ is also a sum of integer squares that divides $(a,b)^2 + (c,d)^2$
 $\Rightarrow$ the set of prime factors of $(a,b)^2 + (c,d)^2$ are contained in the set $\{2, p_1, \dots, p_m, q_1, \dots, q_n\}$ with exponents $\leq a_1, b_1, \dots, a_m, b_m$ with the exponent of primes $\equiv 3 \pmod{4}$ are even.
 $\Rightarrow n = 2^k p_1^{a_1} \dots p_m^{a_m} q_1^{b_1} \dots q_n^{b_n}$ is a sum of integer squares since $a_1, \dots, a_m, b_1, \dots, b_n$ are even exponents.

(6) $R = \mathbb{Z}[\sqrt{n}]$, n is a squarefree integer > 2 . $\Rightarrow$ units are ± 1

(a) Suppose $2 = \alpha\beta$, $\alpha, \beta \in \mathbb{Z}[\sqrt{n}]$
 $4 = N(\alpha)N(\beta)$ $N(\alpha) = 1, N(\beta) = 4$ or $N(\alpha) = 4, N(\beta) = 1$
 or
 $N(\alpha) = 2$ and $N(\beta) = 2$.
 $\nexists N(\alpha) = 1$ as α is a unit $\Rightarrow 2$ is irreducible.
 $\nexists N(\alpha) = 2$ then $a^2 + nb^2 = 2$ for $a, b \in \mathbb{Z}$
 $\nexists b \neq 0$ as $a^2 + nb^2 \geq n \geq 3$ thus $b = 0$
 $\Rightarrow a^2 = 2$ which is impossible.

$\Rightarrow 2$ is irreducible in R .

Suppose $\sqrt{n} = \alpha\beta$, $\alpha, \beta \in R$.

$n = N(\alpha)N(\beta)$
 $0 < N(\alpha), N(\beta) \leq n$
 If $\alpha = a + b\sqrt{n}$, $\beta = c + d\sqrt{n}$
 $\nexists b \neq 0$ then $N(\alpha) = a^2 + nb^2 > n$ which is impossible.
 $\nexists d \neq 0$ then $N(\beta) = c^2 + nd > n$ if $\alpha \neq 0 \Rightarrow a = 0 \Rightarrow N(\alpha) = n$ and $N(\beta) = 1$
 $\nexists b = 0$ then $N(\alpha) = a^2$ and $N(\beta) = d^2$
 $\Rightarrow n = (a,d)^2$ which is impossible since n is squarefree.
 $\Rightarrow N(\alpha) = n$ and $N(\beta) = 1 \Rightarrow N(\alpha) = 1$ and $N(\beta) = n$.
 which implies either α or β is a unit and $\sqrt{n}$ is irreducible.

Suppose $1 + \sqrt{n} = \alpha\beta$, $\alpha, \beta \in \mathbb{Z}[\sqrt{n}]$

$N(1 + \sqrt{n}) = 1 + n = n$
 $1 < N(\alpha), N(\beta) \leq n$
 $\nexists \alpha \neq 1$ and $\beta \neq 0$ $N(\alpha) = a^2 + nb^2 > 1 + n$ (impossible).
 $\nexists \alpha = 1$ and $\beta \neq 0$ $N(\beta) = 1 + n$ as $N(\beta) = 1$ as β is a unit.
 $\nexists \alpha \neq 1$ and $\beta = 0$ $N(\alpha) = a^2 \Rightarrow a = \pm 1$ as $1 + \sqrt{n}$ is irreducible.
 $N(\beta) = d^2 \Rightarrow \beta = \pm d$
 $\Rightarrow 1 + \sqrt{n} = \pm d$ which is impossible.
 $\Rightarrow 1 + \sqrt{n}$ is irreducible!

(b) $\mathbb{Z}[\sqrt{n}] \cong \mathbb{Z}/n\mathbb{Z}$ is an integral domain as n is prime.

$\mathbb{Z}[\sqrt{n}] \cong \mathbb{Z}/(n)\mathbb{Z}$ as n is prime.

Suppose $\sqrt{n}$ is prime, as n is prime (and prime) since $n > 2$
 $\Rightarrow n+1$ is even and $(1+\sqrt{n})$ is not prime.
 Suppose $n+1$ is prime, then n is even so $\sqrt{n}$ is not prime for $n > 2$
 $\Rightarrow$ either $\sqrt{n}$ or $1+\sqrt{n}$ is prime. $\Rightarrow R$ is not U.F.D.
 $\mathbb{Q} = \mathbb{Q}[\sqrt{n}]$ $\mathbb{Q} \subseteq \mathbb{Z}[\sqrt{n}] \subseteq \mathbb{Q}$ $N(a+b\sqrt{n}) = a^2 + nb^2$ has $\mathbb{Q}$ as not U.F.D.
 for odd n , we know that $\mathbb{Z}[\sqrt{n}] \subseteq \mathbb{Q}$ and $\mathbb{Q}[\sqrt{n}]$ is not U.F.D. and therefore U.F.D. for $n \leq 17$.

(c) Consider maximal ideal containing nonprime ideals $(\sqrt{n})$ or $(1+\sqrt{n})$
 then, let $M \supseteq (\sqrt{n})$ or $M \supseteq (1+\sqrt{n})$ $\nexists M \supseteq (\sqrt{n})$ then $(\sqrt{n})$ or $(1+\sqrt{n}) \in \mathcal{P}$
 impossible since $\sqrt{n}$ and $1+\sqrt{n}$ are irreducible.
 Example: $(2, 1+\sqrt{n})$ when $1+\sqrt{n}$ nonprime.
 $(2, \sqrt{n})$ when $\sqrt{n}$ nonprime.

(d) $\mathbb{Z}[i]/(1+i)$

$\mathbb{Z}[i]$ is Euclidean $\Rightarrow \mathbb{Z}[i]$ is a PID.
 $\Rightarrow \mathbb{Z}[i]/(1+i)$ is a field.
 $1+i \mid 2 \pmod{1+i} = 0 \Rightarrow \text{char}(\mathbb{Z}[i]/(1+i)) = 2$
 $\mathbb{Z}[i]/(1+i) \cong \mathbb{F}_2$
 $a+bi \mapsto (a+b) \pmod{2}$ (isomorphism)
 $\text{map: } \mathbb{Z}[i] \rightarrow \mathbb{F}_2$
 $\Rightarrow \mathbb{Z}[i]/(1+i) \cong \mathbb{F}_2$

$\Rightarrow \mathbb{Z}[i]$ prime quotient $\Rightarrow (a)$ is maximal in $\mathbb{Z}[i]$
 $\Rightarrow \mathbb{Z}[i]/(a)$ is a field.

(e) $\text{char}(\mathbb{Z}[i]/(a)) = 9$

$\mathbb{Z}[i]/(a) \cong \mathbb{Z}_9 \oplus \mathbb{Z}_3$ as $n=2$.
 $\Rightarrow |\mathbb{Z}[i]/(a)| = 9^2$

(f) For $n \equiv 1 \pmod{4}$, $p \equiv 1 \pmod{4}$, n, p are irreducible $\Rightarrow p, \sqrt{n}$ prime and $\mathbb{Q}[\sqrt{n}]$ is not U.F.D.
 $\Rightarrow (p, \sqrt{n})$ are prime. $\text{gcd}(p, \sqrt{n}) = 1 \Rightarrow (p) + (\sqrt{n}) = (\mathbb{Z}) = \mathbb{Q}[\sqrt{n}]$

$$\begin{aligned} \text{by c.k.1} \\ \mathbb{Z}[i]/(i^2) &\cong \mathbb{Z}[i]/(i) \times \mathbb{Z}[i]/(1-i) \\ \mathbb{Z}[i]/(i^2) &\cong \mathbb{Z}/(i) \oplus \mathbb{Z}/(1-i) \\ \Rightarrow |\mathbb{Z}[i]/(i^2)| = p^2 &= |\mathbb{Z}[i]/(i)| \cdot |\mathbb{Z}[i]/(1-i)| = p \cdot p \\ \mathbb{Z}[i]/(i^2), \mathbb{Z}[i]/(1-i) &\text{ are fields since } (i), (1-i) \text{ are maximal.} \\ \Rightarrow |\mathbb{Z}[i]/(i^2)| &= |\mathbb{Z}[i]/(1-i)| = p \end{aligned}$$

7. (a) $\chi(\pi^n) = \chi(\pi^n) \subseteq (\pi^n)$ for any $n \geq 0$.

$$\begin{aligned} \text{Let } \phi: \mathbb{Z}[i]/(i^2) &\longrightarrow (\pi^n)/(\pi^{n+1}) \\ \bar{a} &\longmapsto \pi^n \bar{a} \end{aligned}$$

$$\begin{aligned} \text{Suppose } a \equiv b \pmod{\pi^n} \\ (a-b) \in \pi^n \pmod{\pi^n} \quad \pi^n a - \pi^n b = \pi^n(a-b) \equiv 0 \\ \Rightarrow \pi^n a \equiv \pi^n b \Rightarrow \phi \text{ is well defined.} \end{aligned}$$

Now check:

$$\begin{aligned} \bar{a}, \bar{b} \in \mathbb{Z}[i]/(i^2) \\ \phi(\bar{a} + \bar{b}) = \pi^n(\bar{a} + \bar{b}) = \pi^n \bar{a} + \pi^n \bar{b} = \pi^n \bar{a} + \pi^n \bar{b} = \phi(\bar{a}) + \phi(\bar{b}) \quad (\text{addition}) \end{aligned}$$

$$\begin{aligned} \text{Suppose } a \equiv b \pmod{\pi^n} \\ \pi^n a \in (\pi^{n+1}) \Rightarrow \pi^n a \equiv 0 \pmod{\pi^{n+1}} \Rightarrow a \equiv 0 \pmod{\pi} \Rightarrow \bar{a} = \bar{0} \in \mathbb{Z}[i]/(i^2) \Rightarrow \phi \text{ is injective.} \end{aligned}$$

Suppose $\bar{a} \in (\pi^n)/(\pi^{n+1})$

$$\bar{a} = \pi^n \bar{a}, \quad \bar{a} \in (\pi^n) \Rightarrow \bar{a} = \pi^n \bar{a} \Rightarrow \bar{a} \in \mathbb{Z}[i]/(i^2) \Rightarrow \phi \text{ is surjective.}$$

$\Rightarrow \phi$ is an isomorphism.

$$\begin{aligned} (b) \quad |\mathbb{Z}[i]/(\pi^n)| &= |\mathbb{Z}[i]/(\pi)| \cdot |\mathbb{Z}[i]/(\pi^2)| \cdots |\mathbb{Z}[i]/(\pi^n)| \\ &= |\mathbb{Z}[i]/(i)|^n \end{aligned}$$

$$(c) \quad d \in \mathbb{Z}[i], \quad d = a_1^2 + \dots + a_n^2 \quad (a_i^2) + (a_j^2) = (1) \quad \text{since } \pi_i, \pi_j \text{ coprime}$$

$$\begin{aligned} \text{Then by c.k.1} \\ \mathbb{Z}[i]/(d) &\cong \mathbb{Z}[i]/(a_1^2) \times \dots \times \mathbb{Z}[i]/(a_n^2) \\ |\mathbb{Z}[i]/(d)| &= |\mathbb{Z}[i]/(a_1^2)| \cdots |\mathbb{Z}[i]/(a_n^2)| \\ &= |\mathbb{Z}[i]/(a_1)|^2 \cdots |\mathbb{Z}[i]/(a_n)|^2 \\ &= N(a_1)^2 \cdots N(a_n)^2 \\ &= N(d) \end{aligned}$$

$$8. \quad R = \mathbb{Z}[i], \quad I_1 = (2, 1+i), \quad I_2 = (2, 1-i), \quad I_3 = (2, 1-i)$$

$$(a) \quad \text{Let } a+bi \in \mathbb{Z}[i], \quad N(a+bi) = a^2 + b^2 \geq 5 \quad \text{if } b \neq 0.$$

Suppose $2 \mid a$

$$\begin{aligned} 4 &= N(a)N(b) \quad b \neq 0 \text{ in } \mathbb{Z}[i] \\ \Rightarrow N(a) &= 4, \quad \text{or } N(b) = 4 \\ N(b) &= 4 \text{ or } N(b) = 5 \\ \Rightarrow d \text{ is a unit or } \beta \text{ is a unit or } d \text{ is irreducible.} \end{aligned}$$

Suppose $3 \mid a$

$$\begin{aligned} 9 &= N(a)N(b) \\ 9 &= 3 \cdot 3 \\ 3 &= 3 \cdot 1 \\ \text{Suppose } N(a) &= 3 \Rightarrow a^2 + b^2 = 3 \text{ which is impossible.} \\ \Rightarrow 4 \text{ or } 9 &= 3 \cdot 3 \Rightarrow 3 \text{ is an irreducible.} \end{aligned}$$

Suppose $5 \mid a$

$$\begin{aligned} N(a+bi) &= 1 + b^2 = 5 \Rightarrow b^2 = 4 \Rightarrow b = \pm 2 \\ 5 &= 1 + 4 = 1 + 2^2 \text{ is irreducible.} \end{aligned}$$

mod. $1-i \in \mathbb{Z}[i] \Rightarrow 1-i$ is irreducible.
The units of $\mathbb{Z}[i]$ are ± 1 . Therefore none of these are associates.
 $\Rightarrow (1-i) \mid (1+i)(1-i)$ are two distinct factorizations.

$$(b) \quad I_1 = (2, 1+i)$$

$$\mathbb{Z}[i]/I_1 \cong (\mathbb{Z}[i]/(2))/(\bar{1} + \bar{i})$$

Consider $\mathbb{Z}[i]/(2)$:

$$\begin{aligned} a+bi \pmod{2} &= \{0, 1\} + \{0, i, \bar{i}\} \quad ((\bar{i})^2 = -1 \pmod{2}) \\ \mathbb{Z}[i]/(2) &= \{0, \bar{i}, i, \bar{i} + i\} \text{ has 4 elements.} \end{aligned}$$

Consider $\mathbb{Z}[i]/(2)$:

$$\begin{aligned} 2d + (1+i)^2 \bar{a} &\equiv (1+i)^2 \bar{a} \pmod{2} \quad \bar{a} \in \mathbb{Z}[i] \\ \bar{a} &\in \mathbb{Z}[i]/(2) \Rightarrow \bar{a} \in \{0, \bar{i}, i, \bar{i} + i\} \\ (1+i)^2 \bar{a} &\equiv 0 \\ (1+i)^2 \cdot \bar{i} &\equiv (1+i)^2 \cdot i \\ (1+i)^2 \cdot \bar{i} &\equiv \bar{i} + \bar{i} + \bar{i} + \bar{i} = 0 \\ (1+i)^2 \cdot i &\equiv i + i + i + i = 0 \\ (1+i)^2 \cdot (\bar{i} + i) &\equiv (\bar{i} + \bar{i})(1+i)^2 \equiv 0 \pmod{2} \\ \Rightarrow \mathbb{Z}[i]/(2) &\text{ has 2 elements.} \end{aligned}$$

$$\Rightarrow |\mathbb{Z}[i]/I_1| = \frac{|\mathbb{Z}[i]/(2)|}{|\mathbb{Z}[i]/(2)|} = \frac{4}{2} = 2.$$

$$\Rightarrow \mathbb{Z}[i]/I_1 \cong \mathbb{Z}/2\mathbb{Z} \text{ as additive abelian groups.}$$

Since $\mathbb{Z}/2\mathbb{Z}$ is again in $\mathbb{Z}[i]$ is again and $\mathbb{Z}[i]/I_1 \cong \mathbb{Z}/2\mathbb{Z}$ as rings
 $\Rightarrow I_1$ is maximal hence prime.

$$\text{Similarly, } I_2 = (2, 1-i)$$

$$\mathbb{Z}[i]/I_2 \cong (\mathbb{Z}[i]/(2))/(\bar{1} + \bar{i})$$

Consider $\mathbb{Z}[i]/(2)$:

$$\begin{aligned} a+bi \pmod{2} &= \{0, 1\} + \{0, i, \bar{i}\} \\ \mathbb{Z}[i]/(2) &\text{ has 4 elements.} \end{aligned}$$

$$(1-i)^2 \bar{a} \equiv (1-i)^2 \bar{a} \pmod{2} \quad \bar{a} \in \mathbb{Z}[i]$$

$$\bar{a} \in \mathbb{Z}[i]/(2) \Rightarrow \bar{a} \in \{0, \bar{i}, i, \bar{i} + i\}$$

$$\Rightarrow \mathbb{Z}[i]/I_2 \cong \mathbb{Z}/2\mathbb{Z} \text{ as additive abelian groups.}$$

$$\Rightarrow \langle \pm i \rangle \cong \langle \pm i \rangle \text{ as cyclic group or } \mathbb{Z}[i]/I_2 \cong \mathbb{Z}/2\mathbb{Z} \text{ as rings.}$$

$$\begin{cases} \phi: \mathbb{Z} \longrightarrow \mathbb{Z}/2 \\ \pi \longmapsto \pi \pmod{2} \\ \text{Suppose } g \in \mathbb{Z}/2. \text{ Then } \mathbb{Z}/2 \text{ is cyclic, } g = \pi \pmod{2} \\ \Rightarrow \phi \text{ is surjective ring hom. with } \ker \phi = 2\mathbb{Z} \text{ since } \mathbb{Z}/2 \\ \mathbb{Z}/2 \cong \mathbb{Z}/2 \text{ as rings.} \end{cases}$$

- $ra+mb \in (a,b)$
 $\Rightarrow (ra+mb) = (a,b) \Rightarrow ra+mb = \gcd(a,b)$
 $\Rightarrow R$ is a Bezout domain $\Rightarrow$ if finitely generated ideal is principal.

Suppose R has a property that every non-unit nonzero element can be factored into irreducibles.

- Suppose a is irreducible and $\nexists d \in R$ with $a \mid d$ such that d is irreducible
 $\Rightarrow a \mid 1 \Rightarrow a + 0 = 1 \Rightarrow a \in R$ $a \nmid 1 \Rightarrow a \nmid 0 \Rightarrow a \nmid d \Rightarrow d$ is prime.
 $\Rightarrow$ by the same method as for 2-irreducibles $\Rightarrow$ every R is a UFD.

(11) $\Rightarrow$ Suppose R is a UFD $\Rightarrow R$ is a UFD and obviously Bezout.

(12) Suppose R is UFD and Bezout.

Suppose $I \neq 0$ is an ideal in R with $\gcd$ of its elements d .

$\Rightarrow d \in I$ obviously.

Suppose $d \in I \Rightarrow d \mid a \Rightarrow a = dk \Rightarrow d \mid a \Rightarrow d$ is the number of irreducible factors $\Rightarrow a \mid d \Rightarrow d \mid a$.

Polynomial Rings

Prop 1. Let R be an integral domain. Then for $R[x]$ following are true:

- (1) $\deg(RD) \leq \deg(R) + \deg(D)$
 (2) $\deg(RD) = \deg(R) + \deg(D)$ if R is a field.
 $\Rightarrow \deg(RD) = \deg(R) + \deg(D)$ if R is a field $\Rightarrow \deg(RD) = \deg(R) + \deg(D)$ if R is a field.
 (3) $R[x]$ is an integral domain.
 $\Rightarrow R[x]$ is an integral domain.

Prop 2. I ideal of $R[x]$ $\Rightarrow I \cap R$ ideal of R .

Then $R[x]/I \cong (R/I)[x]$.

$\Rightarrow$ if I is prime ideal of R then $I[x]$ is prime ideal of $R[x]$.

Prop 3. $R[x] \rightarrow (R/I)[x]$ (reduction of coefficients modulo I)

$\Rightarrow$ is a ring homomorphism.

Then $R[x]/I[x] \cong (R/I)[x]$.

If I is a prime ideal of R then $(R/I)[x]$ is an integral domain.

$\Rightarrow R[x]/I[x]$ is an integral domain.

$\Rightarrow I[x]$ is a prime ideal.

Examples

Ex 1. $R = \mathbb{Z}$.

$\mathbb{Z}[x]/(2) \cong (\mathbb{Z}/2\mathbb{Z})[x]$.

If n is composite, then $\mathbb{Z}[x]/(n) \cong (\mathbb{Z}/n\mathbb{Z})[x]$ is not integral domain.

If n is prime, then $\mathbb{Z}[x]/(n) \cong (\mathbb{Z}/n\mathbb{Z})[x]$ is an integral domain.

Ex 2. $R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n]$.

$R[x_1, \dots, x_n]$ is a free sum of elements of the form $a_0 x_1^{d_1} \dots x_n^{d_n}$.

where $a_i \in R$, $x_1^{d_1} \dots x_n^{d_n}$ is called a monomial.

$d_1, \dots, d_n$ is the degree of the monomial term.

$(d_1, \dots, d_n)$ is the multi-degree of the term.

$R[x_1, \dots, x_n]$ is called a homogeneous or a form of all monomial terms.

If f is a nonzero polynomial in n variables, the sum of all monomial terms in f of degree k is called the homogeneous component of f of degree k .

If f has a degree d , then f may be uniquely written as the sum $f_0 + f_1 + \dots + f_d$.

where f_k is a homogeneous component of f of degree k for each k .

Example $\mathbb{Z}[x, y] = \left\{ \sum_{i,j} a_{ij} x^i y^j \mid a_{ij} \in \mathbb{Z} \right\}$.

$$f(x, y) = 2x^2 + 3xy - y^2$$

$$g(x, y) = -3xy + 2y^2 + x^2y$$

$$f(x, y) + g(x, y) = 2x^2 + 3xy - y^2 - 3xy + 2y^2 + x^2y = 2x^2 + y^2 + x^2y$$

$$R[x, y] \cdot R[x, y] = \text{a polynomial of degree } 3.$$

Examples

$$(1) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(2) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(3) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(4) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(5) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(6) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(7) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(8) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(9) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(10) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(11) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(12) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(13) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(14) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(15) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(16) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(17) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$(18) f(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$$g(x, y) = x^2y - xy^2 = x^2y^2 - xy^3$$

$\Rightarrow \lambda^2(x^2)/(x^2-x^2) = 0$ is an integral domain $\Rightarrow \lambda^2(x^2)$ is a prime ideal

(16) let $f(x_1, \dots, x_n)$ is a homogeneous polynomial of degree k in $R[x_1, \dots, x_n]$

$$\begin{aligned} \text{let } \lambda \in R \quad f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) &= \sum_{i_1, \dots, i_n} c_{i_1, \dots, i_n} (\lambda x_1)^{i_1} \dots (\lambda x_n)^{i_n} \\ &= \sum_{i_1, \dots, i_n} c_{i_1, \dots, i_n} \lambda^{i_1 + \dots + i_n} (x_1)^{i_1} \dots (x_n)^{i_n} \\ &= \sum_{i_1, \dots, i_n} c_{i_1, \dots, i_n} \lambda^k (x_1)^{i_1} \dots (x_n)^{i_n} = \lambda^k f(x_1, \dots, x_n) \end{aligned}$$

(17) let $f, g \in R[x_1, \dots, x_n]$ be homogeneous of degree k_1, k_2 respectively

consider a monomial term of degree k_1 $x_1^{i_1} \dots x_n^{i_n}$ with $i_1 + \dots + i_n = k_1$

$$\begin{aligned} f(x_1, \dots, x_n) x_1^{j_1} \dots x_n^{j_n} &= \sum_{i_1, \dots, i_n} c_{i_1, \dots, i_n} x_1^{i_1} \dots x_n^{i_n} (x_1^{j_1} \dots x_n^{j_n}) \\ &= \sum_{i_1, \dots, i_n} c_{i_1, \dots, i_n} x_1^{i_1+j_1} \dots x_n^{i_n+j_n} \end{aligned}$$

has degree $k_1 + k_2$ and is homogeneous

Then fg is a linear sum of monomial terms of degree k_2

we have fg is homogeneous of degree $k_1 + k_2$

(18) Ideal I in $R[x_1, \dots, x_n]$ is a linear sum of homogeneous components $I = \sum_{k=0}^{\infty} I_k$

[let $f_1, f_2, \dots, f_r$ are I and $h_1, h_2, \dots, h_s \in I$]

I is homogeneous iff it is generated by homogeneous polynomials

(19) Suppose I is homogeneous. let $J \subseteq I$ be the ideal generated by all homogeneous elements of I

Suppose $f \in I \Rightarrow f_1, \dots, f_d \in I$ and $f_1, \dots, f_d$ are homogeneous

$\Rightarrow f_1, \dots, f_d \in J$, thus I is an ideal of J

(20) Suppose I is generated by homogeneous polynomials

$$f \in I \quad f = q_1 f_1 + \dots + q_r f_r \quad \text{where } q_1, \dots, q_r \text{ are elements in } R[x_1, \dots, x_n]$$

$f_1, \dots, f_r$ are homogeneous elements of the generating set

then $f \in R$ $f = f_1 + \dots + f_d$ $f_1, \dots, f_d$ are homogeneous

forall we have $q_i = \sum_{j=0}^d q_{i,j} x^j$ where $q_{i,j}$ is homogeneous with degree j

$$f = \sum_{j=0}^d f_j x^j = \sum_{j=0}^d q_{j,0} x^j \quad [\text{product of homogeneous is homogeneous}]$$

$$\Rightarrow (f_0 - q_{0,0}) + \dots + (f_d - \sum_{i=1}^d q_{i,0} x^i) = 0$$

$$\Rightarrow f_1, \dots, f_d \in I \Rightarrow I \text{ is homogeneous.}$$

(21) let R be arbitrary ring. let $\text{Func}(R) = \{f: R \rightarrow R\}$ ring.

if $f, g \in \text{Func}(R)$ let $f, g \in \text{Func}(R)$ $\Rightarrow f, g: R \rightarrow R$ $f(x) = f(x)$

(*) $\text{Func}(R) \rightarrow R$ $x \in R$

$$f(x) + g(x) = f(x) + g(x) = f(x) + g(x) = f(x) + g(x)$$

$$f(x) \cdot g(x) = f(x) \cdot g(x) = f(x) \cdot g(x) = f(x) \cdot g(x)$$

$\Rightarrow \text{Func}(R)$ is a ring hom.

(22) consider $R = \mathbb{R}$ ring of real polynomials

if $q: R[x] \rightarrow \text{Func}(R)$ is a ring hom.

Then $q(x) + q(x) = 2q(x)$ is also a ring hom.

let $p(x) = x^2 + 1 \in R[x]$ then

$$q(p(x)) = q(x^2 + 1) = q(x^2) + q(1)$$

$$q_j(f(x) + g(x)) = q_j(f(x)) + q_j(g(x)) = (j+1)q_j(f(x)) + (j+1)q_j(g(x)) = j^2 + 1 + j + 1 = j^2 + 2j + 2 \neq 0, \text{ then } j(x) = j(x)$$

$$\text{However, } q_j \in \text{Func}(R) = f_j(x) = f_j(x) = j^2 + 1 \neq 0 \quad \frac{1}{2}$$

[in non-commutative rings $q(x) = q(x)$ as polynomials $\neq q(x) = q(x)$ as functions.]

(23) let $a \in R$ fixed. let $q: R[x] \rightarrow \text{Func}(R)$ and $q(x) = f(x) \rightarrow R$

(24) Suppose $q(x) + q(x)$ is a ring hom.

let $R = (x-x)/(x+x) \in R[x]$

$$(x-x)/(x+x) = x^2 + x - x^2 + x = 2x \in R[x]$$

$$\Rightarrow q(x) + q(x) = q(x) + q(x) = q(x) + q(x) = q(x) + q(x)$$

$$\Rightarrow q \in C(R)$$

(25) Suppose $q \in C(R)$

let $f, g \in R[x]$

$q(x) + q(x)$ is a ring hom as add/mul action (trivial)

$f \cdot q(x) = f(x) \cdot q(x)$ since $q \in C(R)$

$\Rightarrow q(x) + q(x)$ is a ring hom.

2.2 Polynomial Rings over Fields

let F be a field

let $N: F[x] \rightarrow \mathbb{N} \cup \{0\}$

$$N(f(x)) = \deg(f(x))$$

$$\text{and } N(0) = 0$$

Theorem 1 $F[x]$ is a Euclidean Domain.

let $a(x), b(x) \in F[x]$ $b(x) \neq 0$

$$\text{if } a(x) = 0 \quad \text{then } a(x) = q(x)b(x) + r(x) \quad \text{with } q(x) = 0, r(x) = 0$$

$$\text{Suppose } a(x) \neq 0 \quad a(x) = a_n x^n + \dots + a_0 \quad b(x) = b_m x^m + \dots + b_0$$

$$\text{if } n < m, \text{ then take } q(x) = 0 \text{ and we have } r(x) = a(x)$$

Suppose $n \geq m$

$$\text{then } a'(x) = a(x) - \frac{a_n}{b_m} b(x) \quad \text{has degree } n-1$$

$\frac{a_n}{b_m}$ is not defined since $b_m \neq 0$ and coefficients belong to the field.

By induction, $a'(x) = q'(x)b(x) + r(x)$ with $r(x) = 0$ or $\deg(r(x)) < \deg(b(x))$

$$\deg(a(x), r(x))$$

$$a(x) = a'(x) + \frac{a_n}{b_m} b(x) = \left(q'(x) + \frac{a_n}{b_m} \right) b(x) + r(x) \quad \text{with } r(x) = 0 \text{ or } \deg(r(x)) < \deg(b(x))$$

for uniqueness, suppose $q'(x), r'(x)$

$$\Rightarrow a(x) = q'(x)b(x) + r'(x) = q(x)b(x) + r(x)$$

$$\Rightarrow (q'(x) - q(x))b(x) + (r'(x) - r(x)) = 0$$

$$\Rightarrow (q'(x) - q(x))b(x) = (r(x) - r'(x))$$

$$\deg(r(x) - r'(x)) = \deg(q'(x) - q(x)) + \deg(b(x)) < \deg(b(x))$$

$$\Rightarrow \deg(q'(x) - q(x)) = 0 \Rightarrow q'(x) - q(x) = a_0 \in F$$

$$\text{since } \deg(r(x) - r'(x)) < \deg(b(x))$$

$$\text{since } \deg(q'(x) - q(x))b(x) < m \Rightarrow q'(x) - q(x) = 0$$

$$x f(x) - f(x) = 0$$

corollary 4. If F is a field, then $F[x]$ is P.I.D, D.F.D.

Example 1

(1) $\mathbb{Z}[x]$ is not P.I.D since $\mathbb{Z}$ is not a field. [(x) prime $\Rightarrow$ maximal P.I.D. $\frac{\mathbb{Z}[x]}{(x)} \cong \mathbb{Z}$ is not principal]

(2) $\mathbb{Q}[x]$ is P.I.D since $\mathbb{Q}$ is a field

(3) If p is prime, $\mathbb{Z}/p\mathbb{Z}[x] \cong \mathbb{F}_p[x]$ is P.I.D since $\mathbb{Z}/p\mathbb{Z}$ is a field.

$$(2, x) \text{ mod } (3) = (x) \text{ in } \mathbb{Z}/3\mathbb{Z}[x] \Rightarrow \text{proper non-unit ideal.}$$

$$\exists \text{ pol. } s \text{ is a unit in } \mathbb{Z}/3\mathbb{Z} \Rightarrow (2, x) \text{ mod } (3) = \mathbb{Z}/3\mathbb{Z}[x]$$

(4) $\mathbb{Q}[x, y]$ is not P.I.D since $\mathbb{Q}[x]$ is not a field.

* $q(x), r(x)$ w.r.t. to div algorithm are independent of field extensions.

suppose $F \subseteq E$ and $a(x) = q(x)b(x) + r(x)$ in $E[x]$.

$$a(x) = q(x)b(x) + r(x) \text{ in } F[x]$$

$$\text{by uniqueness in } E[x] \quad q(x) = q(x) \quad r(x) = r(x)$$

In particular, $\exists \text{ } b(x) | a(x)$ in $E[x]$ $\Leftrightarrow \exists b(x) | a(x)$ in $F[x]$.

$$\gcd(a(x), b(x)) \text{ in } E[x] = \gcd(a(x), b(x)) \text{ in } F[x] \quad [\text{remainders are same}]$$

Exercises

(1) $f(x) \in F[x]$ of $\deg n \geq 1$.

then let $g(x) \in F[x]$. since $F[x]$ is euclidean, $q(x) = q(x)f(x) + r(x)$ $r(x) = 0$ or $\deg(r) < \deg(f)$

$$\Rightarrow \overline{q(x)} = \overline{r(x)}$$

$r(x)$ is the universal residue rep. of degree $\leq n-1$, also unique.

Furthermore, $1, x, \dots, x^{n-1}$ are unique residues

in $F[x]/(f(x))$. $\Rightarrow F[x]/(f(x)) \cong \text{span}_F(1, x, \dots, x^{n-1})$ since every $\overline{q(x)} \in \text{span}_F(1, x, \dots, x^{n-1})$

(2) let F be a finite field of order p . $f(x) \in F[x]$ of $\deg n \geq 1$.

$$\text{then } F[x]/(f(x)) \cong \text{span}_F(1, x, \dots, x^{n-1}) \Rightarrow |F[x]/(f(x))| = |F|^n = p^n$$

(3) $f(x) \in F[x]$ $F[x]/(f(x))$ is a field $\Leftrightarrow \deg(f)$ is maximal.

$\Leftrightarrow f(x)$ is prime in P.I.D $F[x]$

$\Leftrightarrow f(x)$ is irreducible in a U.F.D $F[x]$.

(4) let F be a finite field.

suppose $F[x]$ has finitely many irreducibles (prime in U.F.D).

Let $f(x) \in F[x]$ $f(x) = p_1(x) \dots p_k(x)$ where p_i are irreducibles.

claim: $f(x) + 1$ is irreducible.

$\exists \text{ } p_i(x) | f(x) + 1$ implies then $p_i(x) | 1$ $\Rightarrow p_i(x)$ is a unit $\frac{1}{p_i}$

$\Rightarrow F[x]$ has infinitely many irreducibles hence prime.

(5) show $F[x]$ is a D.F.D

$$f_1(x), f_2(x), \dots, f_n(x)$$

$$(f_1(x)) \subseteq (f_2(x)) \subseteq \dots \subseteq (f_n(x)) \quad \text{so } n \geq 1$$

$(f_1(x)) \subseteq (f_2(x)) \subseteq \dots \subseteq (f_n(x))$ are all the ideals in $F[x]/(f_n(x))$

(6) (i) $\mathbb{Z}[x]/(x^2) \cong \mathbb{Z}/x^2\mathbb{Z}$ char 2

(ii) $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$ char 0

(iii) $\mathbb{Z}[x]/(x^2) \cong \{a+bx \mid a, b \in \mathbb{Z}\}$ char 0

(iv) $\mathbb{Z}[x, y]/(x^2+y^2-1) \cong \{a+by+cx^2 \mid a, b, c \in \mathbb{Z}/2\mathbb{Z}\}$ char 2

$$(a+by+cx^2)^2 = (a^2 + (by)^2 + (cx^2)^2 + 2abx + 2acyx + 2bcx^2y + (by)^2 = 2acd + 2bcd + 2bcx^2y = d \in \mathbb{Z}/2\mathbb{Z}$$

$$+ (ax+by+cx^2)^2 = 0$$

$$+ (ax+by+cx^2+1)^2 = 1.$$

(7) $\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}/x^2\mathbb{Z} / (x^2+1)$

$$(x^2+1) = (x+1)(x^2+x+1) \text{ in } \mathbb{Z}/x^2\mathbb{Z}$$

$\Rightarrow (1, x+1), (2, x^2+x+1)$ are ideals in $\mathbb{Z}[x]/(x^2+1)$

(8) $a(x) = x^2-2$ $b(x) = x+1$ in $\mathbb{Q}[x]$

$$x^2-2 = q(x)(x+1) - 3$$

$$x^2-2 = x^2+x-3$$

$$= (x^2-x+1)(x+1) - 3$$

$$(x+1) = 3(\frac{x^2-x+1}{3}) + 0$$

$$(3) = (1) \Rightarrow \gcd(x^2-2, x+1) = 1$$

$$1 = \frac{1}{3}(x+1)(x^2-x+1) - \frac{1}{3}(x^2-2)$$

(9) $a(x) = x^6 + 2x^3 + x^2 + x + 1$

$$b(x) = x^2 + x^2 + 2x^2 + 2x^2 + 2x + 2$$

$$x^6 + x^4 + 2x^3 + 2x^2 + 2x + 1 \div \frac{x^6 + 2x^3 + x^2 + x + 1}{-(x^4 + x^2 + 2x^2 + 2x + 1)}$$

$$-x^4 - x^2 - x \div \frac{x^4 + x^2 + 2x^2 + 2x + 1}{-x^4 - x^2 - x}$$

$$x^4 + x^2 + 1 \div \frac{x^4 + x^2 + 2x^2 + 2x + 1}{-x^4 - x^2 - x}$$

$$a(x) = 1 \cdot b(x) + (-x^4 - x^2 - x)$$

$$b(x) = -(x+1)(-x^4 - x^2 - x) + (x^6 + x^2 + x + 1)$$

$$(-x^4 - x^2 - x) = (-x) \cdot (-x^3 - x + 1) + 0$$

$$\Rightarrow \gcd(a(x), b(x)) = (-x^3 - x + 1)$$

$$a(x) - b(x) = (-x^4 - x^2 - x)$$

$$-(x+1)(a(x)-b(x)) = -(x+1)(-x^2-x^2-x) = b(x) - (x^2+x+1)$$

$$\bullet \quad (x+1)a(x) + (x+1)b(x) = x^3+x+1$$

$$(2) \quad f(x), g(x) \neq 0 \in \mathbb{Q}[x] \quad \gcd(f(x), g(x)) = d(x)$$

$$(a) \quad \text{let } h(x) \in \mathbb{Q}[x]$$

$$\text{suppose } \exists a(x), b(x) \in \mathbb{Q}[x] \text{ s.t. } a(x)f(x) + b(x)g(x) = h(x) \\ \Rightarrow h(x) \in (f(x), g(x)) = (d(x)) \quad (\text{Euclidean}) \\ \Rightarrow d(x) \mid h(x).$$

$$\text{suppose } d(x) \mid h(x) \Rightarrow h(x) \in (f(x), g(x)) \Rightarrow \exists a(x), b(x) \in \mathbb{Q}[x] \quad h(x) = a(x)f(x) + b(x)g(x).$$

$$(c) \quad \text{suppose } a_0(x), b_0(x) \text{ are particular solutions to the eqn } a(x)f(x) + b(x)g(x) = h(x)$$

$$(a(x)-a_0(x))f(x) = (b_0(x)-b(x))g(x) \\ \mathbb{Q}[x] \text{ is Euclidean, } \left(\frac{f(x)}{d(x)}, \frac{g(x)}{d(x)}\right) = (1) \quad (\text{Ex 4 rec 2.1})$$

$$(a(x)-a_0(x))\frac{f(x)}{d(x)} = (b_0(x)-b(x))\frac{g(x)}{d(x)}$$

$$\text{Then, } \frac{f(x)}{d(x)} \mid (b_0(x)-b(x))\frac{g(x)}{d(x)} \Rightarrow \frac{f(x)}{d(x)} \mid (b_0(x)-b(x)) \\ \Rightarrow b(x) = b_0(x) - m(x)\frac{f(x)}{d(x)}$$

$$\text{and} \\ a(x) = a_0(x) + m(x)\frac{f(x)}{d(x)} \quad \text{where } m(x) \text{ ranges over the } \mathbb{Q}[x]$$

$$(2) \quad \text{let } F[x_1, x_2, \dots] \text{ be an infinite polynomial ring over field } F. \\ \text{let } I = (x-y_1^2, x-y_2^2, \dots, x-y_n^2, \dots) \text{ ideal in ring.} \\ \text{Def } R = F[x_1, x_2, \dots] / I:$$

$$y_i - y_{i+1}^2 \equiv 0 \pmod I \Rightarrow y_i = y_{i+1}^2 \\ y_i^2 = x \pmod I \\ x = y_i^2 \\ \text{Quotient } \tilde{y}_i = x^{1/2^i} \\ R_1 = \text{subring generated by } F \text{ and } x^{1/2}$$

$$(3) \quad R_1 = \text{subring generated by } F \text{ and } x^{1/2}. \quad F \subseteq R_2 \quad x^{1/2} \cdot (x^{1/2}) (x^{1/2}) \in R_2 \Rightarrow R_1 \subseteq R_2 \\ \text{suppose } R_1 \subseteq R_2 \subseteq \dots \subseteq R_n \\ R_{n+1} = \text{subring generated by } F \text{ and } x^{1/2^{n+1}}, \quad (x^{1/2^{n+1}})^2 = x^{1/2^n} \Rightarrow R_n \subseteq R_{n+1}$$

$$\Rightarrow R_1 \subseteq R_2 \subseteq \dots \\ \text{Def } [x^{1/2}, x^{1/4}, \dots] = \bigcup_{n=1, \dots, \infty} \langle x^{1/2^n} \rangle \\ \text{since } \langle x^{1/2^n} \rangle \subseteq R_n \text{ for some } n \\ \Rightarrow R \subseteq \bigcup R_n.$$

$$(4) \quad \text{let } S_n = F[x_1, \dots, x_n] / (x-y_1^2, x-y_2^2, \dots, x-y_{n-1}^2) \\ \text{then, } S_n \cong F[x_1, \dots, x_n] / (x-y_1^2, \dots, x-y_{n-1}^2) \\ = F[x_{n-1}, y_n] / (y_{n-1}-y_n^2) \cong F[y_n]. \quad (\text{cumulative second set of eqns})$$

$$\text{Suppose } f(y_n) = 0 \text{ in } R_n. \text{ then, } f(y_n) = 0 \text{ involves finitely many variables } \{ f(y_n) = a_n y_n^{m_1} + \dots + a_0 = 0 \\ a_n y_n^{m_1} + \dots + a_0 = 0 \text{ in } S_N \quad N \times n \}$$

$$\text{Then, } f(\tilde{y}_n) = 0 \text{ in } S_N \text{ for some } N > n \\ \text{Then, } f(y_n) = 0 \text{ in } F[y_n] \Leftrightarrow f = 0 \Rightarrow R_n \cong F[y_n]$$

$$\text{since } F[y_n] \text{ is P.I.D. } \Rightarrow R_n \text{ is P.I.D. } \Rightarrow R_n \text{ is Bezout.}$$

$$(5) \quad \text{suppose } I = (x, x^{1/2}, x^{1/4}, \dots) \text{ is finitely generated by } r_1, \dots, r_n \in R \\ I = (r_1, \dots, r_n) \\ \text{then } r_i \text{ is contained in some } R_N \subseteq R \\ \Rightarrow I \subseteq R_N. \text{ then } R_N \text{ is P.I.D. } I = (a) \text{ for some } a \in R_N \\ \text{let } x^{1/2^{m+1}} \in I: \text{ however } x^{1/2^{m+1}} \notin R_N = x^{1/2^m} R_N \Rightarrow \frac{1}{2}$$

$$(6) \quad \text{let } F \text{ be field of char } p \text{ with } F^p \subseteq F \\ \text{let } a \in F(x) \\ \text{exp } \sum_{i=0}^n a_i x^i \text{ with } a_i \in F \\ \text{with } \sum_{i=0}^n a_i x^i + \sum_{i=0}^n b_i x^i = \sum_{i=0}^n (a_i + b_i) x^i \\ \text{and } \left(\sum_{i=0}^n a_i x^i\right) \left(\sum_{j=0}^m b_j x^j\right) = \sum_{k=0}^{n+m} \left(\sum_{i+j=k} a_i b_j\right) x^k \\ [x a = a' x \quad a \in F]$$

$$\text{let } h(x) = \log(x) \text{ be the term on } R. \\ (a) \quad x^a = e^{a \log x} = x^{a'} a' x = (a')^{p+1} x^a \quad (\text{induction}) \\ \text{monomial distributivity} \\ a x^a ((b+c) x^b) \\ = a(b+c)^{p^k} x^{a+b} \\ = a(b^p + c^p) x^{a+b} \\ = a b^p x^{a+b} + a c^p x^{a+b} \\ = (a x^a)(b x^b) + (a x^a)(c x^b)$$

$$\text{monomial associativity} \\ a x^a (b^p x^{c p^m}) = a x^a (b^p c^{p^m}) \\ = a(b^p c^{p^m}) x^{a+p^m} \\ = a x^a c^{p^m} x^{a+p^m} \\ = a x^a c^{p^m} x^{a+p^m} \\ = a x^a c^{p^m} x^{a+p^m} \\ = a x^a c^{p^m} x^{a+p^m} \\ = a x^a c^{p^m} x^{a+p^m}$$

$$\Rightarrow R \text{ is a ring.}$$

$$(b) \quad \text{suppose } h \in R. \\ p q = \sum_{i=0}^n \left(\sum_{j=0}^m a_j b_j^p\right) x^i \Rightarrow \deg(pq) = n + m = \deg(p) + \deg(q)$$

Suppose $p \neq 0$

$$N(p) = \deg(p) = \deg(1) + \deg(1) = N(1) = 0$$

$$\Rightarrow \deg(1) = 0 \neq \deg(1) = 0$$

$$\Rightarrow p, q \in F \text{ which is a field so } p=q=0.$$

$\Rightarrow R$ has no zero divisors.

$$(c) \text{ if } g \in R, g \neq 0 \quad \frac{1}{g} f = 0 \text{ then } f = 0 \cdot g = 0$$

$$\text{Suppose } \deg f < \deg g \Rightarrow f = 0 \cdot g + r(x) \text{ where } r(x) = \{0\} \quad \deg f(x) < \deg g(x)$$

Suppose $\deg f > \deg g$.

$$f = \sum_{i=0}^n a_i x^i \quad g = \sum_{j=0}^m b_j x^j$$

Then $f(x) = g(x) \cdot q(x) + r(x)$ where $r(x) = 0$ or $\deg r(x) < \deg g(x)$
 and $r(x) = 0$.

Then by induction, $f(x) = q(x)g(x) + r(x)$ with $r(x) = 0$ or $\deg r(x) < \deg g(x)$

$$\text{Then, } f(x) = q(x)g(x) + \sum_{i=0}^m b_i x^i + r(x) \\ = (q(x) + \sum_{i=0}^m b_i x^i)g(x) + r(x) \text{ with } r(x) = 0 \text{ or } \deg r(x) < \deg g(x) \\ = g(x)g(x) + r(x)$$

$\Rightarrow R$ is right euclidean.

Suppose R is left ideal of R .

Then R is euclidean, $\forall g \in R, g \neq 0$ $\exists q, r \in R$ with $f = qg + r$ with $r(x) = 0$ or $\deg r(x) < \deg g(x)$

Suppose $\deg r = 0$ or $\deg r < \deg g$ then R is left ideal.

Then $\deg(r) < \deg(g) \Rightarrow \deg(r) = 0$ then we are done. Suppose $\deg(r) = 0$.

Then $\deg(r) = 0 \Rightarrow \deg(r) = 0$ then we are done. Suppose $\deg(r) = 0$.

Then, $\deg(r) = 0 \Rightarrow \deg(r) = 0$ then we are done. Suppose $\deg(r) = 0$.

Then, $\deg(r) = 0 \Rightarrow \deg(r) = 0$ then we are done. Suppose $\deg(r) = 0$.

Then, $\deg(r) = 0 \Rightarrow \deg(r) = 0$ then we are done. Suppose $\deg(r) = 0$.

(d) if $f \in R$ for some $R \in F$ with $R \in F$, $\deg f = 0$.

Suppose $\deg f = 0$ then $f = 0 \cdot g + r$ with $r(x) = 0$ or $\deg r(x) < \deg g(x)$

$$\frac{1}{g} f = 0 \Rightarrow f = 0 \cdot g = 0$$

$$\Rightarrow f = 0 \cdot g = 0$$

$$\frac{1}{g} f = 0$$

$$\Rightarrow \deg f = 0 \text{ or } \deg f < \deg g$$

$$\Rightarrow (f \cdot g) = 0 \text{ or } (f \cdot g) = 0$$

$\Rightarrow R$ is not left euclidean.

Suppose R is right ideal generated by $r, 0 \in R$ is principal generated by $r \in R$.

$$r \cdot q = 0 \quad q, r \in R$$

$$(r \cdot 1) = 0 \Rightarrow r = 0$$

$$\frac{1}{g} f = 0 \Rightarrow f = 0 \cdot g = 0$$

$$\frac{1}{g} f = 0 \Rightarrow f = 0 \cdot g = 0$$

$$\frac{1}{g} f = 0 \Rightarrow f = 0 \cdot g = 0$$

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$$\frac{1}{g} f = 0 \Rightarrow f = 0 \cdot g = 0$$

Then, R is not principal.

$\Rightarrow R$ is not right PID $\Rightarrow R$ is not left euclidean with respect to any norm.

9.3 Polynomial rings that are U.F.D

Prop 5 (Gauss' Lemma) Let R be a U.F.D with $F = \text{Frac}(R)$.

Let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x] \Rightarrow p(x)$ is reducible in $R[x]$.

More precisely, if $p(x) = A(x)B(x)$ for some non-constant polynomials $A(x), B(x) \in F[x]$.

Then $\exists r, s \in F$ s.t. $rA(x) = a(x)$ and $sB(x) = b(x)$ are in $R[x]$ and

$$p(x) = a(x)b(x) \text{ is a factorization in } R[x].$$

Prf Suppose $p(x)$ is reducible in $F[x] \Rightarrow \exists A(x), B(x) \in F[x] \quad \deg A(x), \deg B(x) \geq 1$.

Let $p(x) = A(x)B(x)$. Coefficients of $A(x), B(x)$ can be written as quotients of elements from R .

Multiplying both sides of a common denominator below get, $a(x) = a'(x)b(x)$ where $a'(x), b(x)$

have coefficients in R s.t. $a'(x)b(x) \in R[x]$.

If $d \in R^* \Rightarrow p(x) = d \cdot a'(x)b(x)$ is a factorization in $R[x]$.

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$\forall d \in R \Rightarrow d \in R^* \dots \dots$ unique factorization into irreducibles (and primes)

Since R is prime in $R[x]$ $\left(\frac{R[x]}{pR[x]} \right)$ is an integral domain.

$\Rightarrow R[x]$ is prime in $R[x]$. $\Rightarrow \frac{R[x]}{pR[x]} = 0$ in $R[x]/pR[x]$

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By Theorem 7, $R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n]$ is a U.F.D. by induction.
 Since a polynomial ring in an arbitrary number of variables can be seen as a union of finitely variable rings, we have that it is a U.F.D.

Example 2:
 (a) $\mathbb{Z}[x]$ is a U.F.D. since $\mathbb{Z}$ is a U.F.D. ($\mathbb{Z}[x]$ is not a U.F.D.).
 (b) $\mathbb{Q}[x]$ is a U.F.D.

Suppose R any arbitrary integral domain.
 Suppose $p(x) \in R[x]$ is irreducible.
 (a) $p(x) = 0$ or $p(x) \in R$.
 (b) $p(x) \in R$ is a unit in R .
 (c) $p(x) \in R$ is a prime in R .

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 (a) $p(x) = 0$ or $p(x) \in R$.
 (b) $p(x) \in R$ is a unit in R .
 (c) $p(x) \in R$ is a prime in R .

that reduce to zero in $\mathbb{Z}[x]$.
These polynomials in $\mathbb{Z}[x]$ are of the form $a_0 + a_1x + \dots + a_nx^n$ and all nonconstant divide by p .

Prop 13 (Eisenstein's criterion) Let R be a prime ideal in integral domain R .
Let $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_0$ ($n \geq 1$) be an element in $R[x]$.
If $a_0, a_1, \dots, a_{n-1} \in R$ and $a_n \notin R$ then $f(x)$ is irreducible in $R[x]$.
If $f(x)$ is reducible then $f(x) = g(x)h(x)$ and $a_0 = g_0h_0$ and $a_n = g_ng_n$.
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Corollary 14 Let p be prime in $\mathbb{Z}$. Let $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$, $n \geq 1$. Suppose $p \mid a_i$ for $i = 0, \dots, n-1$ and $p \nmid a_n$. Then $f(x)$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.
If $f(x)$ is prime, $a_0 \notin (p)^2$, follows from Prop 13 $\Rightarrow f(x)$ is irreducible in $\mathbb{Z}[x] \Rightarrow f(x)$ is irreducible in $\mathbb{Q}[x]$.

Examples

(1) $x^2 + 10x + 5 \in \mathbb{Z}[x]$. $10, 5 \in (5)$. $5 \nmid 1$. Thus $x^2 + 10x + 5$ irreducible in $\mathbb{Z}[x]$ by Eisenstein.

(2) If a poly integer divisible by a prime p but not divisible by p^2 , then $x^n - a$ is irreducible in $\mathbb{Z}[x]$ for $a \in \mathbb{Z}^+$. In particular, $x^n - 2$ is irreducible in $\mathbb{Z}[x]$.
is by corollary 14 is irreducible in $\mathbb{Q}[x]$. $\Rightarrow$ it has no rational roots for $n \geq 2$.
(Check, i.e. if $x^n = 2$ for $n \geq 2$.)

(3) $f(x) = x^3 + 1$.
consider $g(x) = f(x+1) = x^3 + 3x^2 + 3x + 2$. is irreducible mod 2 $\Rightarrow$ irreducible in $\mathbb{Z}[x]$.
 $g(x)$ and $f(x)$ have the same factorization. $\Rightarrow f(x)$ is irreducible by Eisenstein.

(4) Let p be a prime.

$$f_p(x) = \frac{x^{p+1}-1}{x-1} = x^p + x^{p-1} + \dots + x + 1$$

$$f_p(x+1) = \frac{(x+1)^{p+1}-1}{x} = x^p + px^{p-1} + \dots + px + 1 \in \mathbb{Z}[x]$$

by Eisenstein, $f_p(x)$ is irreducible $\Rightarrow f_p(x)$ is irreducible in $\mathbb{Z}[x]$.

(5) Let $R = \mathbb{Z}[x]$ and n be any positive integer.
consider $X^n - x \in R[X]$.

$R = \mathbb{Z}[x] \cong \mathbb{Z}[x]$ is a PID $\Rightarrow (x)$ is prime.

$x \in (x)$ $x \notin (x)^2$ by Eisenstein $X^n - x$ is irreducible in $R[X]$.

Prop 13.2

- (1) $\Rightarrow$ $f(x) \in \mathbb{Z}[x]$ has a factor of degree $\leq n$ has a root in $\mathbb{Z}$ [on Euclidean algorithm]
- (2) deg $f(x)$ over $\mathbb{Z}$ is irreducible and it has a root in $\mathbb{Z}$ [follows from (1)]
- (3) $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$.
If $\frac{a_i}{a_n} \in \mathbb{Z}$ is a root in $\mathbb{Q}[x]$ then $r \mid a_0$ and $r \mid a_n$.
particular, $\frac{a_i}{a_n} \in \mathbb{Z}$ for all i , $\Rightarrow f$ has no roots in $\mathbb{Z}$.
- (4) $\mathbb{Z}$ is a local PID nonconstant monic.
then, $f(x)$ irreducible in $\mathbb{Z}[x]$ irreducible. [Eisenstein's lemma: non constant monic and prime p divides all coefficients except the leading one $\Rightarrow$ irreducible in $\mathbb{Z}[x]$]
- (5) Eisenstein. $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$.
 p prime in $\mathbb{Z}$, $a_0, \dots, a_{n-1} \in (p)$ but $a_n \notin (p)^2$ $\Rightarrow f(x)$ is irreducible in $\mathbb{Z}[x]$.

Exercises

(1) (a) $x^2 + x + 1 \in \mathbb{F}_3[x]$.

$$\deg f = 2 > 1 \Rightarrow \text{irreducible}$$

$$x^2 + x + 1 \in \mathbb{F}_3[x] \text{ same as above}$$

(b) $x^2 + x + 1 \in \mathbb{F}_3[x]$ same as above.

(c) $x^2 + 1 \in \mathbb{F}_3[x]$

$$x^2 + 1 = (x^2 + 2)(x^2 + 2) \text{ and are degree 2 with no roots in } \mathbb{F}_3 \text{ is irreducible}$$

(d) $x^2 + 10x + 5 \in \mathbb{Z}[x]$

$$x^2 + 10x + 5 = (x^2 + 5 + 5x)(x^2 + 5 + 5x) \text{ neither factor has a root in } \mathbb{Z}$$

$$\Rightarrow f(x) \text{ irreducible in } \mathbb{Z}[x]$$

(e) (a) $x^2 - 4x + 4$

$$\text{consider } f(x) = x^2 - 4x + 4 \in \mathbb{Z}[x] \Rightarrow \text{irreducible}$$

$$(b) x^2 + 3x + 2 = (x+1)(x+2)$$

$$\text{consider } f(x) = x^2 + 3x + 2 \in \mathbb{Z}[x], -1 \notin (3)$$

$$(c) x^2 + 2x + 1 = (x+1)^2$$

$$(d) x^2 + 2x + 1 = (x+1)^2$$

$$(e) x^2 + 2x + 1 = (x+1)^2$$

$$(f) x^2 + 2x + 1 = (x+1)^2$$

$$(g) x^2 + 2x + 1 = (x+1)^2$$

$$(h) x^2 + 2x + 1 = (x+1)^2$$

$$(i) x^2 + 2x + 1 = (x+1)^2$$

$$(j) x^2 + 2x + 1 = (x+1)^2$$

$$(k) x^2 + 2x + 1 = (x+1)^2$$

$$(l) x^2 + 2x + 1 = (x+1)^2$$

$$(m) x^2 + 2x + 1 = (x+1)^2$$

$$(n) x^2 + 2x + 1 = (x+1)^2$$

$$(o) x^2 + 2x + 1 = (x+1)^2$$

$$(p) x^2 + 2x + 1 = (x+1)^2$$

$$(q) x^2 + 2x + 1 = (x+1)^2$$

$$(r) x^2 + 2x + 1 = (x+1)^2$$

$$(s) x^2 + 2x + 1 = (x+1)^2$$

$$(t) x^2 + 2x + 1 = (x+1)^2$$

$$(u) x^2 + 2x + 1 = (x+1)^2$$

$$(v) x^2 + 2x + 1 = (x+1)^2$$

$$(w) x^2 + 2x + 1 = (x+1)^2$$

$$(x) x^2 + 2x + 1 = (x+1)^2$$

$$(y) x^2 + 2x + 1 = (x+1)^2$$

$$(z) x^2 + 2x + 1 = (x+1)^2$$

$$(aa) x^2 + 2x + 1 = (x+1)^2$$

$$(ab) x^2 + 2x + 1 = (x+1)^2$$

$$(ac) x^2 + 2x + 1 = (x+1)^2$$

$$(ad) x^2 + 2x + 1 = (x+1)^2$$

$$(ae) x^2 + 2x + 1 = (x+1)^2$$

$$(af) x^2 + 2x + 1 = (x+1)^2$$

$$(ag) x^2 + 2x + 1 = (x+1)^2$$

$$(ah) x^2 + 2x + 1 = (x+1)^2$$

$$(ai) x^2 + 2x + 1 = (x+1)^2$$

$$(aj) x^2 + 2x + 1 = (x+1)^2$$

$$(ak) x^2 + 2x + 1 = (x+1)^2$$

$$(al) x^2 + 2x + 1 = (x+1)^2$$

$$(am) x^2 + 2x + 1 = (x+1)^2$$

$$(an) x^2 + 2x + 1 = (x+1)^2$$

$$n=4 \quad (n-1)(n-2)(n-3)(n-4) + 1$$

$$= (x^4 - 5x + 5)^2 \rightarrow \text{reducible}$$

note: max constant term $n! + 1$ is never a perfect square

$\Rightarrow f(x)$ is irreducible $\forall n \geq 4$

$\Rightarrow f(x)$ is irreducible $\forall n \geq 1, n \neq 4$

(5). $F_2[x]$

$$F_2[x]/(x) \cong F_2 \rightarrow x \text{ is prime, so } x \text{ is irreducible.}$$

$$\deg 1: x, x+1$$

$$\deg 2: x^2 + x + 1$$

$$\deg 3: x^3 + x + 1, x^3 + x^2 + 1$$

$$F_2[x]$$

$$\deg 1: x, x+1 \quad \text{no } F_2 \text{ roots}$$

$$\deg 2: x^2 + ax + b \quad \text{no } F_2 \text{ roots} \quad \begin{cases} (x+1)(x+2) = x^2 + 2x + 2 \\ (x+1)^2 = x^2 + 2x + 1 \\ (x+2)^2 = x^2 + 4x + 4 \end{cases}$$

$$\text{no } x^2+1, x^2+x+2, x^2+2x+2$$

$$\deg 3: (x^2+1)(x+a) = x^3 + ax^2 + x + a \quad \text{no } F_2$$

$$(x^2+x+2)(x+a) = x^3 + ax^2 + x^2 + ax + 2x + 2a = x^3 + (a+1)x^2 + (a+2)x + 2a$$

$$(x^2+2x+2)(x+a) = x^3 + ax^2 + 2x^2 + 2ax + 2x + 2a = x^3 + (a+2)x^2 + 2(a+1)x + 2a$$

$$x^3+x^2+x+1, x^3+2x^2+x+2, x^3+1, x^3+2$$

$$x^3+2x^2+2, x^3+x+1, x^3+x+2, x^3+x^2+1 \quad \text{are reducible}$$

$$x^3+2x^2+x+1, x^3+2x^2+2x+1, x^3+x^2+2x+2, x^3+x^2+x+2$$

$$x^3+2x^2+1, x^3+2x+1, x^3+2x+2, x^3+x^2+2x \quad \text{are irreducible}$$

$$(6) |F_2[x]/(\text{irreducible in } F_2[x] \text{ of degree } n)| = 2^n = 2 \quad (n=1)$$

$$\text{eg } 1: 2^1 = 2 \quad |F_2[x]/(\deg 1 \text{ irreducible})| = 2^1 = 2$$

$$\text{eg } 2: 2^2 = 4 \quad |F_2[x]/(\deg 2 \text{ irreducible})| = 2^2 = 4$$

$$\text{eg } 3: 2^3 = 8 \quad |F_2[x]/(\deg 3 \text{ irreducible})| = 2^3 = 8$$

$$(7) \mathbb{R}[x]/(x^2+1)$$

$$\bar{x}^2 = -1 \pmod{x^2+1}$$

$$\begin{array}{ccc} \mathbb{R}[x] & \xrightarrow{\varphi} & \mathbb{C}[x] \xrightarrow{\psi} \mathbb{C} \\ \downarrow & \nearrow & \\ \mathbb{R}[x]/(x^2+1) & & \end{array}$$

$$(8) R_1 = F_2[x]/(x^2+1), R_2 = F_2[x]/(x^2+xy+2)$$

y^2+xy+2, x^2+1 are irreducible in $F_2[x]$, but they don't have roots in $F_2(y), F_2(x)$ respectively

$\Rightarrow R_1, R_2$ are fields of order $11^2 = 121$

$$\text{Suppose } f(x) = f_1(x)$$

$$ax^2 + b = cx^2 + d \quad a=c, b=d \in R_1$$

$$\text{then } \varphi(f(x)) = a\bar{x}^2 + b = c\bar{x}^2 + d = \varphi(f_1(x)) \quad [R_1 \text{ a field}]$$

$\Rightarrow$ well defined.

$$\begin{aligned} \varphi(f_1(x) + f_2(x)) &= \varphi((a+c)x^2 + (b+d)) = (a+c)(\bar{x}^2 + 1) + (b+d) \\ &= \varphi(f_1(x)) + \varphi(f_2(x)) \end{aligned}$$

$$\varphi(f_1(x) \cdot f_2(x)) = \varphi(f_1(x)) \cdot \varphi(f_2(x))$$

$\Rightarrow$ ring homomorphism (sends units to units)

$$(9) x^2 - \sqrt{2}$$

$$\mathbb{Q}[\sqrt{2}] \text{ is a UFD}$$

$$x^2 - \sqrt{2} = (x + \sqrt{\sqrt{2}})(x - \sqrt{\sqrt{2}}) \quad \text{is irreducible in } \mathbb{Q}[\sqrt{2}][x]$$

$$\Rightarrow x^2 - \sqrt{2} \text{ is irreducible in } \mathbb{Q}[\sqrt{2}][x]. \quad (\text{corollary 4})$$

$$(10) f(x) = x^3 - 4x^2 + 8x + 2 \in \mathbb{Q}[\sqrt{2}][x]$$

$$\text{Suppose } \frac{p}{q} \in \mathbb{Q}[\sqrt{2}] \text{ is a root.}$$

$$\text{Then } p|2, q|1 \Rightarrow \frac{p}{q} \in \mathbb{Z}[\sqrt{2}]$$

$$\Rightarrow p = \pm 1, \pm 2, \pm \sqrt{2}, \pm 2\sqrt{2}$$

However, $f(x) \neq 0 \Rightarrow f$ has no roots in $\mathbb{Q}[\sqrt{2}]$ as f has no linear factor.

Suppose p, q are monic quadratics in $\mathbb{Q}[\sqrt{2}][x]$

$$\text{so } p, q \text{ are monic quadratics in } \mathbb{Z}[\sqrt{2}][x]$$

$$p = (x^2 + ax + b)(x^2 + cx + d) \quad \text{since } x^2 \text{ coefficient is 1}$$

$$\begin{cases} b+d-a^2-c^2 = 2 \\ a(a+b) = 8 \\ b+d = 2 \end{cases}$$

$$d = \frac{2}{b} \Rightarrow a\left(\frac{2}{b} - b\right) = 8, \quad b + \frac{2}{b} - a^2 = -4$$

$$\Rightarrow a = 4 \cdot \left(\frac{b^2 - 2}{b^2}\right), \quad b + \frac{2}{b} - 4\left(\frac{b^2 - 2}{b^2}\right) = -4 \quad (*)$$

$$b|2 \text{ in } \mathbb{Z}[\sqrt{2}] \Rightarrow b = \pm 1, \pm \sqrt{2}, \pm 2$$

plugging them in to $(*)$, we find contradiction.

$$(11) x^2 + y^2 + 1 \text{ in } \mathbb{Q}[x, y] = \mathbb{Q}[x][y] \quad (\mathbb{Q}[x] \text{ integral domain})$$

$$\mathbb{Q}[x, y]/(x) \cong \mathbb{Q}[y] \quad y^2 + 1 \in \mathbb{Q}[y] \quad y^2 + 1 \pmod{x}$$

$$(y+i)(y-i) = y^2 + 1 \quad \text{is irreducible in } \mathbb{Q}[y] \Rightarrow x^2 + y^2 + 1 \text{ is irreducible in } \mathbb{Q}[x, y]$$

$$(12) (\Leftarrow) \mathbb{Q}_p \text{ is irreducible } (\mathbb{Q}_p(x+1) \text{ is irreducible by Eisenstein})$$

$$(\Rightarrow) \text{ let } n \text{ be an integer}$$

Suppose $\mathbb{Q}_n(x)$ is irreducible in $\mathbb{Q}[x]$ Suppose n is not a prime.

$$n = pq \text{ for some } p, q \in \mathbb{Z} \text{ non-unit.}$$

$$\text{Then } \mathbb{Q}_n = \frac{(x^p)^q - 1}{x^p - 1}$$

$$x^p = \mathbb{Q}_p(x-1) + 1$$

$$\text{Then, } \mathbb{Q}_n = \frac{(\mathbb{Q}_p(x-1) + 1)^q - 1}{x^p - 1}$$

$$= \frac{\sum_{k=0}^q \binom{q}{k} (\mathbb{Q}_p(x-1))^k - 1}{x^p - 1}$$

$$= \frac{\sum_{k=1}^q \binom{q}{k} (\mathbb{Q}_p(x-1))^k}{x^p - 1} = \mathbb{Q}_p \left(\sum_{k=1}^q \binom{q}{k} (\mathbb{Q}_p(x-1))^{k-1} \right) \nmid$$

$f(x) = x^2 - 4$ $D_x(f) = 2x \neq 0$
 $(f, D_x(f)) = 0 \Rightarrow f$ has a multiple root.
 (2.3) $f(x) = x^2 + 12x + 16$ in $\mathbb{Z}/25\mathbb{Z}$ find roots. $(x+4)(x+4) = x^2 + 8x + 16 \equiv x^2 + 12x + 16 \pmod{25}$
 $x = -4$ is a root in $\mathbb{Z}/25\mathbb{Z}$
 (2.4) $\mathbb{Z}/25\mathbb{Z} \cong (\mathbb{Z}/5\mathbb{Z})[\alpha]$ α is a root of $x^2 + 12x + 16$ in $\mathbb{Z}/5\mathbb{Z}$ and $\mathbb{Z}/25\mathbb{Z}$.
 $\Rightarrow$ requirement that R is an integral domain is necessary as otherwise it would break prop 12.
 (2.5) $\mathbb{Z}/25\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ by C.R.T.
 $f(x) = x^2 + 12x + 16$
 $q(x) = q(x)q(x) = (x_1, x_2)(x_1, x_2) = (x_1^2, x_2^2)$
 $x = f(x) \pmod{25} = q(x)h(x)$ x is irreducible in $\mathbb{Z}/25\mathbb{Z}$ $\Rightarrow q(x) = \pm 1$ and $h(x) = x$
 $q(x) = \pm 1$ and $h(x) = \pm x$
 x is irreducible in $\mathbb{Z}/25\mathbb{Z} \Rightarrow q(x) = \pm 1$ and $h(x) = \pm x$
 $q(x) = \pm 1$ and $h(x) = \pm x$
 $\Rightarrow f = (3x+2)(2x+3)$ is the only factorization upto multiplication by units.

(2.6) $(3, x)$ in $\mathbb{Z}/6\mathbb{Z}$ is principal ideal $\Leftrightarrow (q(x), q(x))$ is a principal ideal in $\mathbb{F}_2[x] \times \mathbb{F}_3[x]$.

$(q(x), q(x)) = ((1, 0), (x, x)) = ((1, x))$ is principal $\Rightarrow (3, x)$ is principal.

(2.7) $\mathbb{Z}/30\mathbb{Z}[x] \cong \mathbb{Z}/2\mathbb{Z}[x] \times \mathbb{Z}/3\mathbb{Z}[x] \times \mathbb{Z}/5\mathbb{Z}[x]$ $(3, x) = (2x+3)$

$f(x) = x = (10x+21)(15x+16)(6x+26)$
 $(10x+21)(15x+16) = 150x^2 + 475x + 336 \pmod{30} \equiv 25x+6$ same degree.

$f(x) = (10x+21)(15x+16)(6x+26)$ $\begin{cases} f(x) \pmod{2} = x & \text{irreducible} \\ f(x) \pmod{3} = x^2 \end{cases}$

(2.8) $n = p_1 \dots p_k$ $\mathbb{Z}/n\mathbb{Z}[x] \cong \mathbb{Z}/p_1\mathbb{Z}[x] \times \dots \times \mathbb{Z}/p_k\mathbb{Z}[x]$.

x has one factorization unique upto units in $\mathbb{Z}/n\mathbb{Z}$ $\Leftrightarrow \prod_{i=1}^k (a_i x + b_i)$ a_i, b_i are zero divisor pairs in $\mathbb{Z}/p_i\mathbb{Z}$.

2.6 Polynomial Rings over Fields

Prop 15 $F[x]/(f(x))$ is a field $\Leftrightarrow f(x)$ is irreducible.

Prop 16 $g(x)$ is constant in $F[x]$
 $g(x) = f_1^{n_1} \dots f_k^{n_k}$ $f_1, \dots, f_k$ irreducibles distinct.
 then $F[x]/(g(x)) \cong F[x]/(f_1^{n_1}) \times \dots \times F[x]/(f_k^{n_k})$

Prop 17 f_1, f_2 are distinct irreducibles $\Rightarrow (f_1, f_2) = 1 \Rightarrow (f_1^{n_1}, f_2^{n_2}) = (1)$
 $\Rightarrow$ comaximal!

$(g(x)) = (f_1^{n_1}) \dots (f_k^{n_k})$
 By C.R.T, we have the desired result.

Def 1 $f(x)$ is divisible by $(x-a)^m$ but not by $(x-a)^{m+1}$ then a is a root of multiplicity m .

Prop 18 f has roots $a_1, \dots, a_k$ in F not necessarily distinct.
 then $f(x)$ has $(x-a_1)(x-a_2) \dots (x-a_k)$ as a factor.
 In particular, a polynomial f of degree n in $F[x]$ has at most n roots in F , even counted with multiplicity.

Prop 19 Suppose f has roots $a_1, \dots, a_k$ in F not necessarily distinct.
 Suppose a_i has multiplicity $m_i \geq 1$ in the distinct index $\{1, \dots, k\}$.

$(x-a_i)^{m_i} \mid f$ for each i
 $f \in ((x-a_1)^{m_1}) \cap ((x-a_2)^{m_2}) \dots ((x-a_k)^{m_k})$
 since they are coprime:
 $f = h(x)(x-a_1)^{m_1} \dots (x-a_k)^{m_k} \Rightarrow (x-a_1) \dots (x-a_k) \mid f$

Suppose $f \in F[x]$ and $\deg(f) = n$. Since $F[x]$ is a U.F.D.,
 f has at most n linear factors $\Rightarrow$ at most n roots.

Definition 1.

Def 1 $f(x) = 0$ then $f(x) = h(x)(x-a_1)$

Suppose now for $2 \leq i$.

$f(x) = h(x)(x-a_1)$, $h(x)$ has roots $a_2, \dots, a_k$.
 by induction, $(x-a_2)(x-a_3) \dots (x-a_k) \mid h(x) \Rightarrow (x-a_1)(x-a_2) \dots (x-a_k) \mid f$

Prop 19 A finite subgroup of $F^\times$ is cyclic. In particular, if F is finite, $F^\times$ is cyclic.

Proof Let G be a finite subgroup of $F^\times$.
 by Lagrange's theorem $|G|$ divides $|F^\times|$.
 $G \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \dots \times \mathbb{Z}_{n_k}$ where $n_1 | n_2 | \dots | n_k$ and $n_k = |G|$.
 then $n_k \mid n_i$ $\Rightarrow$ each $\mathbb{Z}_{n_i}$ contains n_i elements of order dividing n_i .
 If $k \geq 2$, then there would be more than n_k such elements.
 $\Rightarrow$ there would be more than n_k roots of $x^{n_k} - 1$ in field F , contradicting prop 18.
 $\Rightarrow k=1$ so the subgroup is cyclic.
 If F is a finite field then $F^\times$ is cyclic.
 Corollary 19 Let p be prime, $\mathbb{Z}/p\mathbb{Z}$ is a finite field $\Rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.

(f) let p be an odd prime in $\mathbb{Z}$. let a be a positive integer.

$$\mathbb{Z}[i] \text{ is a U.D. } \Rightarrow \mathbb{Z}[i] \text{ is a U.D.}$$

So similarly, $\text{norm}(P)$ is prime ideal of $\mathbb{Z}[i]$

$P \nmid (P^2)$ is by Eisenstein $\Rightarrow x^2 - p$ is irreducible over $\mathbb{Z}[i]$.

So similarly, $P \nmid (P^2)$ is prime in $\mathbb{Z}[i]$.

$\Rightarrow \mathbb{Z}[i]$ is by Eisenstein $\Rightarrow x^2 - p$ is irreducible over $\mathbb{Z}[i]$.

(g) $x^2 + 12x^4 + 18x + 6$

$$x = (1+i)(1-i)$$

$$a, b, c \in \mathbb{Z}(1+i) \quad b = 1-i \notin \mathbb{Z}(1+i)$$

By Eisenstein, $x^2 + 12x^4 + 18x + 6$ is irreducible over $\mathbb{Z}[i]$.

5. let q be a prime number. $\sum_{d|n} q(d) = n$

Suppose $n = p^m$ with $p \in \mathbb{Z}^+$

$$\sum_{d|n} q(d) = \sum_{i=0}^{m-1} q(p^{m-i}) = p^m - p^{m-1} + p^{m-2} - p^{m-3} + \dots - (p-1) + (1) = p^m$$

$$\left[\begin{array}{l} q(p^{m-i}) = p^{m-i} - p^{m-i-1} \\ q(p^{m-i-1}) = p^{m-i-1} - p^{m-i-2} \end{array} \right]$$

Suppose $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ with $p_i \neq p_j$

$$\text{Then } \sum_{d|n} q(d) = \sum_{d_1|p_1^{a_1}} q(d_1) \sum_{d_2|p_2^{a_2}} q(d_2) \dots \sum_{d_k|p_k^{a_k}} q(d_k) \quad (d_i, d_j \text{ relatively prime})$$

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$$

$$\sum_{d|n} q(d) = \sum_{d_1|p_1^{a_1}} q(d_1) \sum_{d_2|p_2^{a_2}} q(d_2) \dots \sum_{d_k|p_k^{a_k}} q(d_k)$$

By induction,

$$= p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} = n$$

let $\mathbb{Z}_n$ be a cyclic group of order n .

Since, $\mathbb{Z}_n$ contains a unique subgroup of order d dividing n .

$$|\mathbb{Z}_n| = |\mathbb{Z}_{d_1}| + |\mathbb{Z}_{d_2}| + \dots + |\mathbb{Z}_{d_k}|$$

$$n = \phi(d_1) + \phi(d_2) + \dots + \phi(d_k) \quad d_i | n$$

(h) let G be a finite subgroup of order n of P^* , P field.

Then, $\phi(d) = \text{ord}(d) \quad \forall d | n$. In particular, $\text{ord}(n) \geq 1$.

Let $\sum_{d|n} \text{ord}(d) \leq n$ has at most n roots in P .

$$\Rightarrow \sum_{d|n} \text{ord}(d) \leq n \quad [x + x^2 + \dots + x^n \text{ is zero of } x^n - 1]$$

$$\Rightarrow \sum_{d|n} \text{ord}(d) \leq \sum_{d|n} \phi(d) \quad [\text{previous exercise}]$$

$$\text{ord}(1) \leq \phi(1) = 1$$

Suppose true for some d . Then for $d' \neq d$

$$\sum_{d|N} \text{ord}(d) \leq \sum_{d|N} \phi(d)$$

$$\sum_{d|N} \text{ord}(d) + \text{ord}(d') \leq \sum_{d|N} \phi(d) + \phi(d')$$

$$\Rightarrow \text{ord}(d') \leq \phi(d')$$

$$\text{Since } \sum_{d|n} \text{ord}(d) = n = \sum_{d|n} \phi(d),$$

$$\text{ord}(d') - \phi(d') \leq 0$$

$$\sum_{d'|n} (\text{ord}(d') - \phi(d')) = 0 \Rightarrow \text{ord}(d') = \phi(d') \quad \forall d' | n$$

$$\Rightarrow \text{ord}(n) \geq 1 \Rightarrow G \text{ is cyclic.}$$

9). Suppose $|F| = n$ then $|F^*| = n-1$.

Then $P \neq F^*$

Suppose $1 \neq -1$ in F

Suppose $q: F^* \rightarrow F^*$ is an isomorphism

$$q(s+t) = q(s) \cdot q(t) = 0$$

$$s+t \neq 0 \text{ in } F^*$$

$$\Rightarrow \text{Ker } q = \{0\}$$

Suppose $1 = -1$ in F

$q: F^* \rightarrow F$ an isomorphism.

$$\forall a \in F^*$$

$$q(a^2) = q(a) \cdot q(a) = 0$$

$\Rightarrow a = a^{-1} \Rightarrow$ consider $\langle 1, a, b \rangle < F^*$ is a finite subgroup that is not cyclic. $\downarrow$